

Semiconductor Process Lab Notebook

Wafer Fabrication & Process Engineering Record

This Notebook Belongs To

Lab / Organization

How to Use This Notebook

1. Record every process run immediately, in ink.
2. Capture the full recipe: temp, pressure, power, gas flow, time.
3. Log in-line data, metrology results and any deviations.
4. Have a witness sign off on critical or non-standard runs.

Semiconductor Process Record

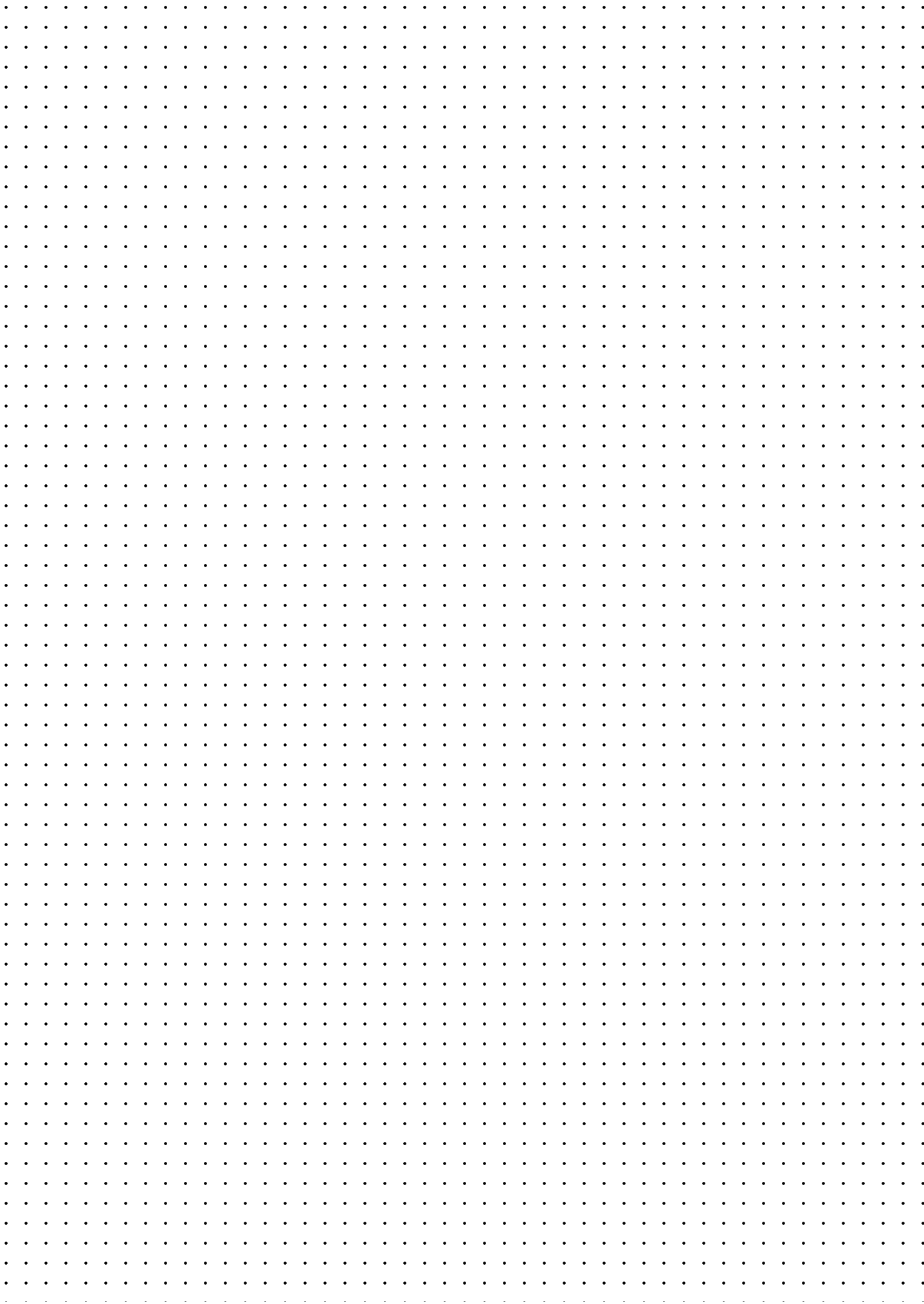
No. SAMPLE — DEMO

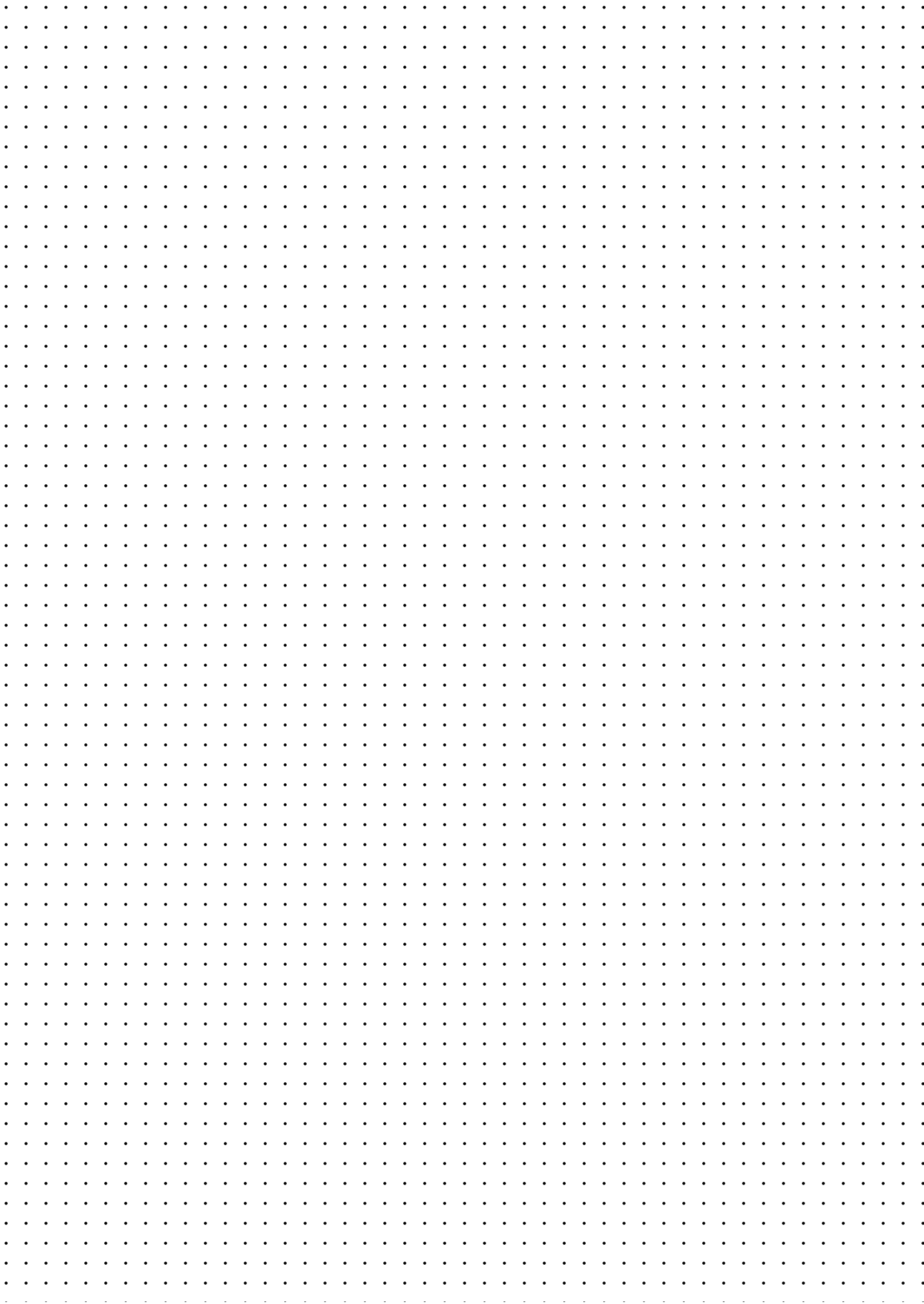
Date 2025-03-12		Operator J. Chen		Lot / Wafer ID LOT-A12 / W03																			
Tool / Station Oxford ICP-RIE #2			Process Step SiO₂ dry etch																				
Objective Etch 200 nm thermal oxide, stop on Si																							
Materials & Consumables 6" Si wafer, 200 nm thermal SiO₂, photoresist mask (AZ-5214)																							
Equipment Settings <table><tr><th>Parameter</th><th>Setting</th><th>Note</th></tr><tr><td>Pressure:</td><td>5 mTorr</td><td> ICP: 800W RF: 50W </td></tr><tr><td>Gas:</td><td>CHF₃/O₂ = 30/5 sccm</td><td> He bckgnd: 10 Torr Temp: 20°C</td></tr><tr><td colspan="3"></td></tr><tr><td colspan="3"></td></tr><tr><td colspan="3"></td></tr></table>						Parameter	Setting	Note	Pressure:	5 mTorr	ICP: 800W RF: 50W	Gas:	CHF ₃ /O ₂ = 30/5 sccm	He bckgnd: 10 Torr Temp: 20°C									
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Gas:	CHF ₃ /O ₂ = 30/5 sccm	He bckgnd: 10 Torr Temp: 20°C																					
Procedure 1. O₂ descum 30s 2. Main etch 120s 3. Overetch 30s 4. N₂ purge 60s 5. Vent & unload																							
Observations & In-line Data Optical end-point detected at 95s. DC bias stable at 180V. No arcing.																							
Metrology / Result Ellipsometry: oxide removed. Step height 205 nm (overetch 5nm into Si).			Deviations / Issues N/A — recipe within spec																				
Witness / Sign-off P. Liu (2025-03-12)																							

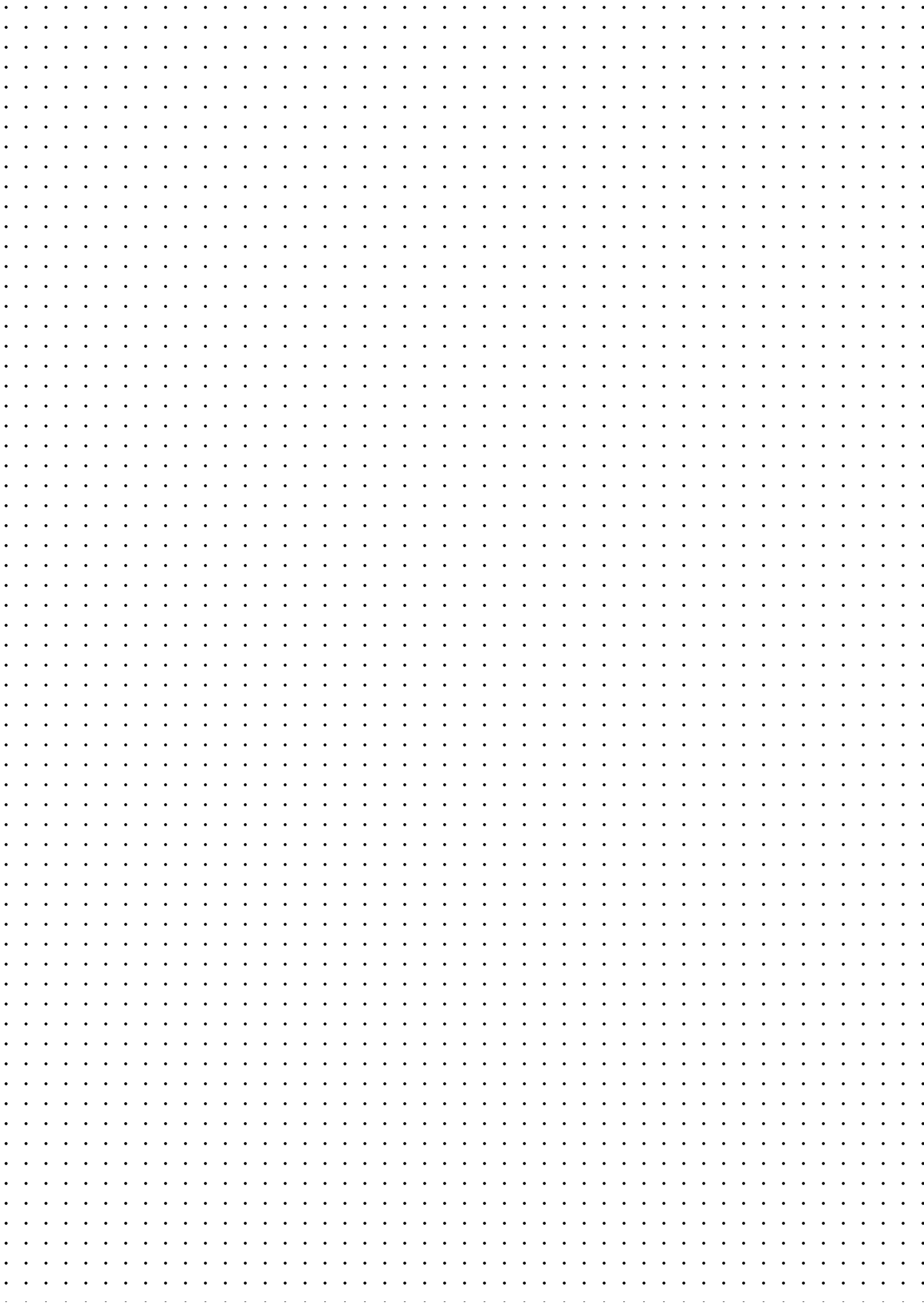
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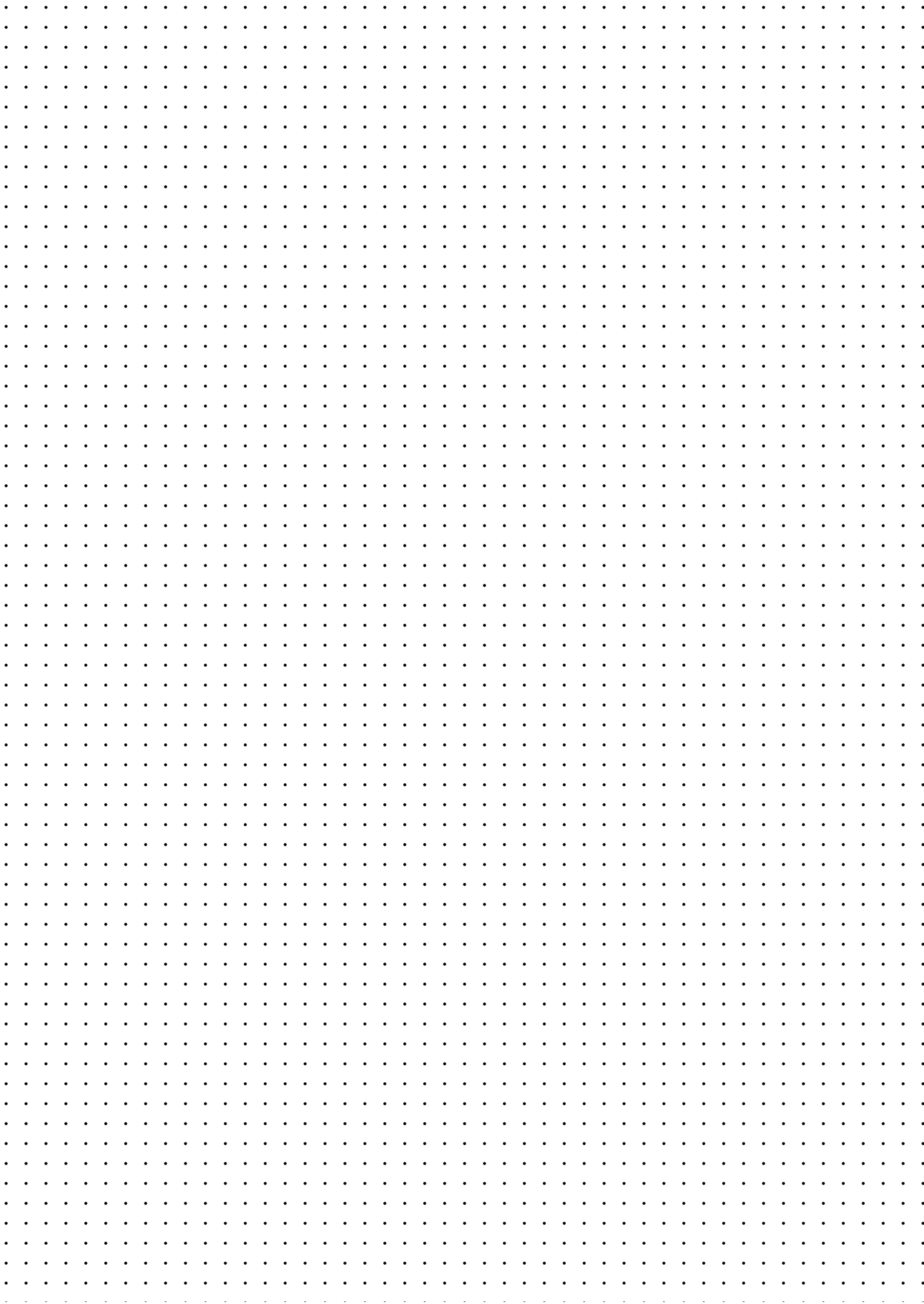


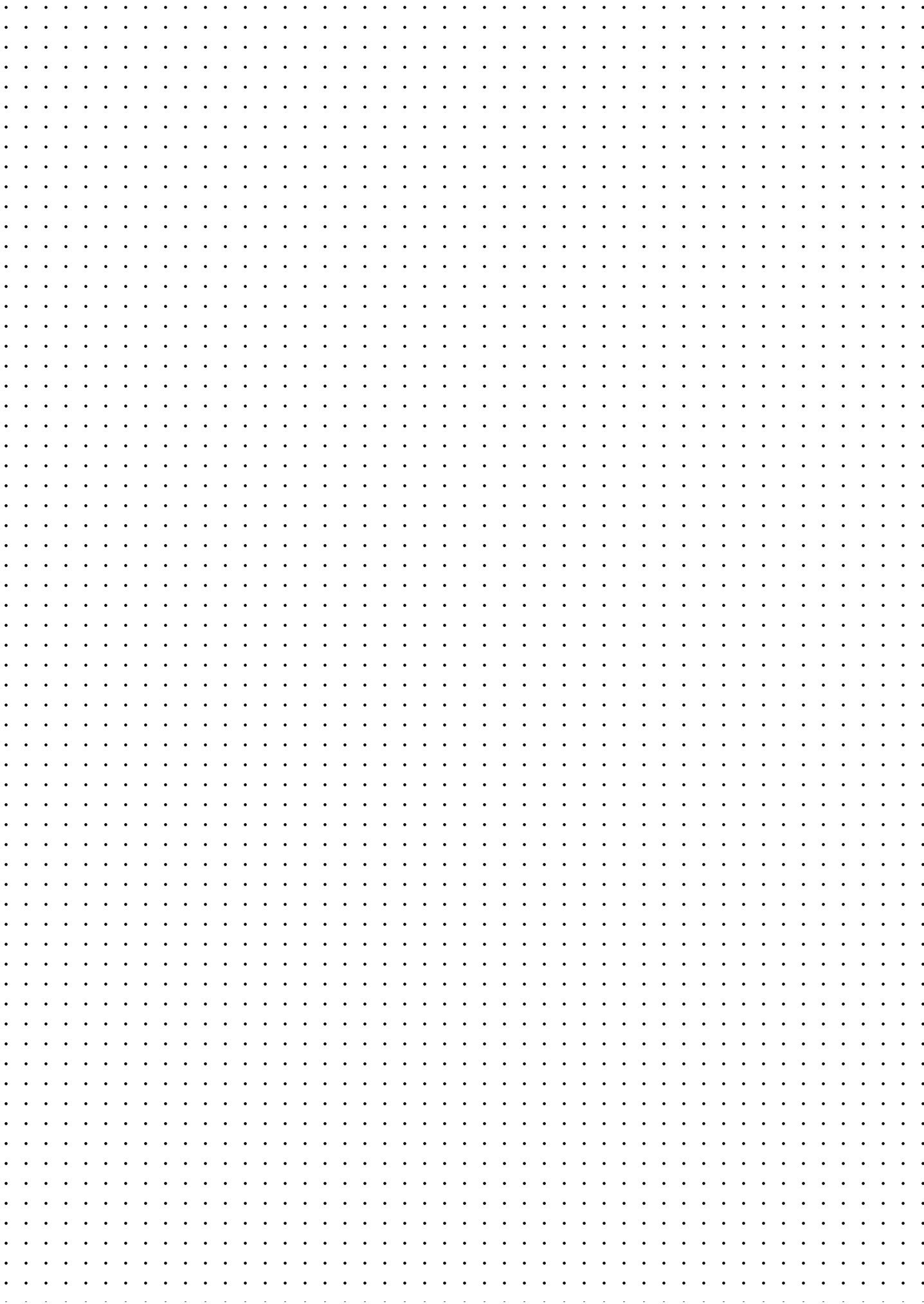


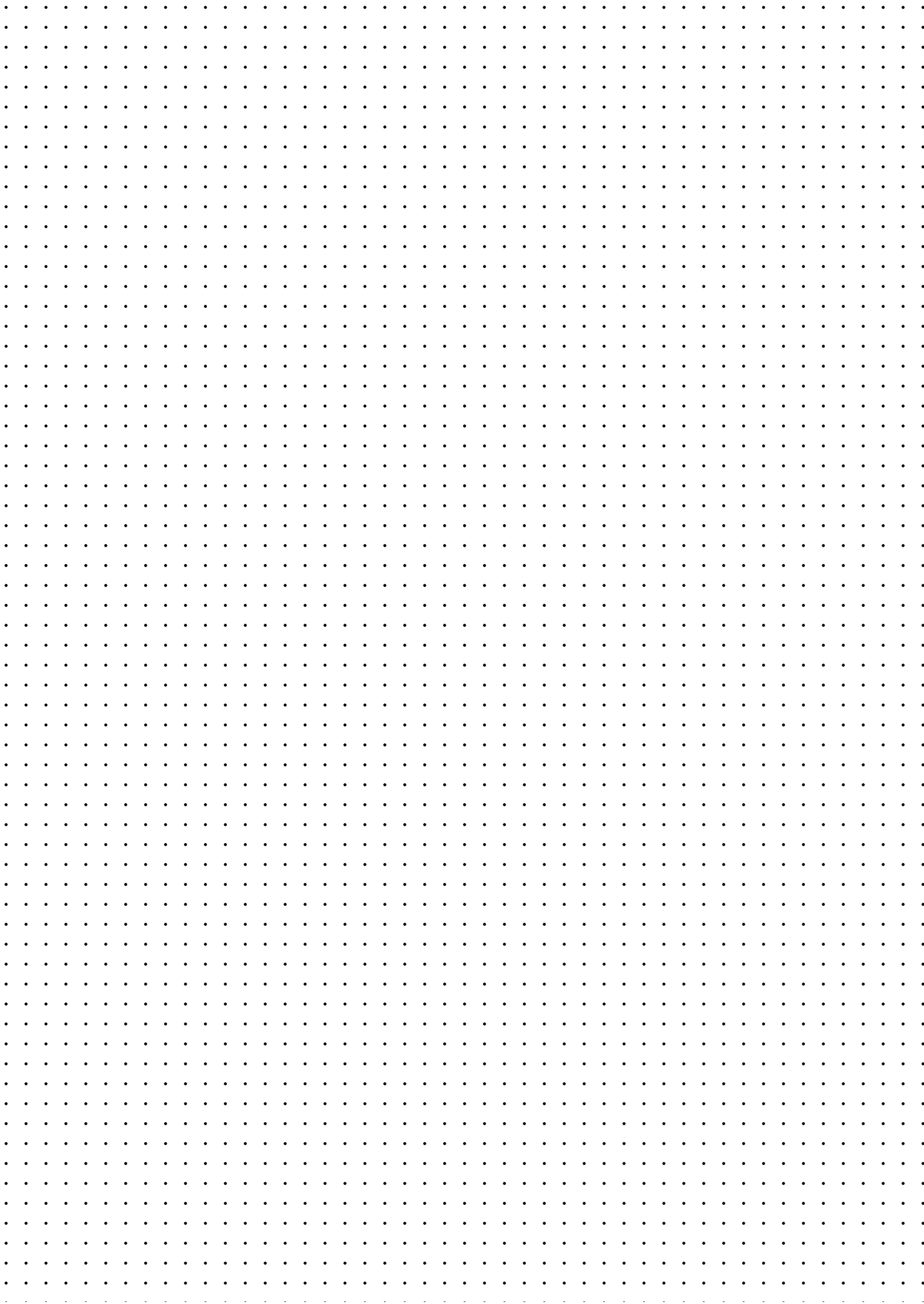
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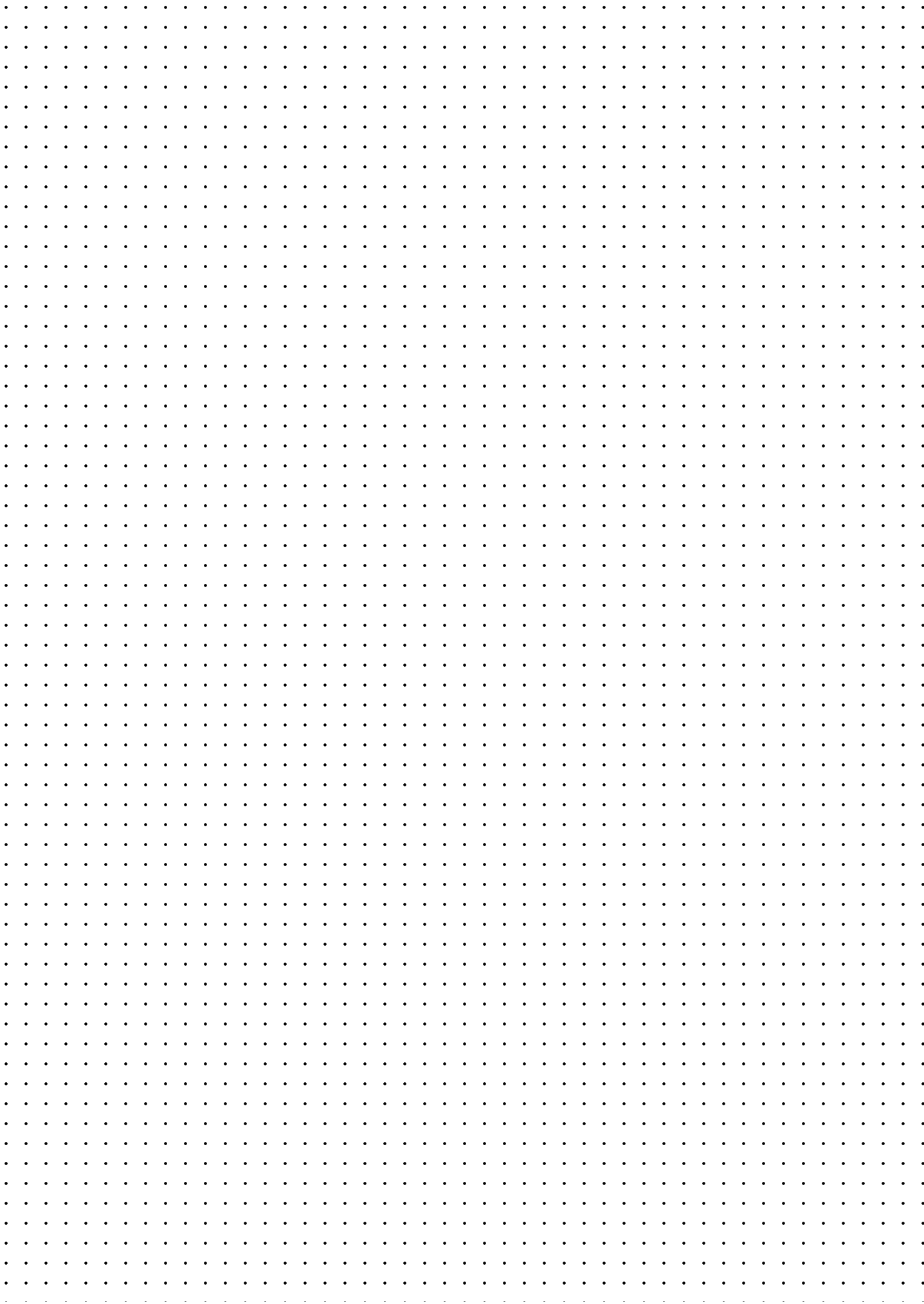
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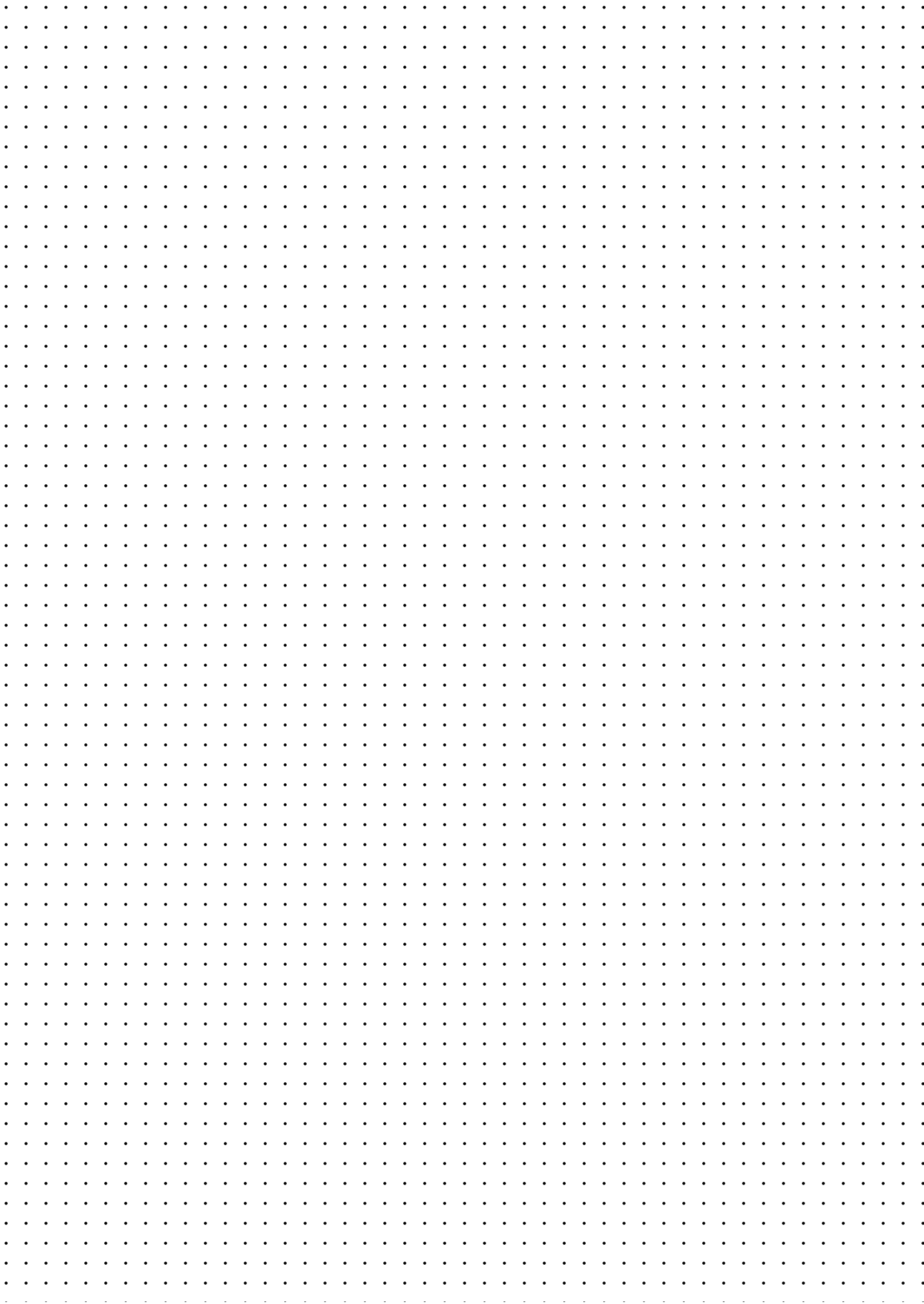
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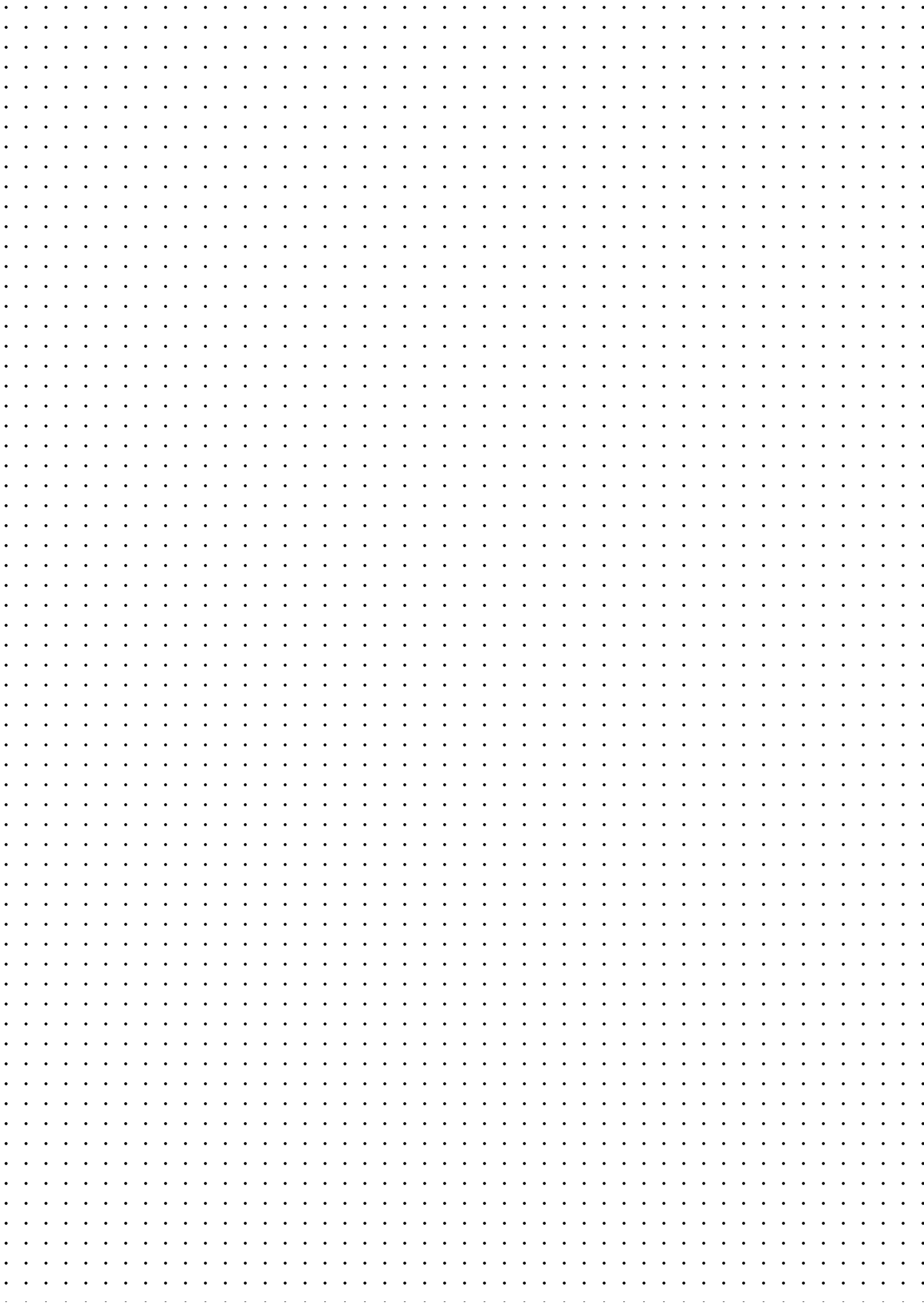
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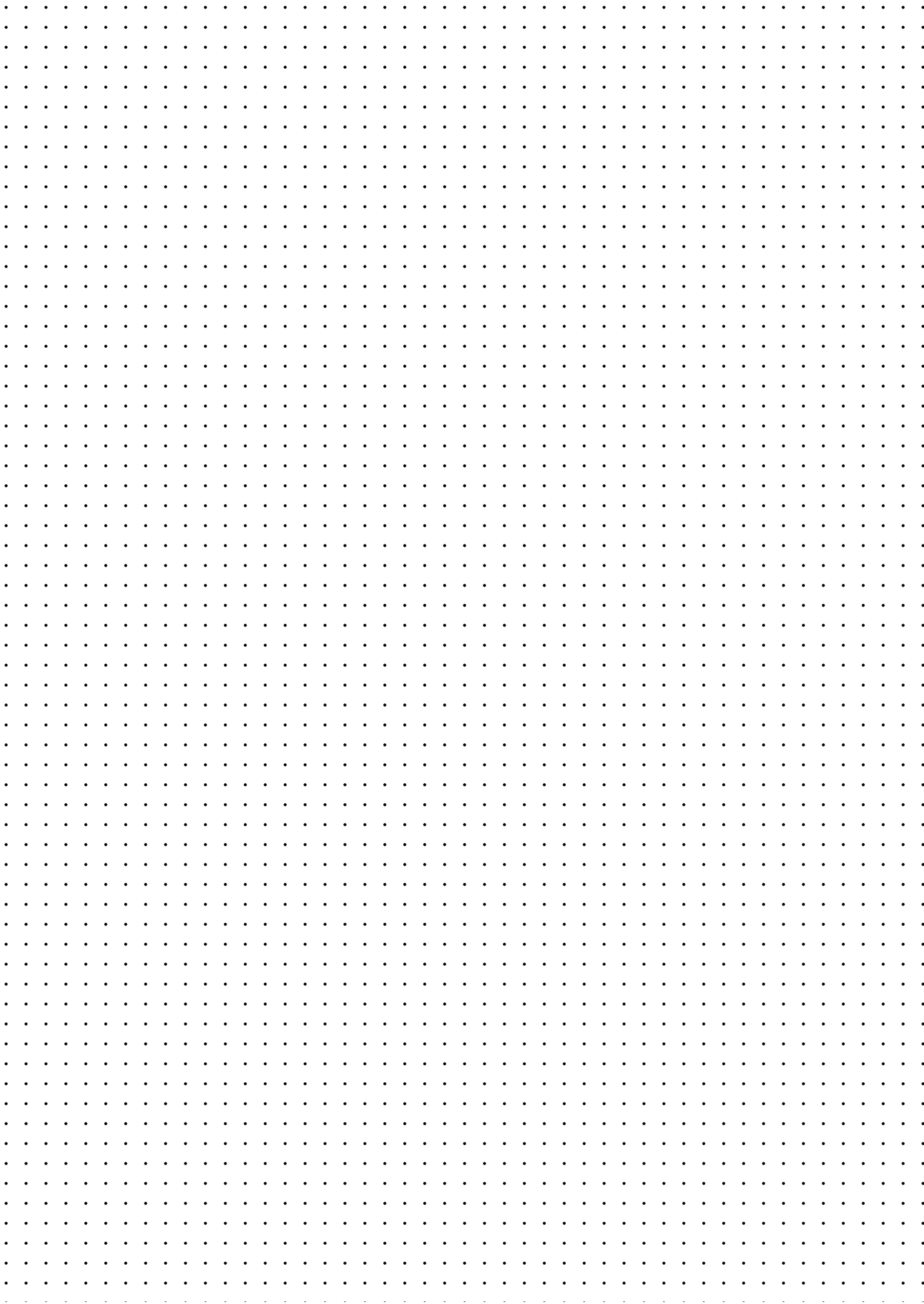


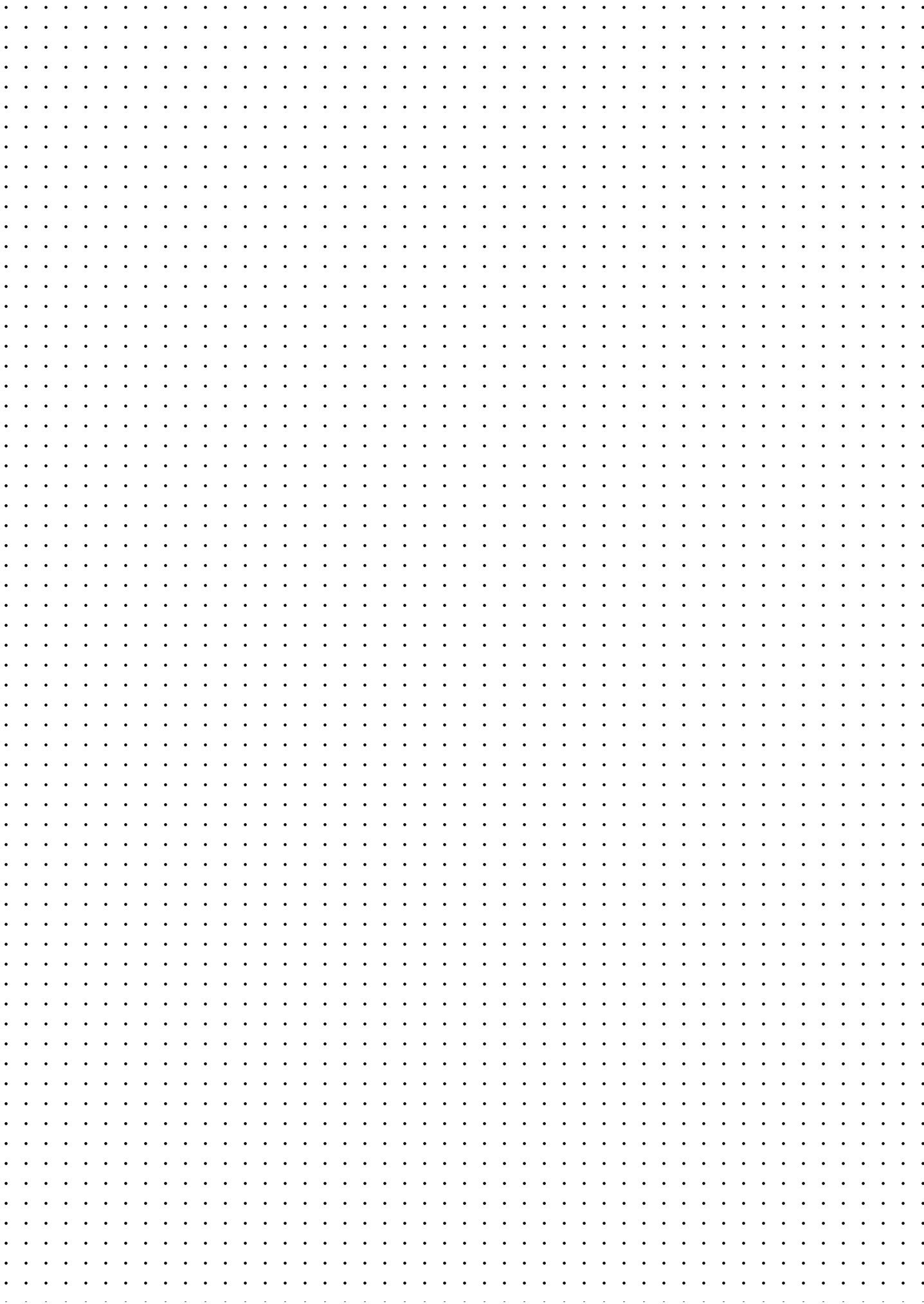


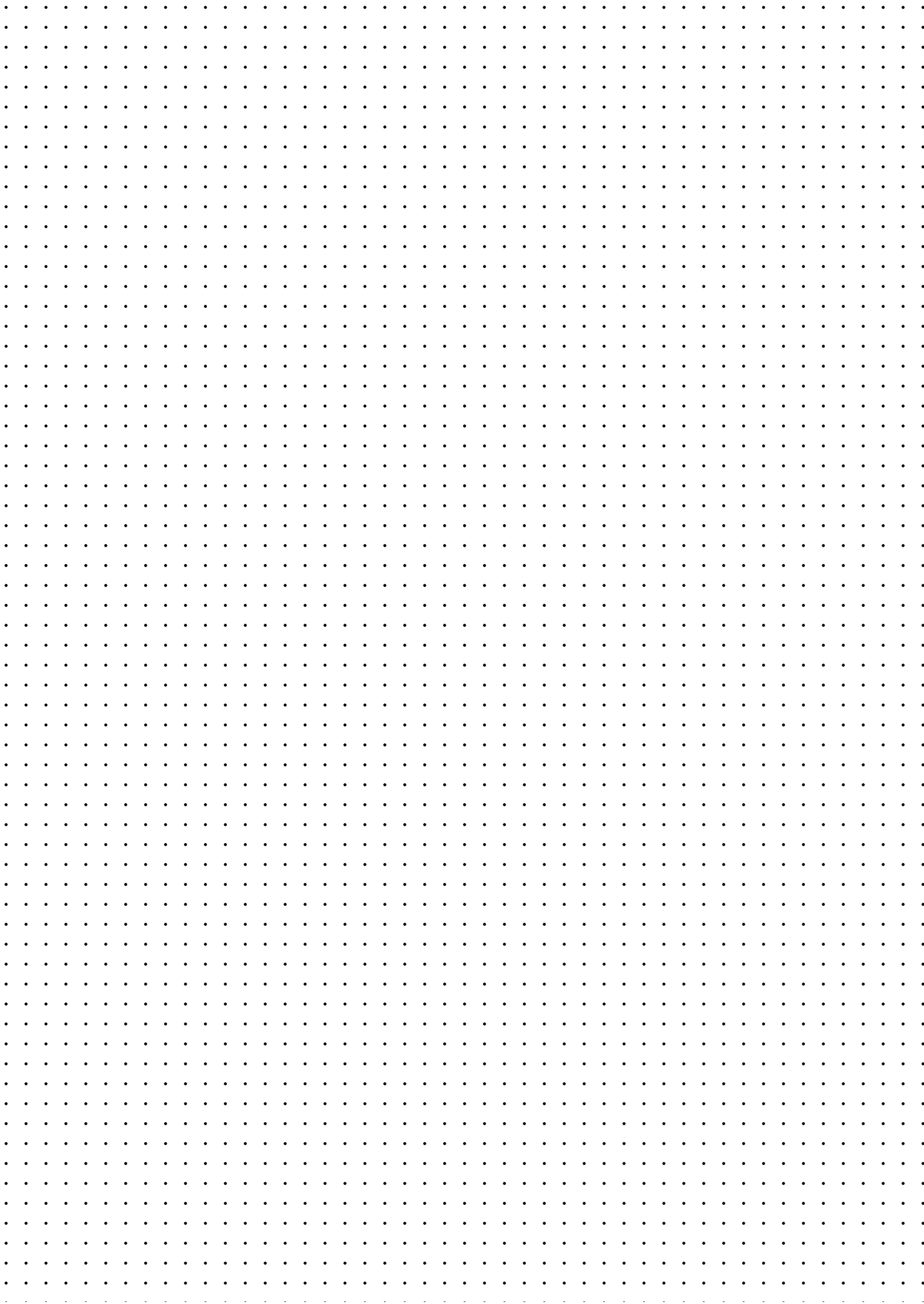
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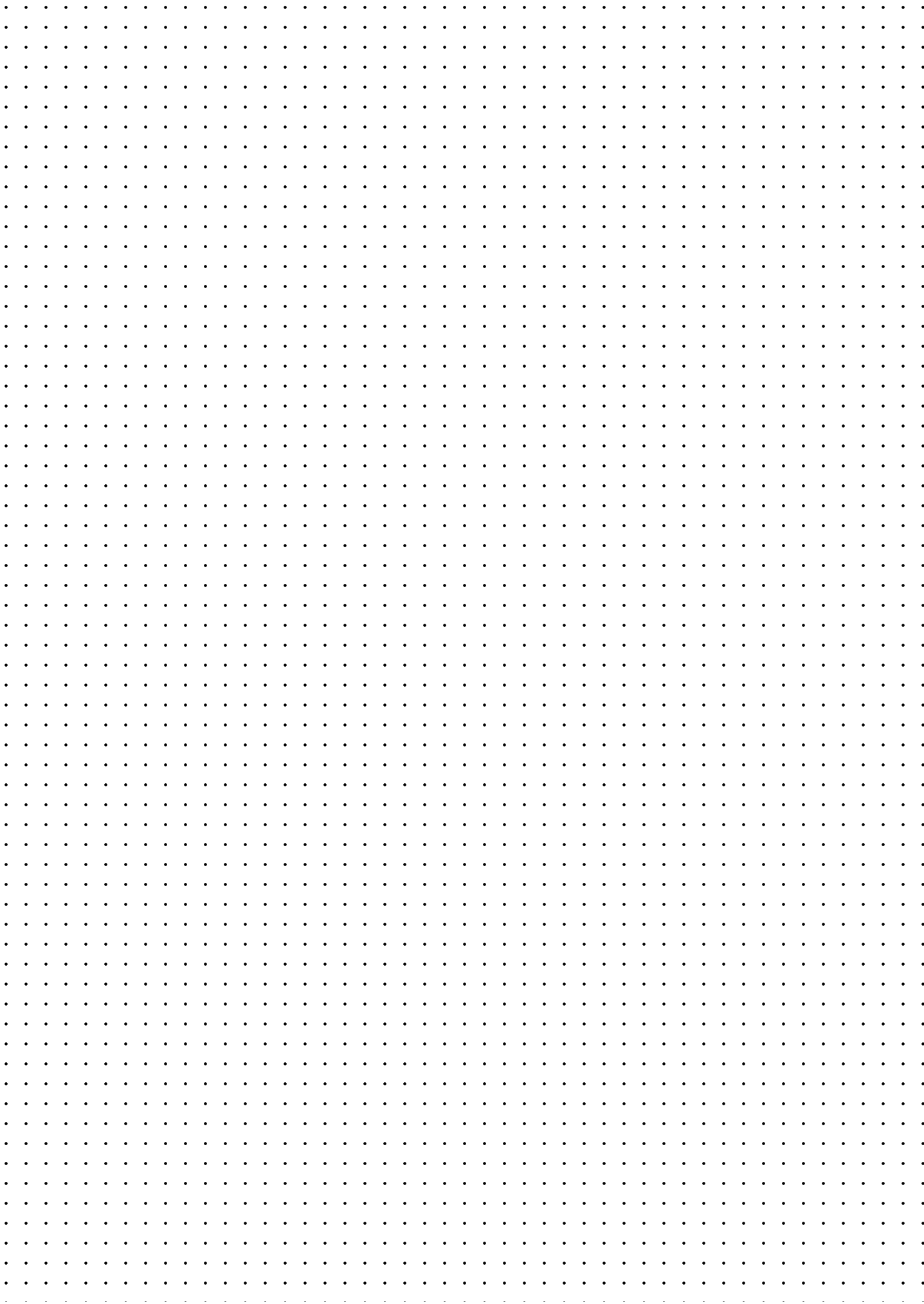


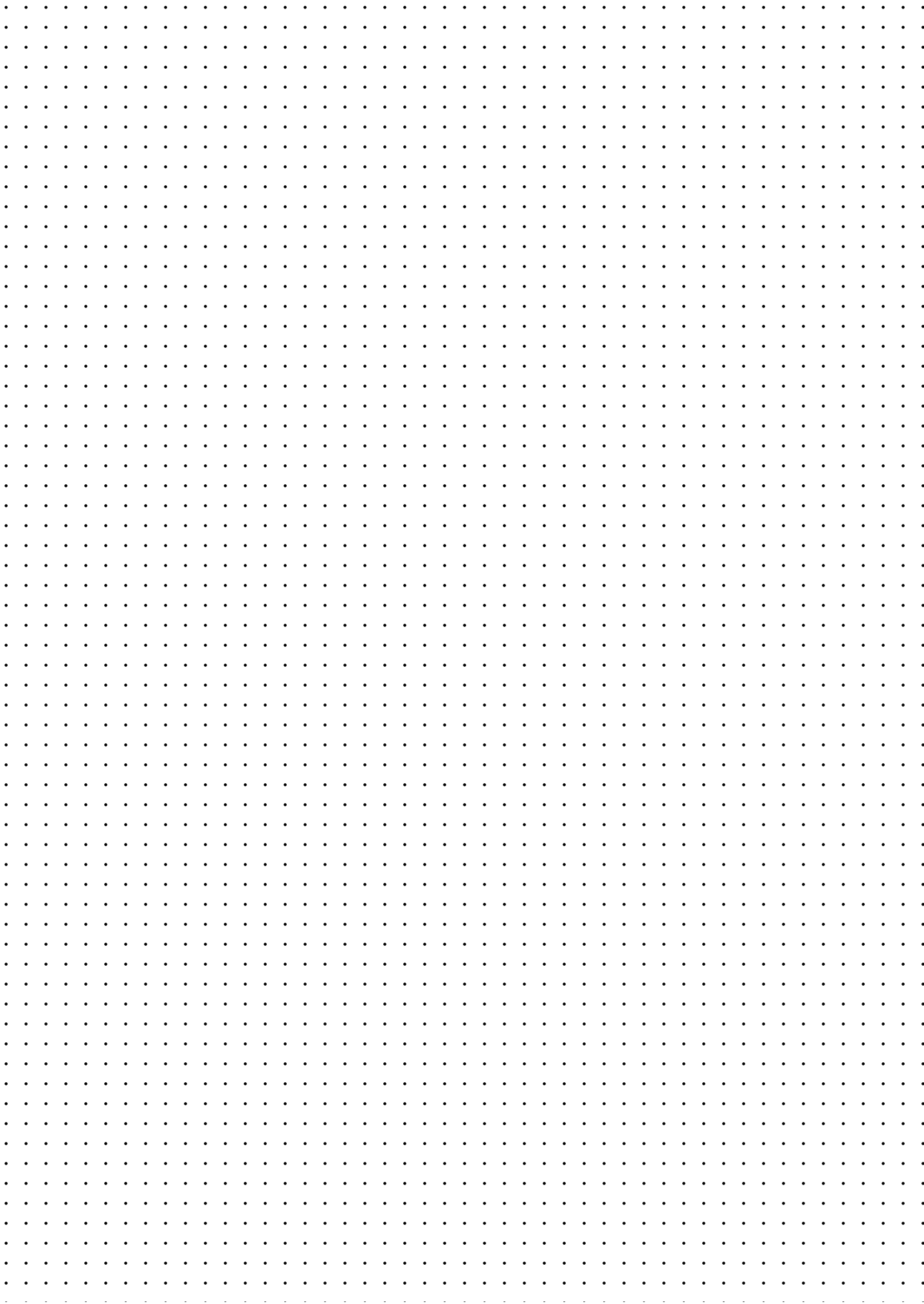


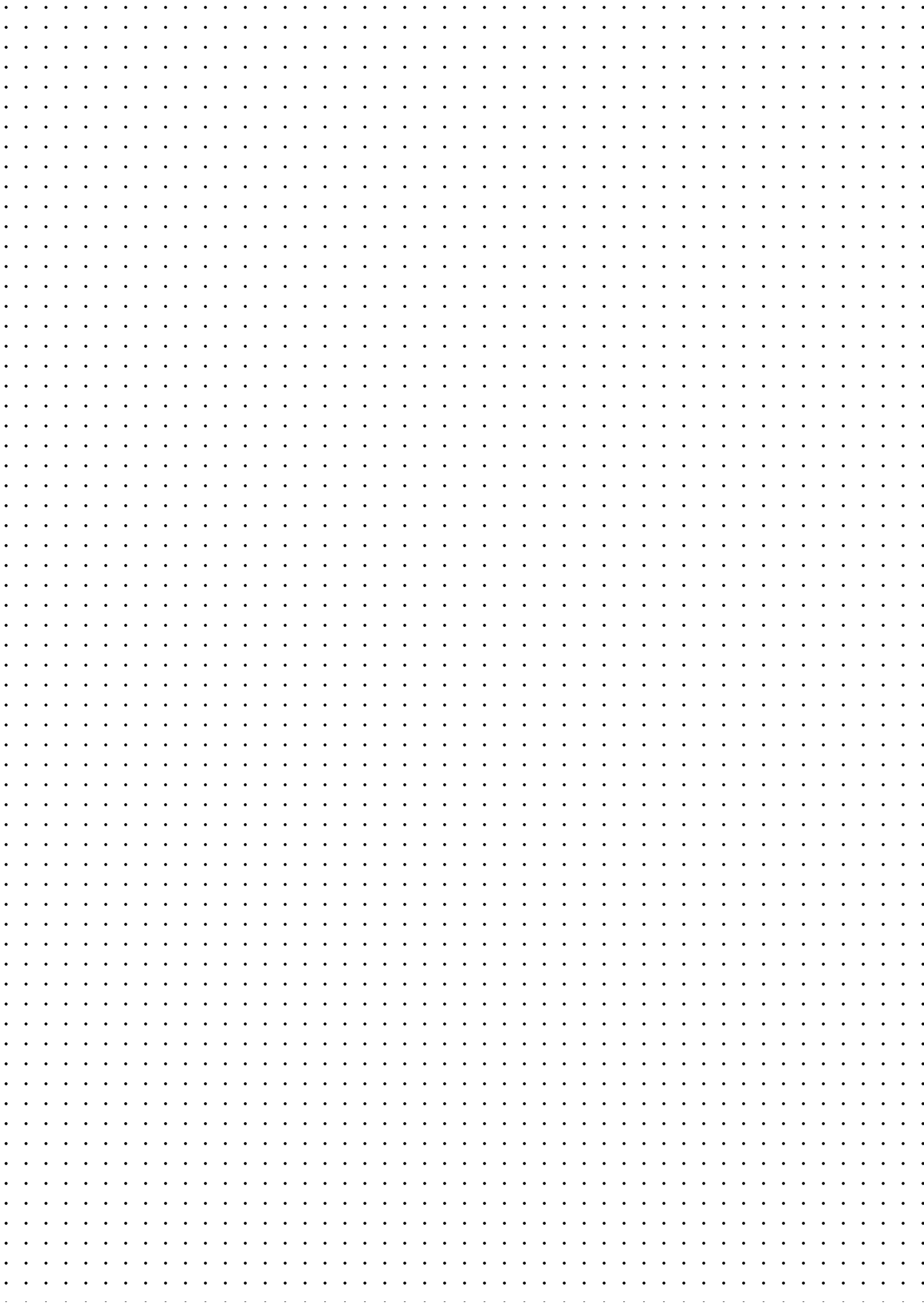
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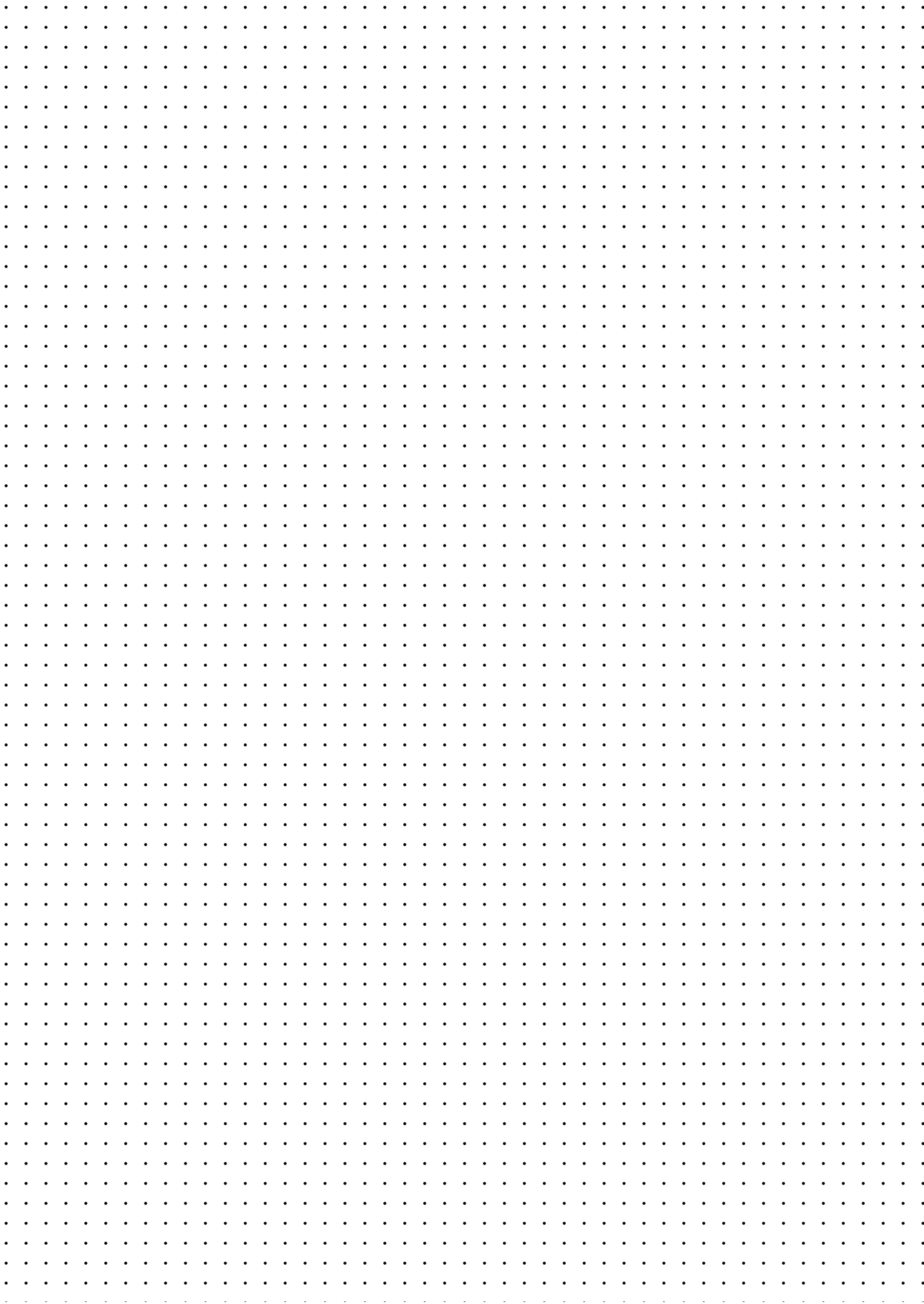
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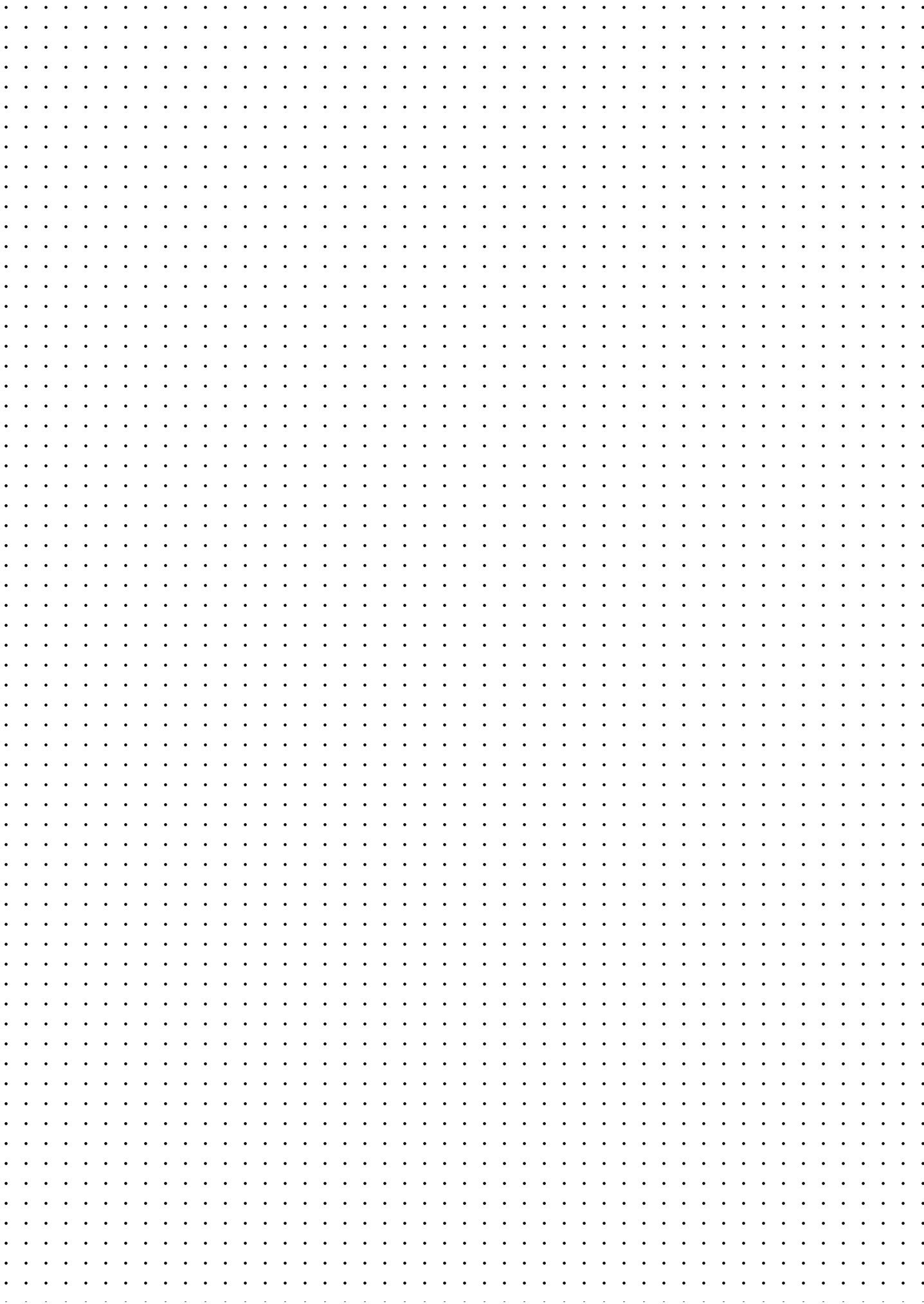
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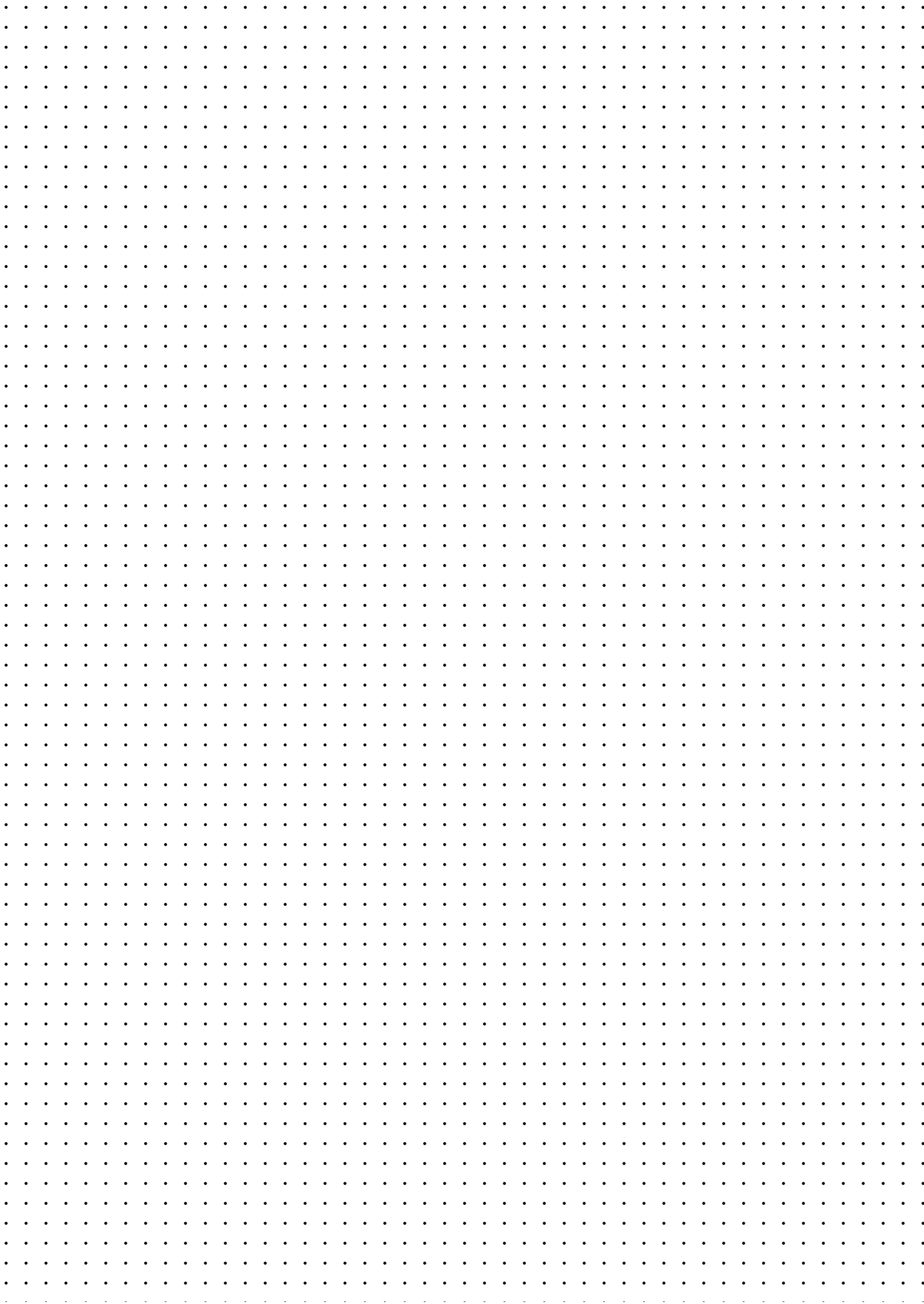
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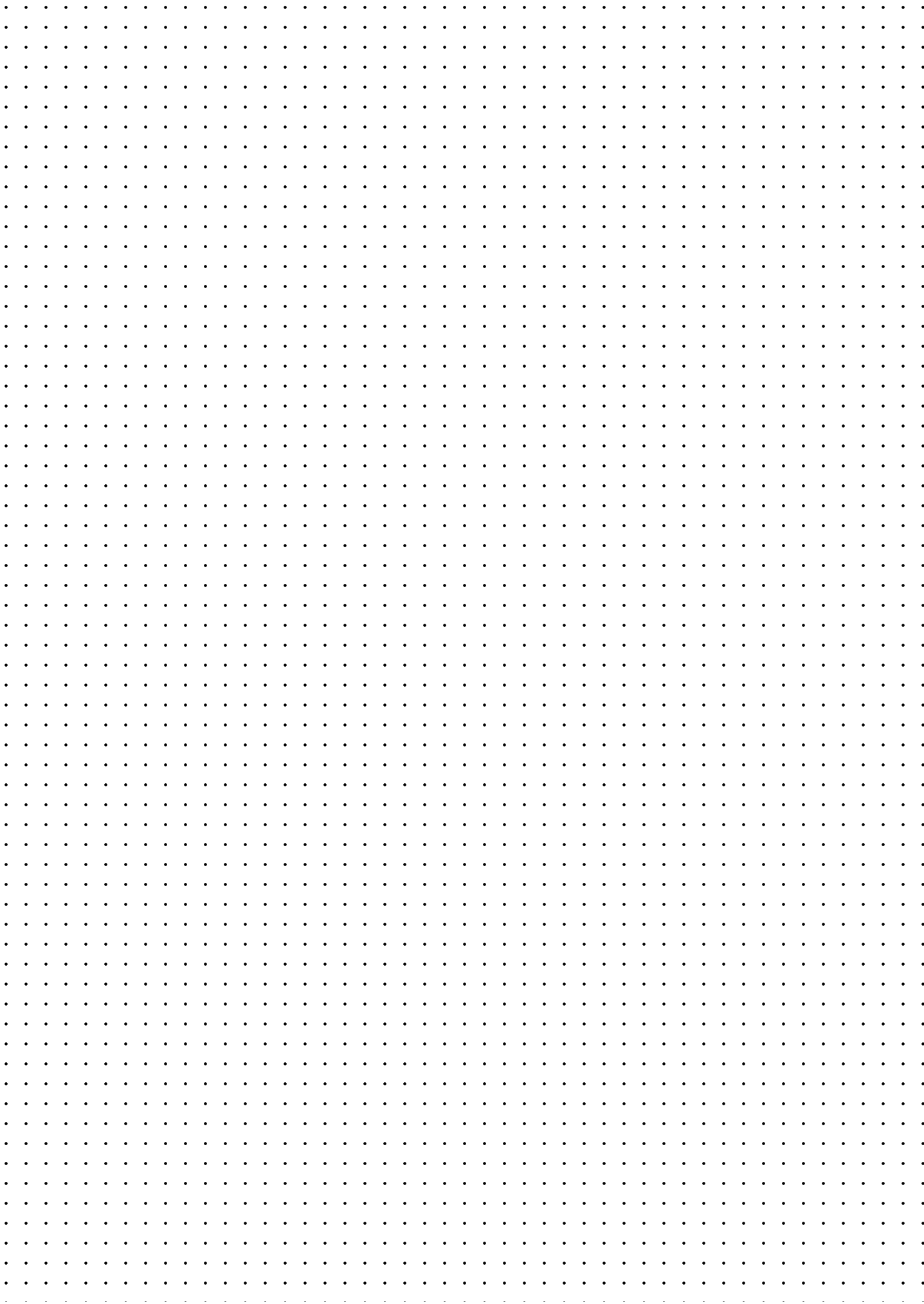
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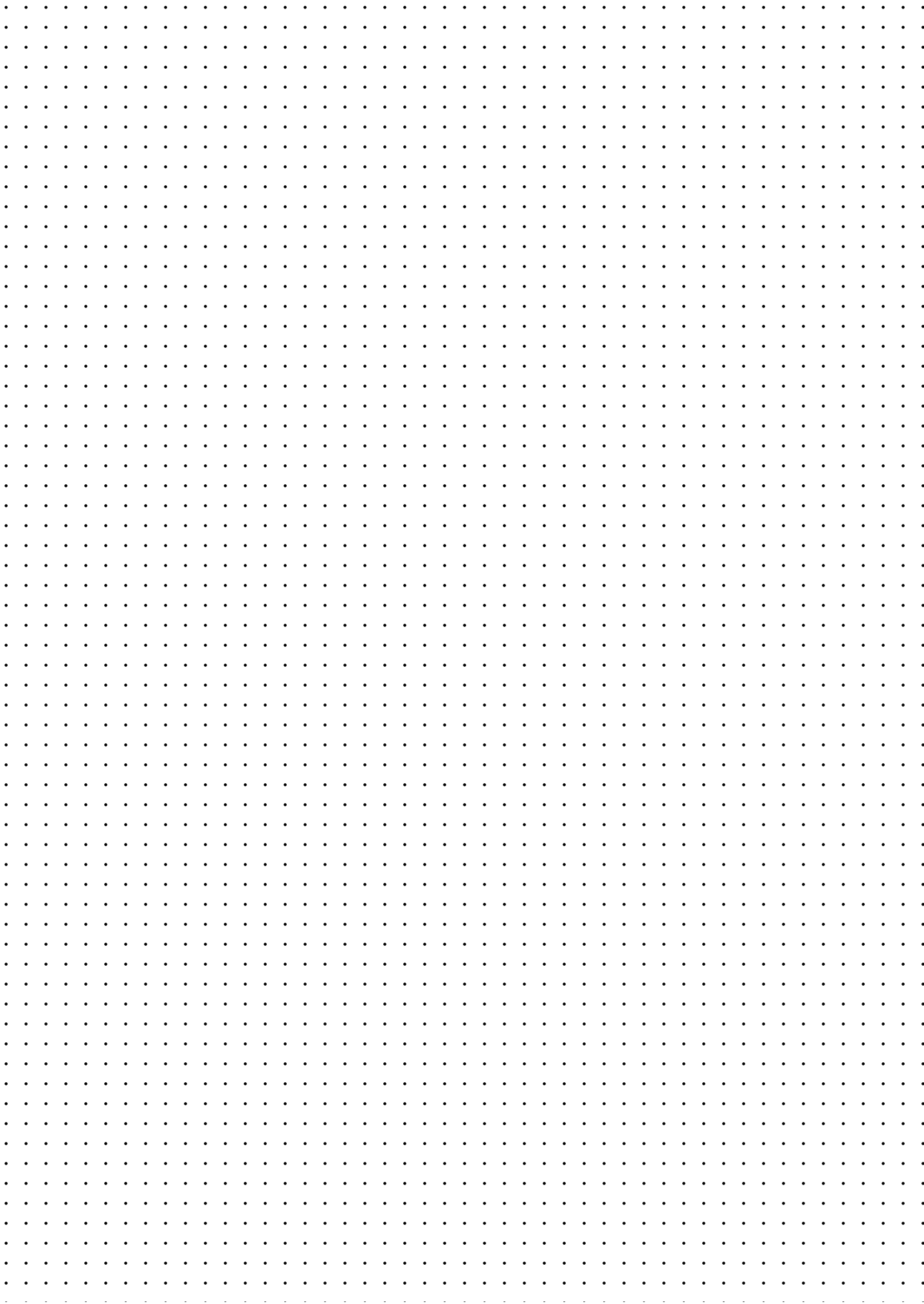
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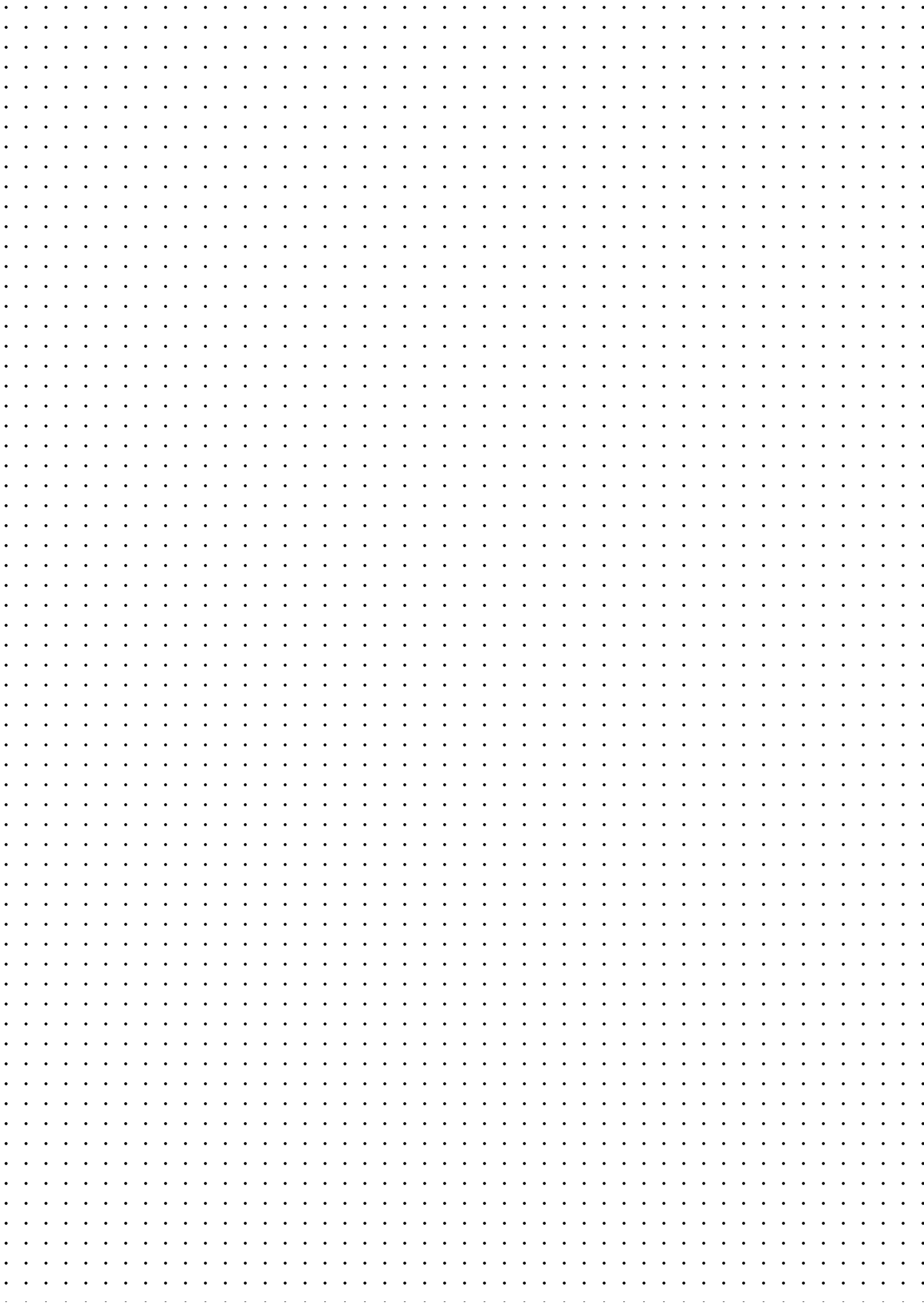
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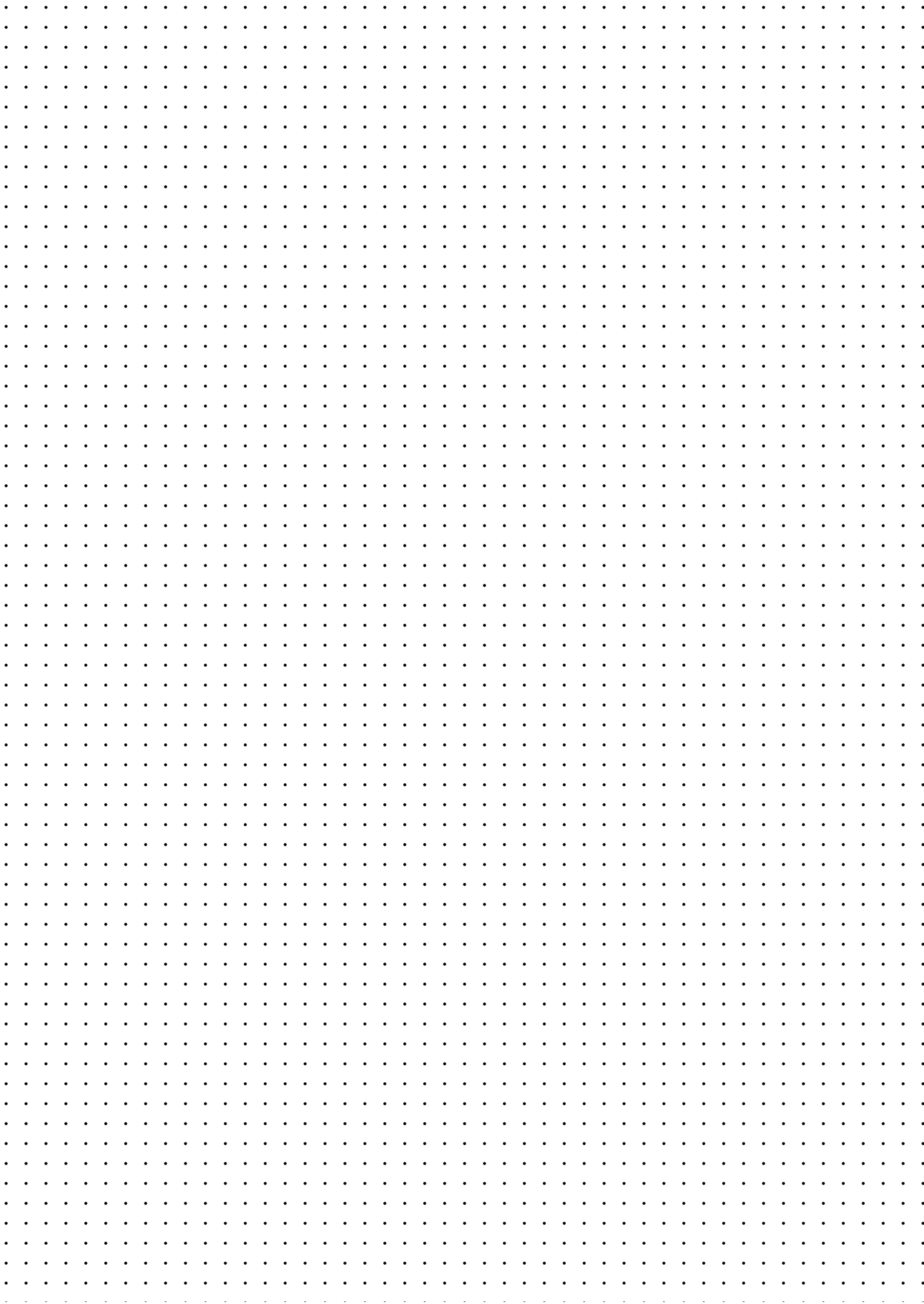


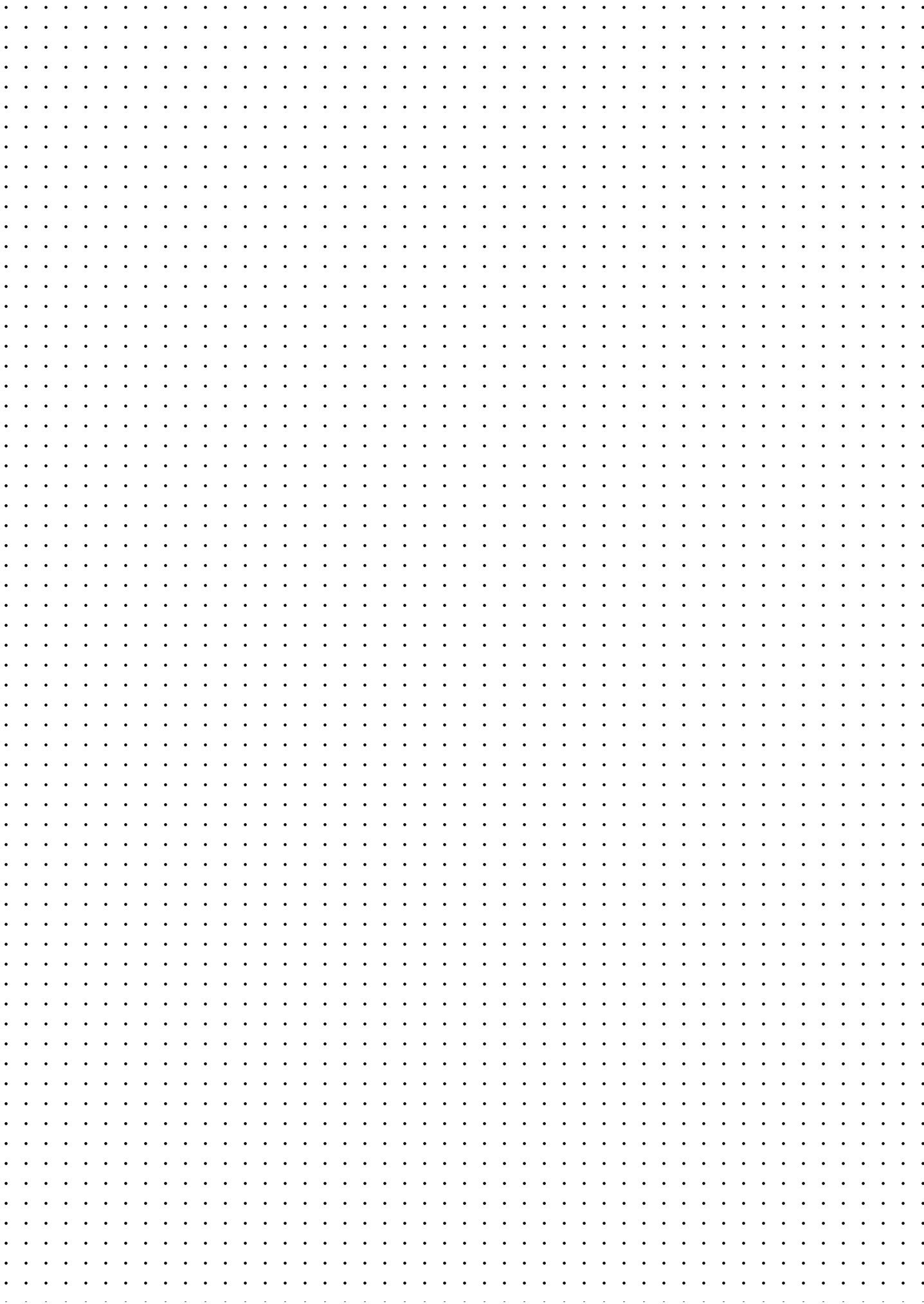


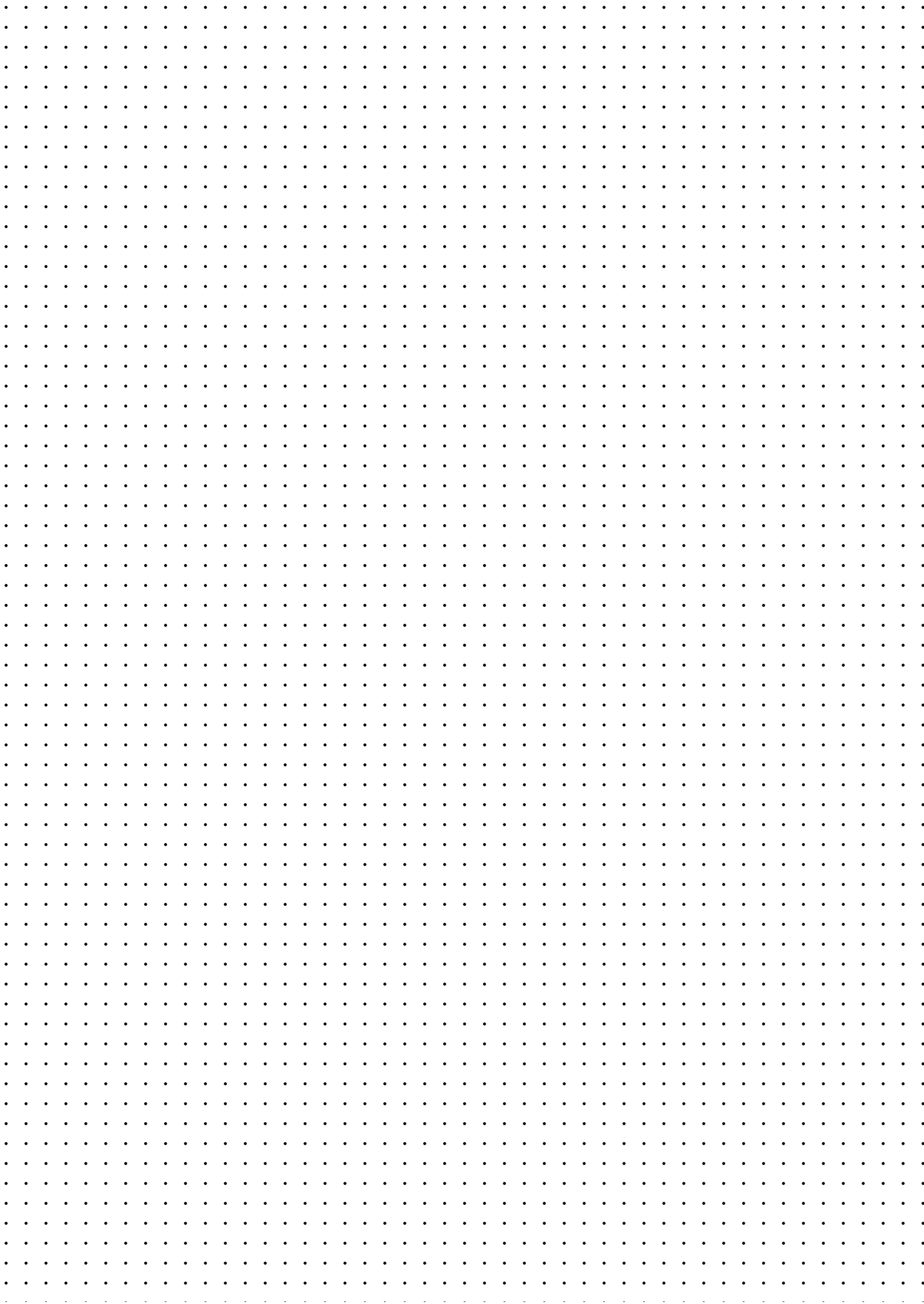
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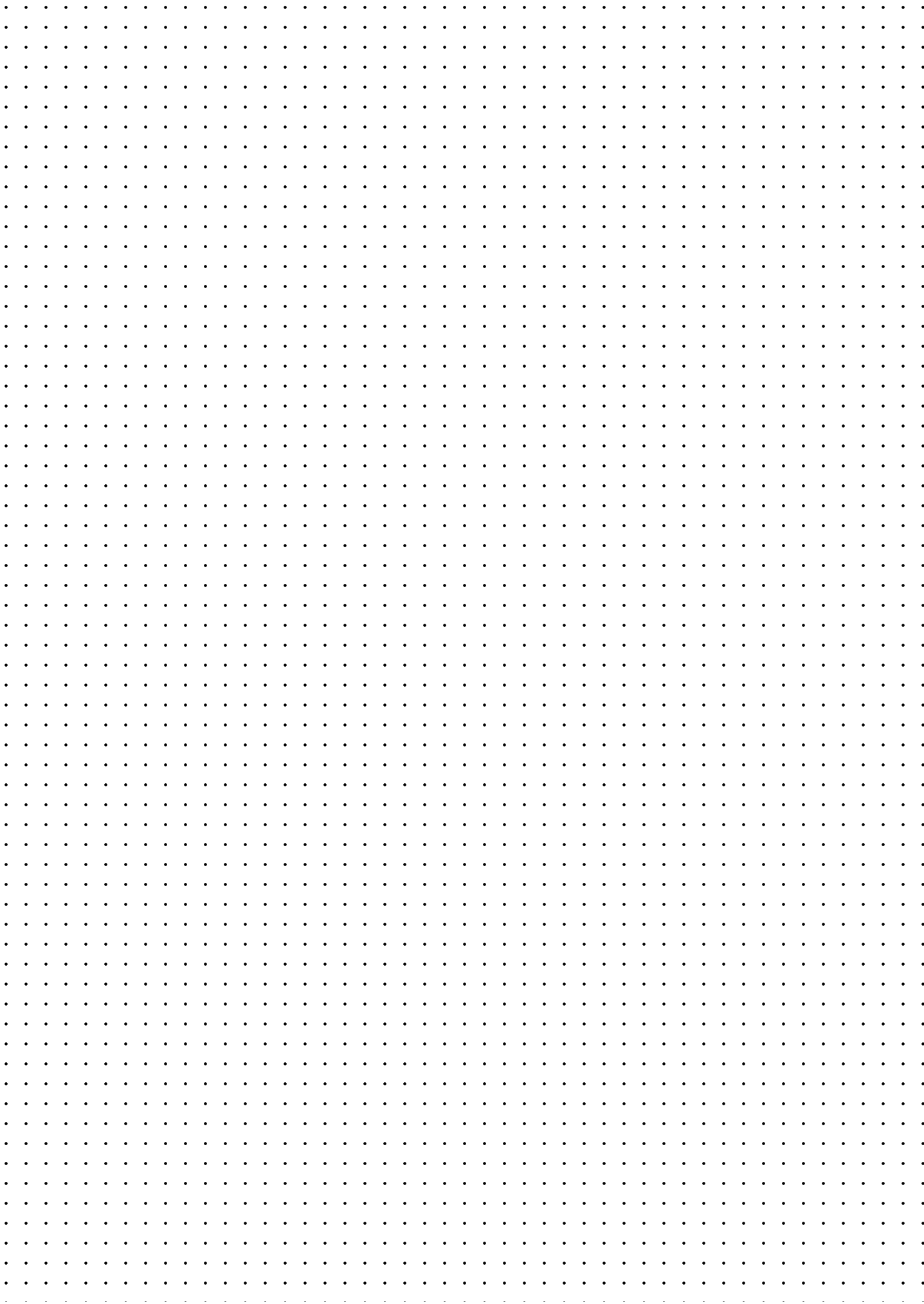
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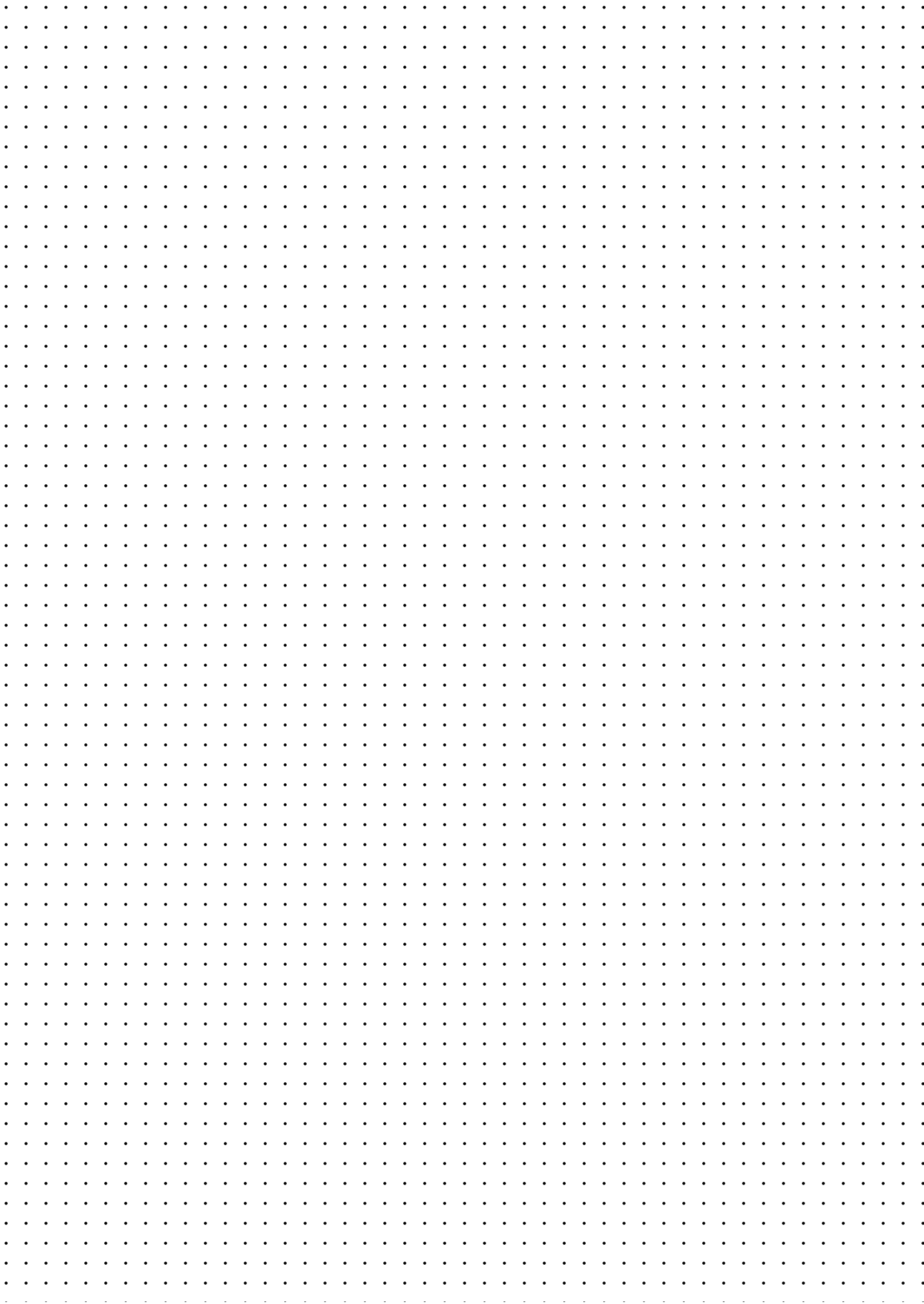
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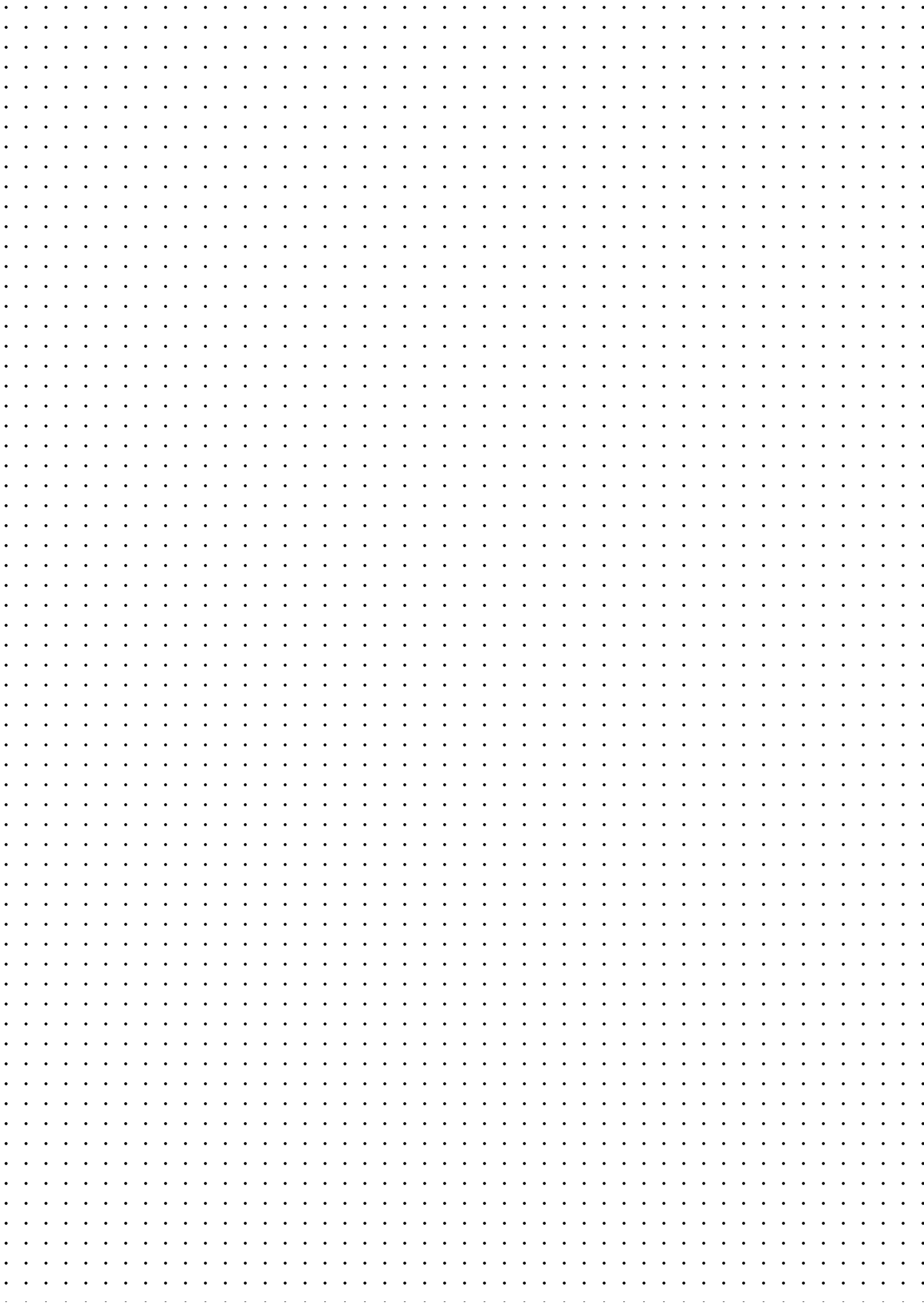
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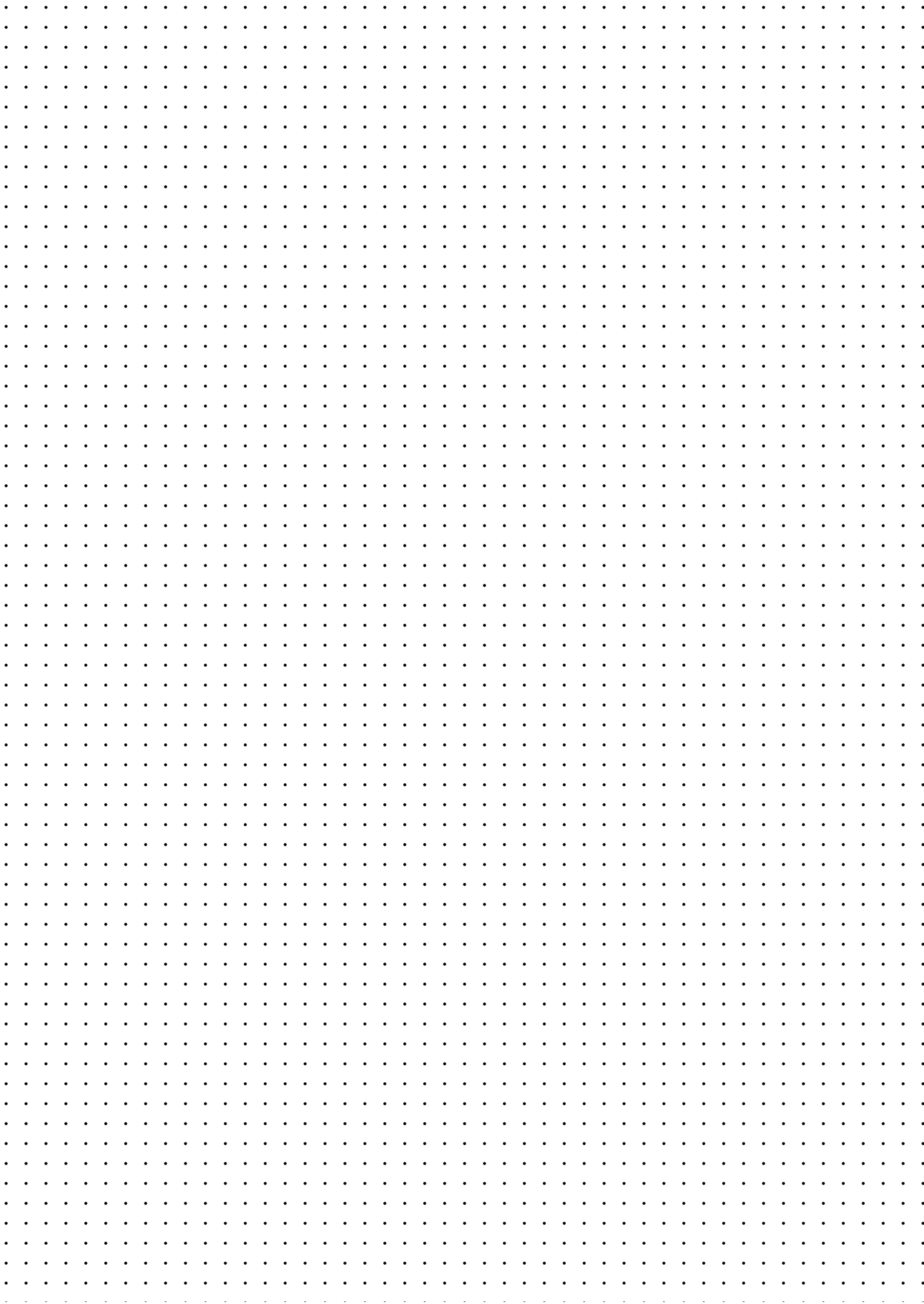


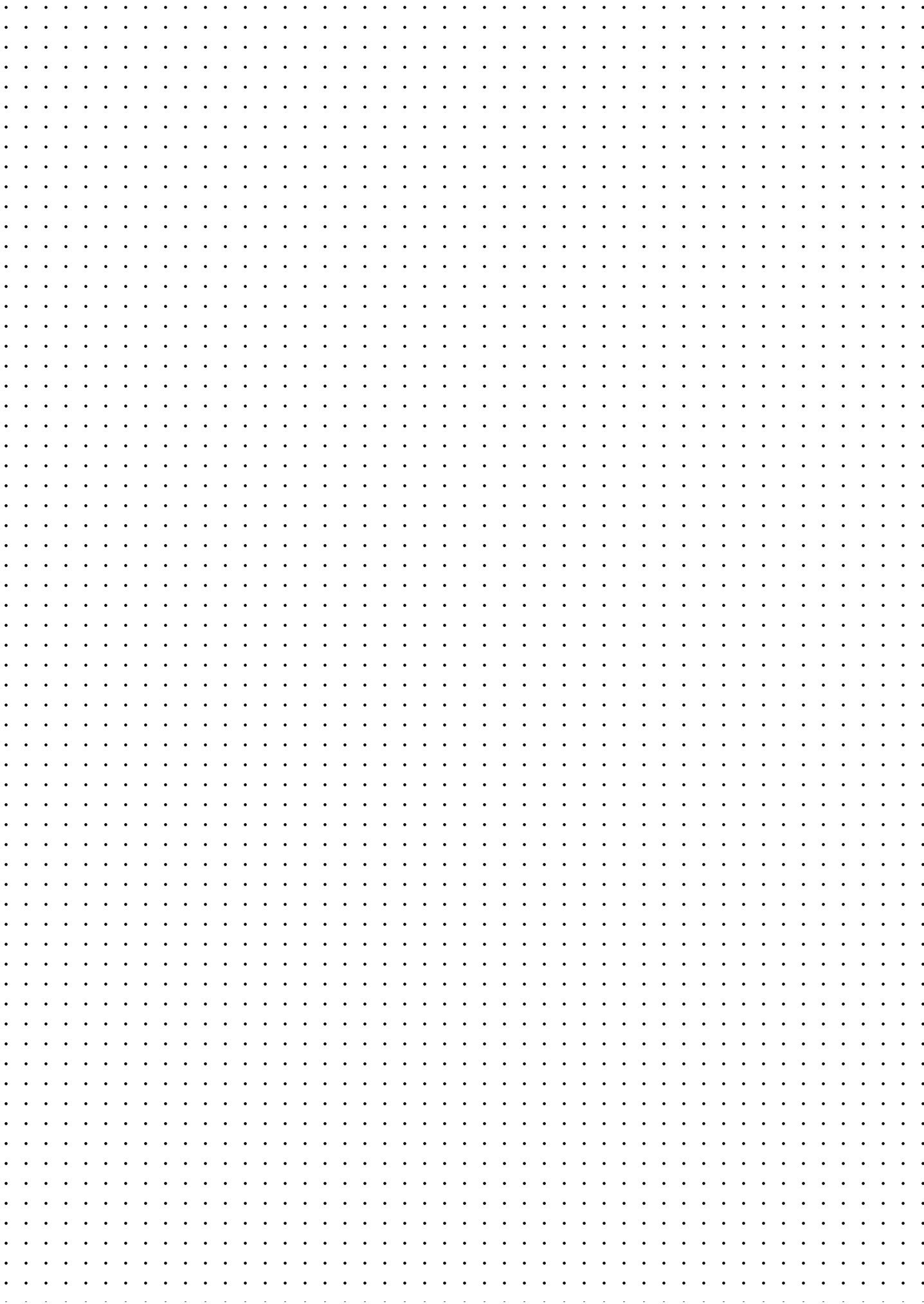


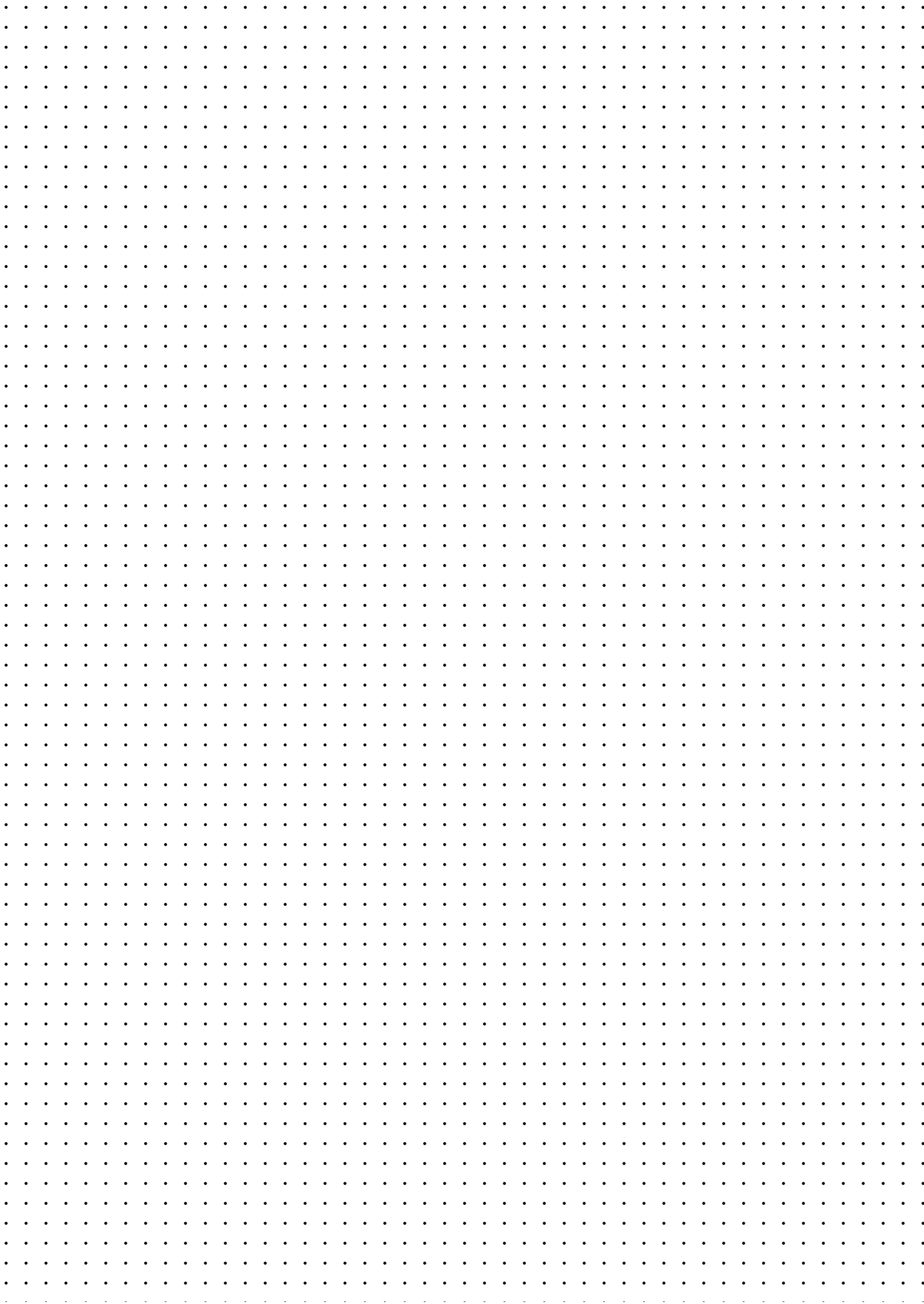
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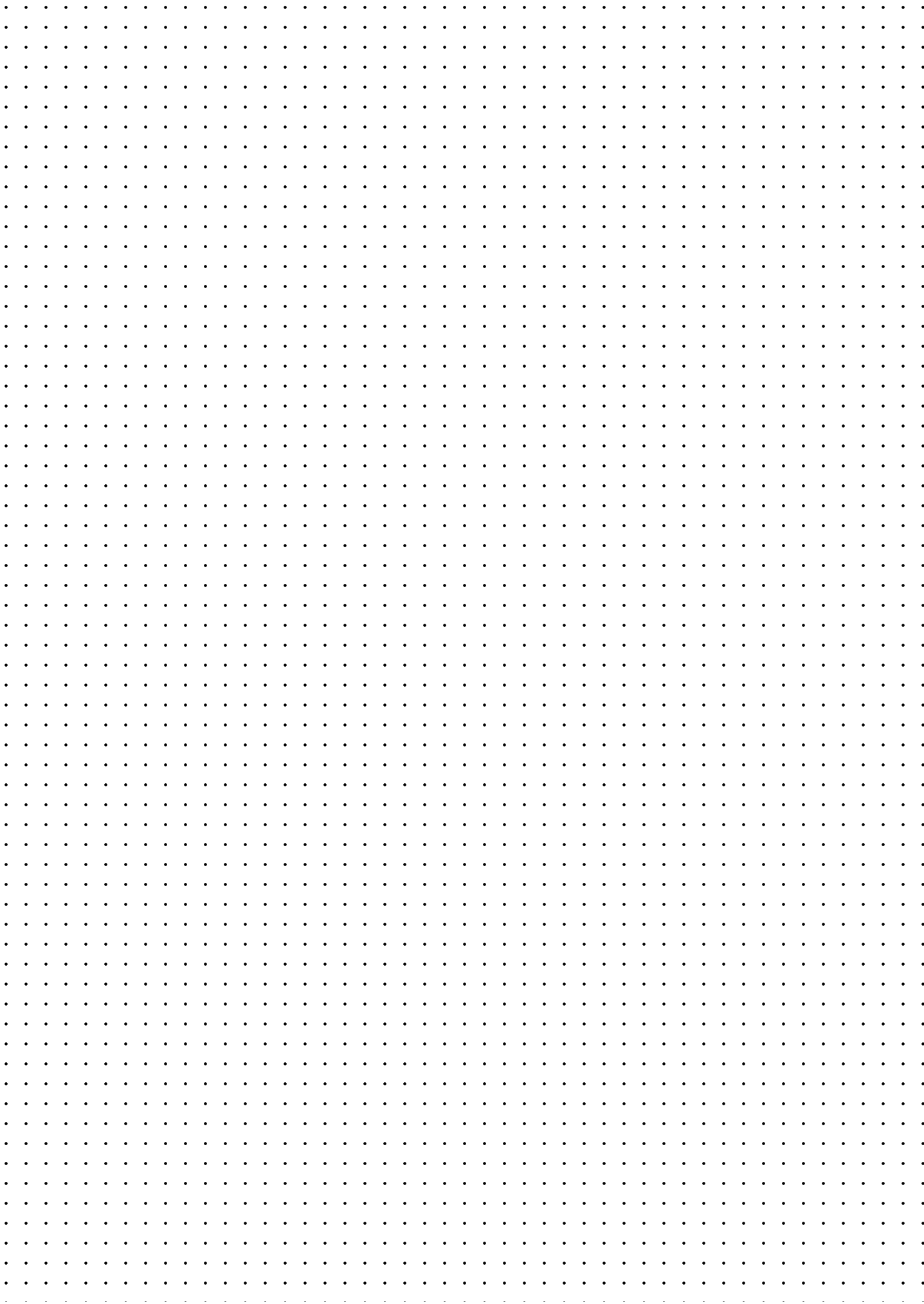


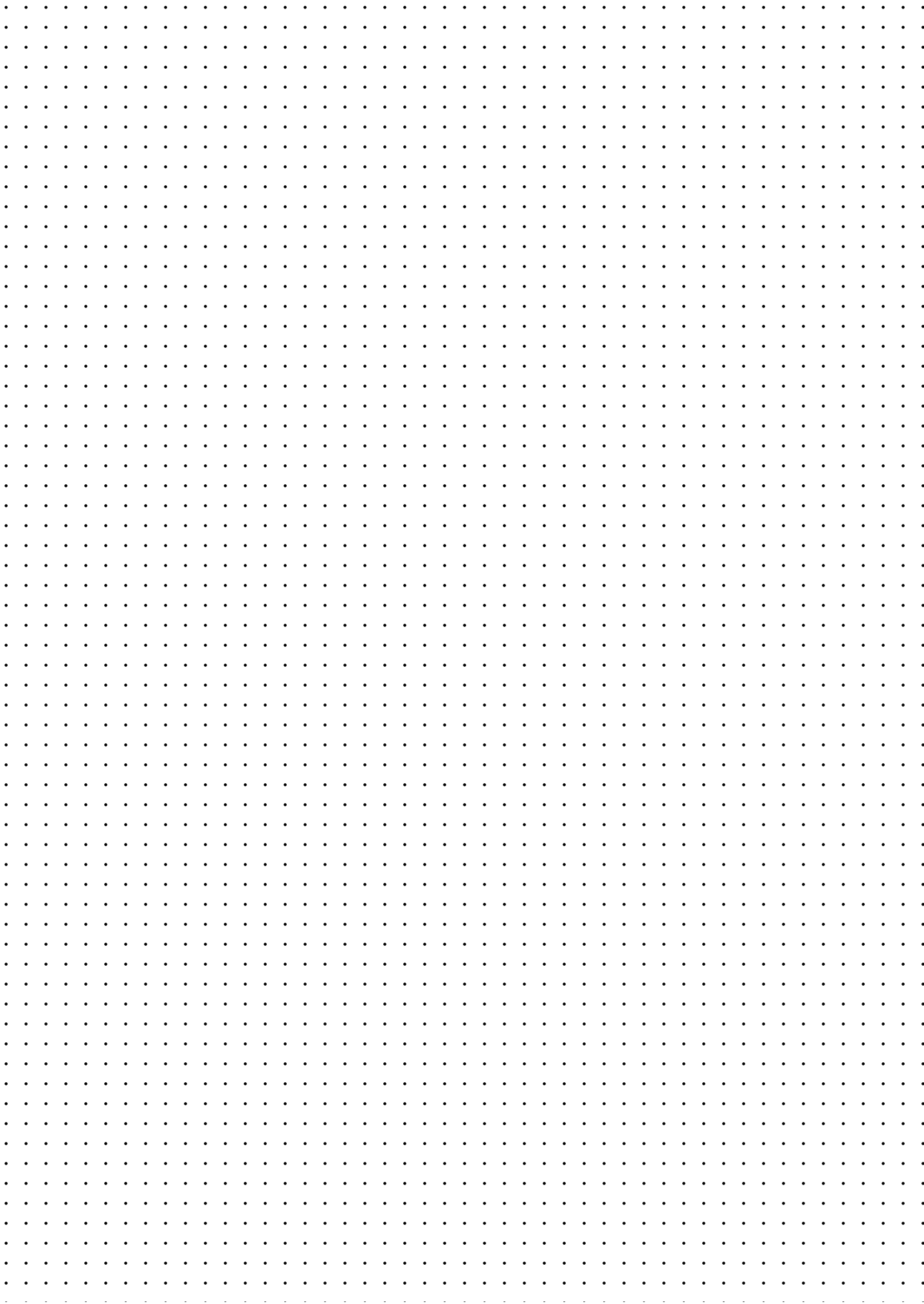


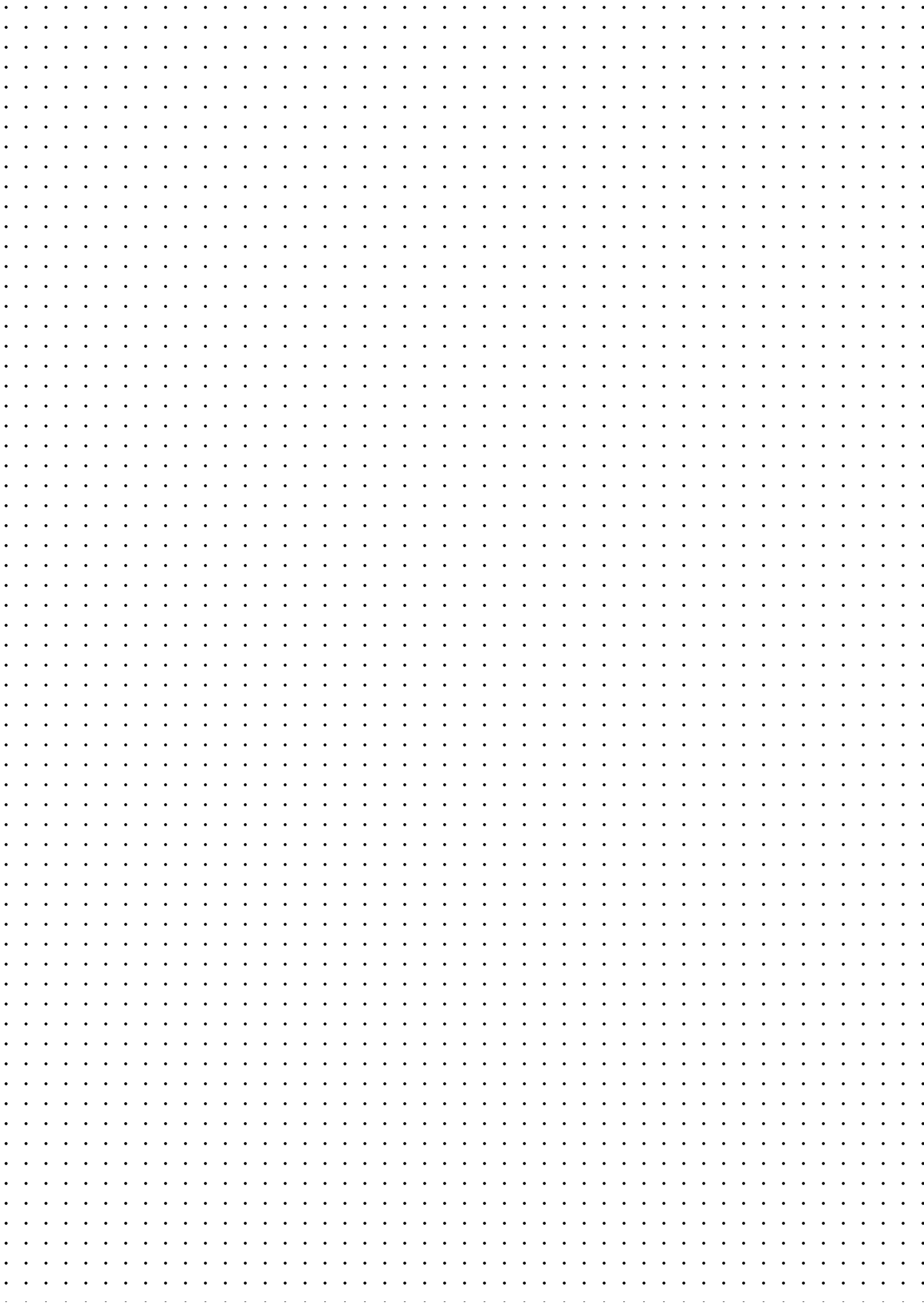
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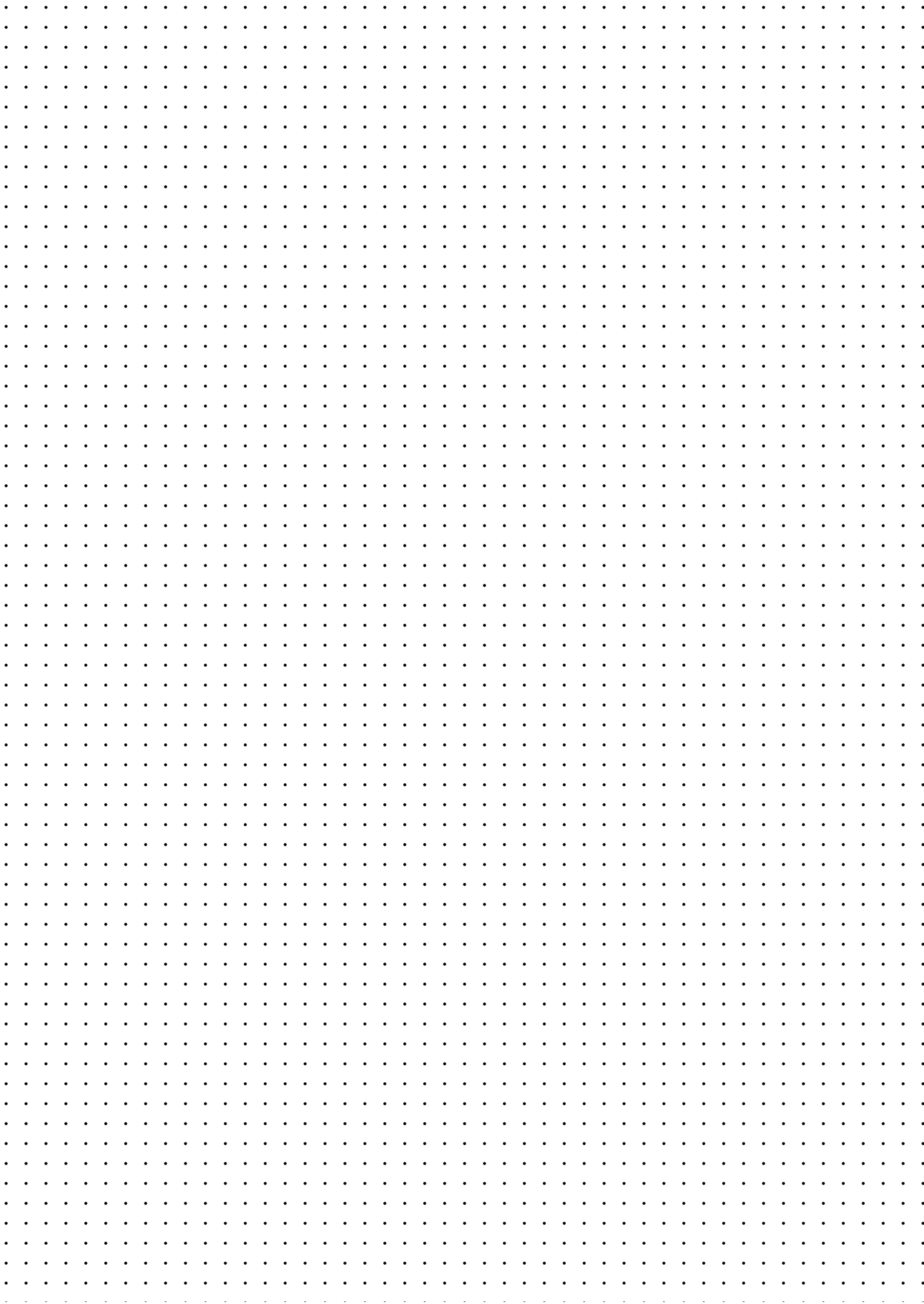


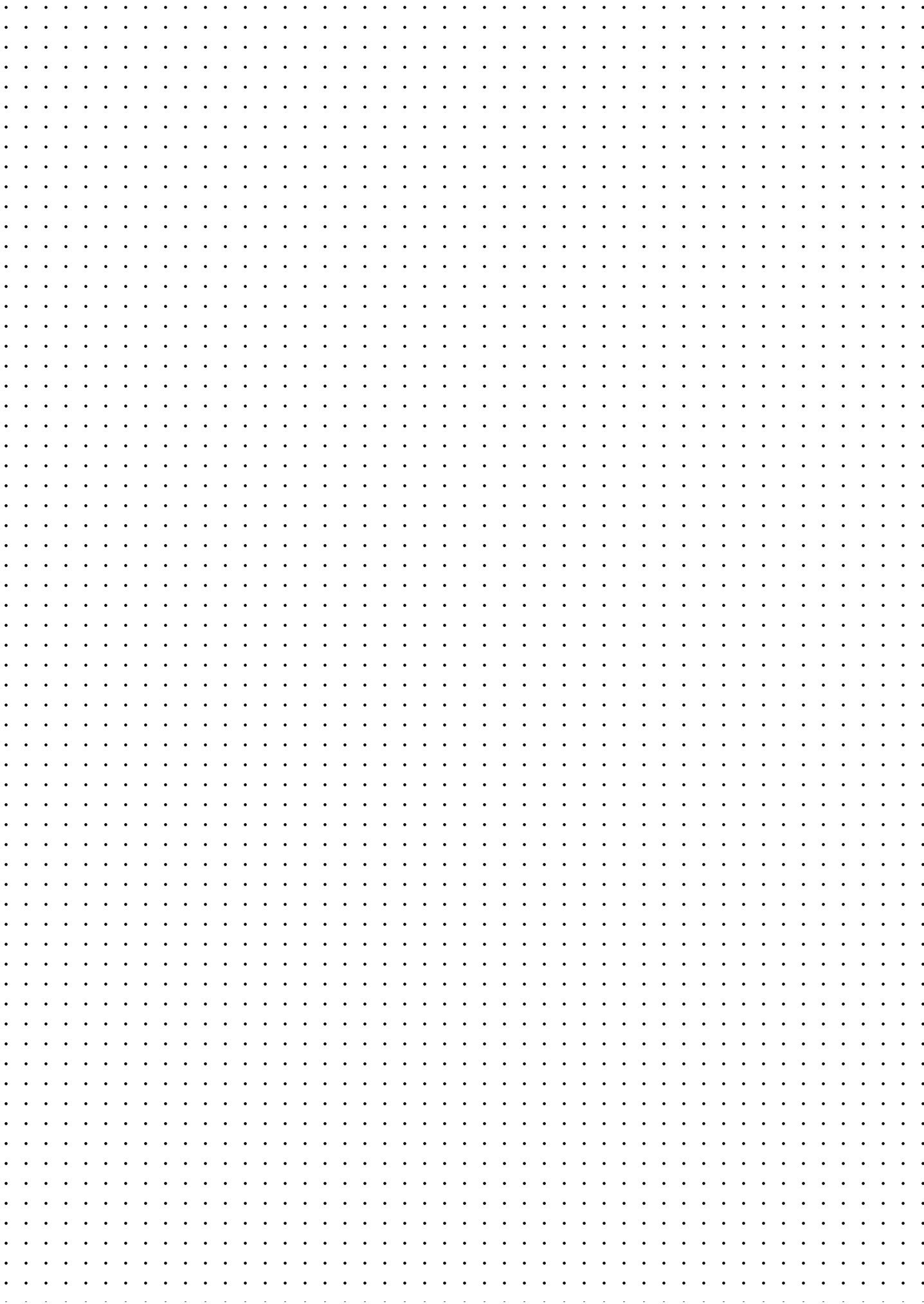


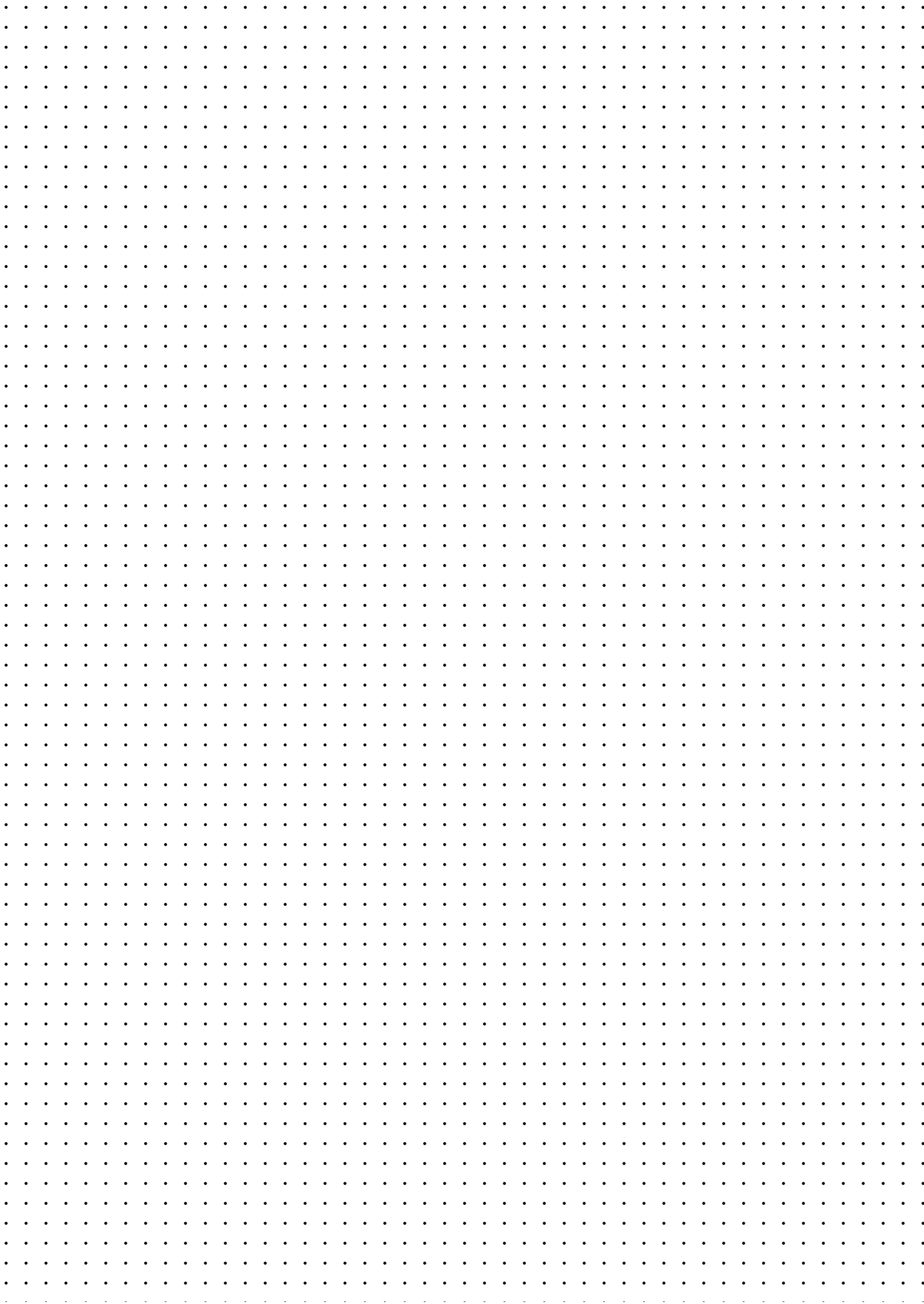
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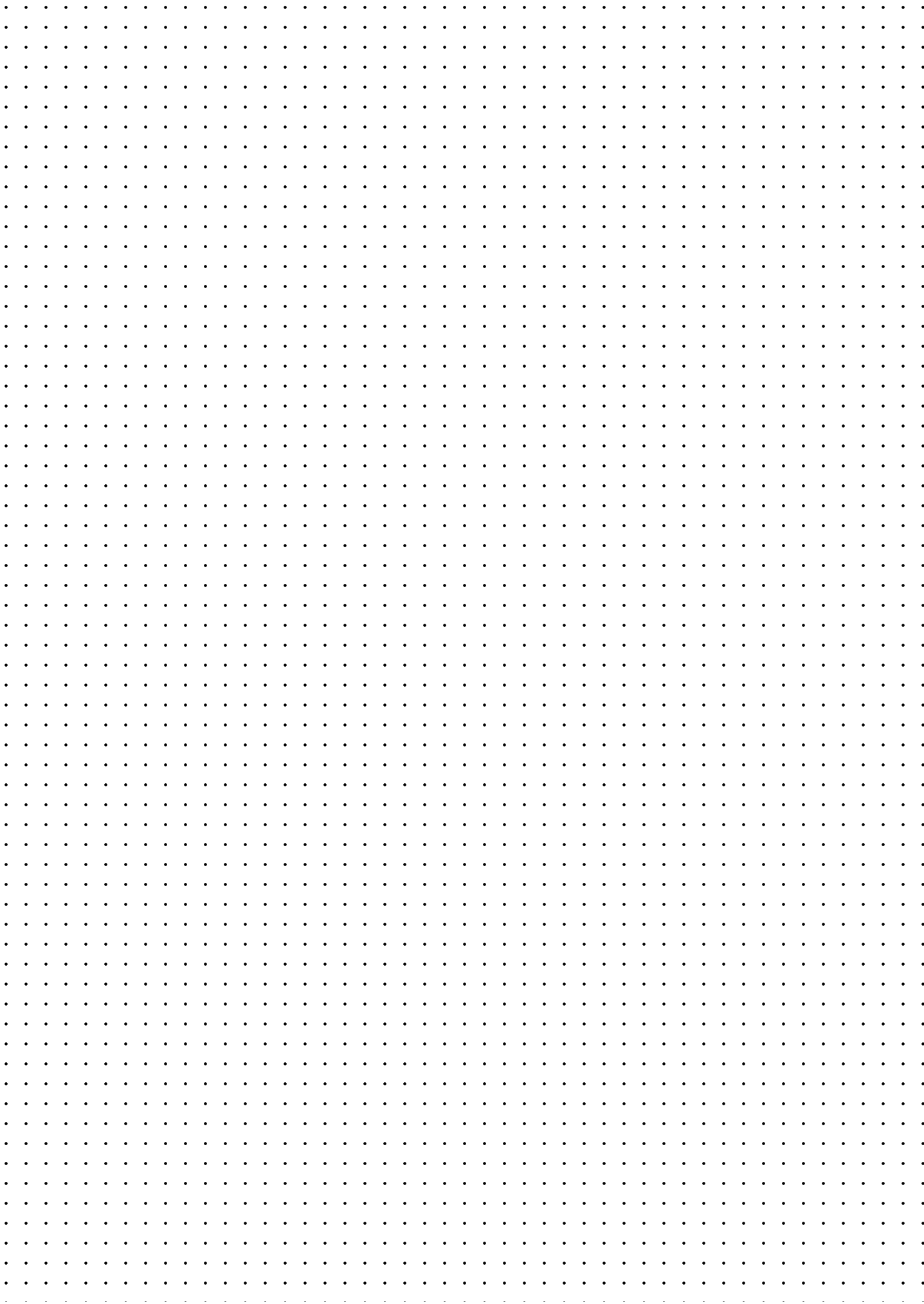
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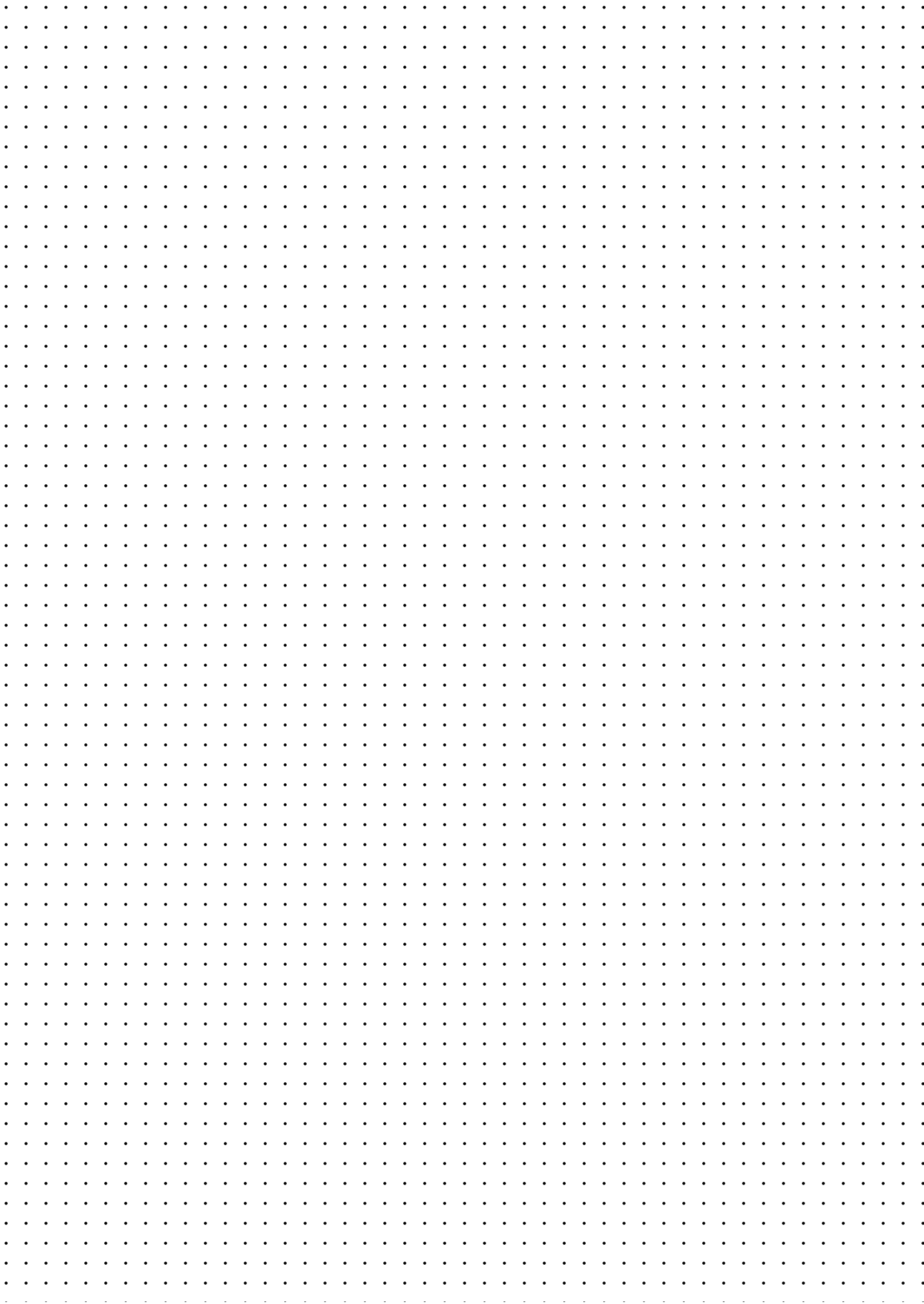
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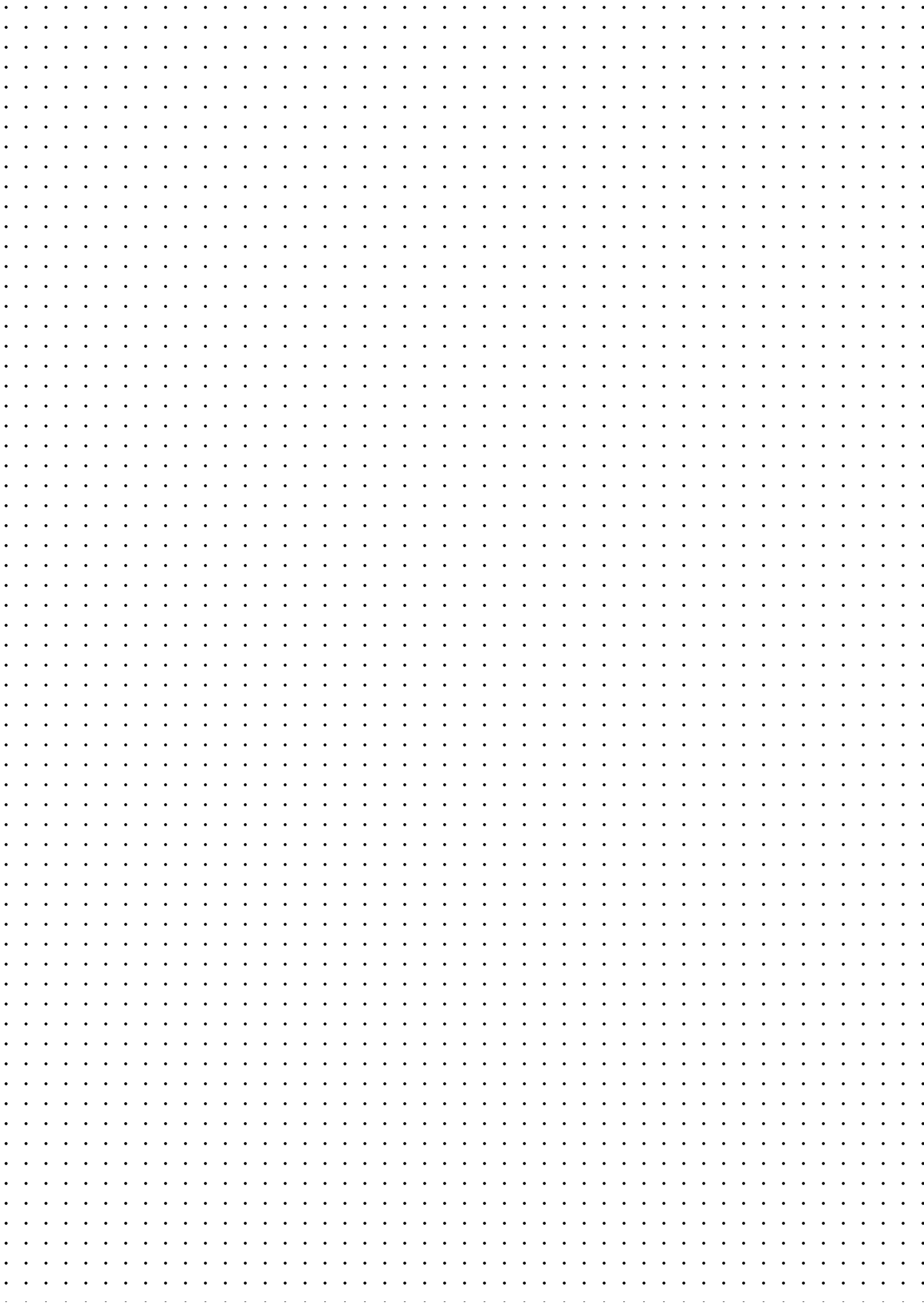
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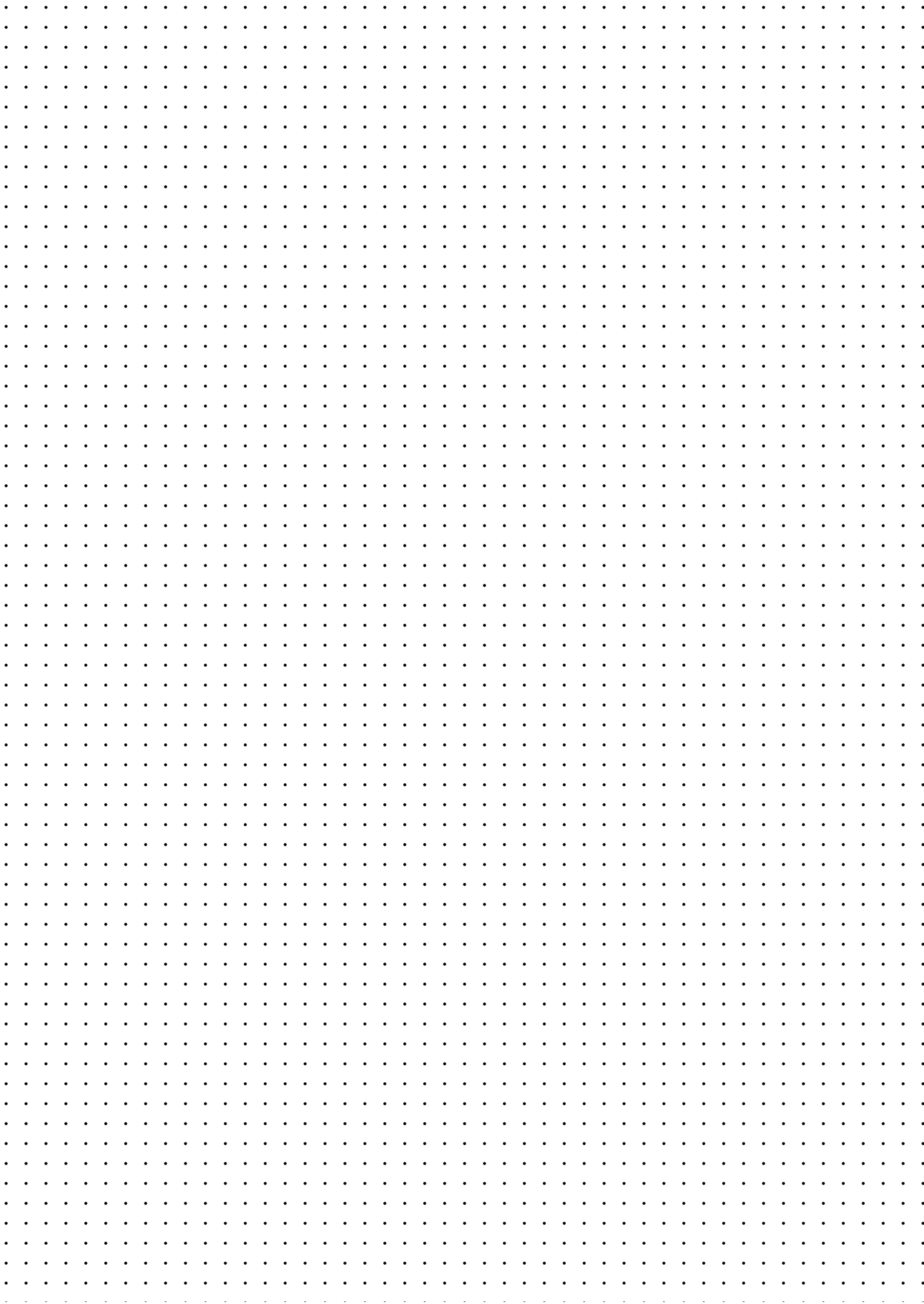


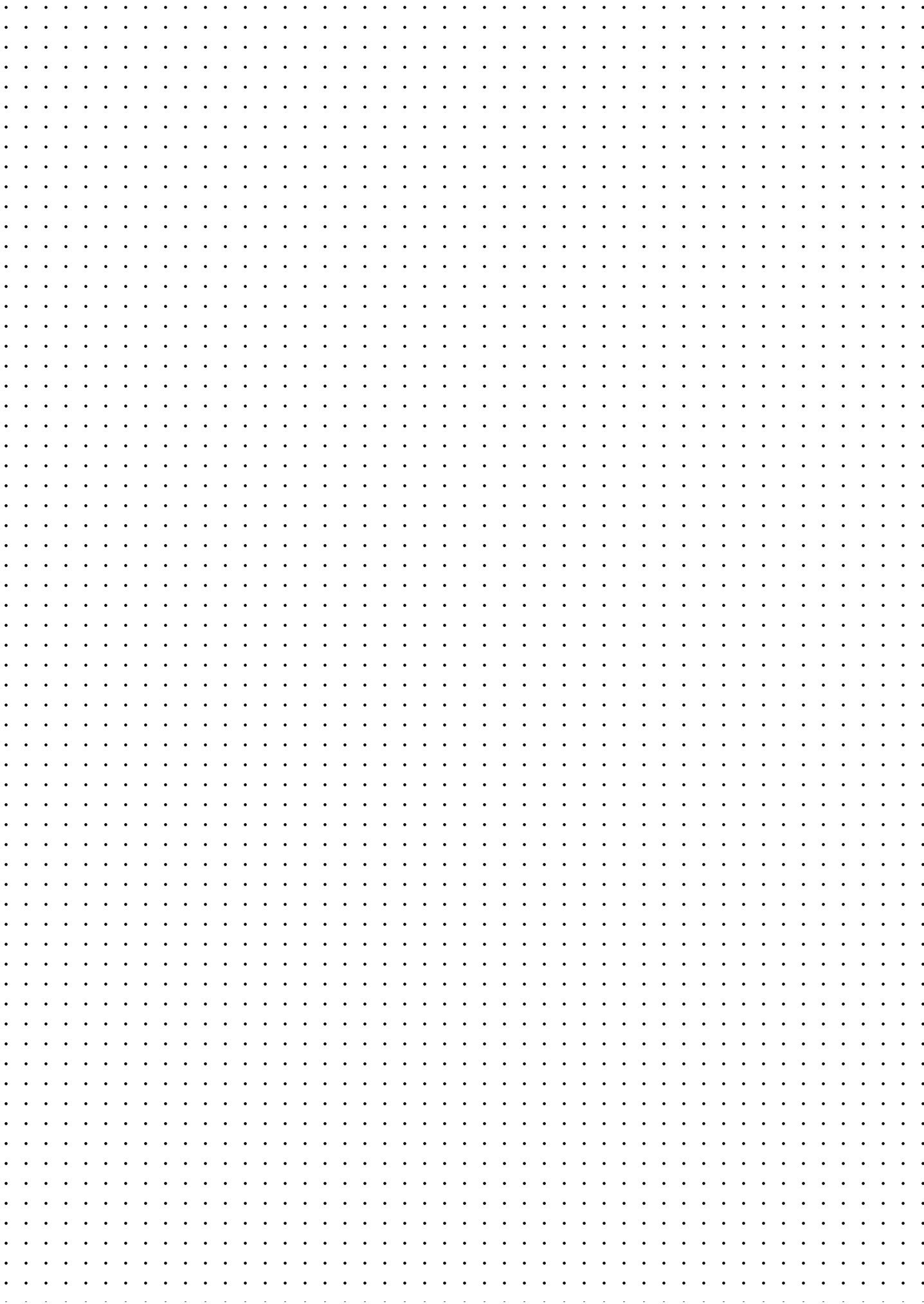


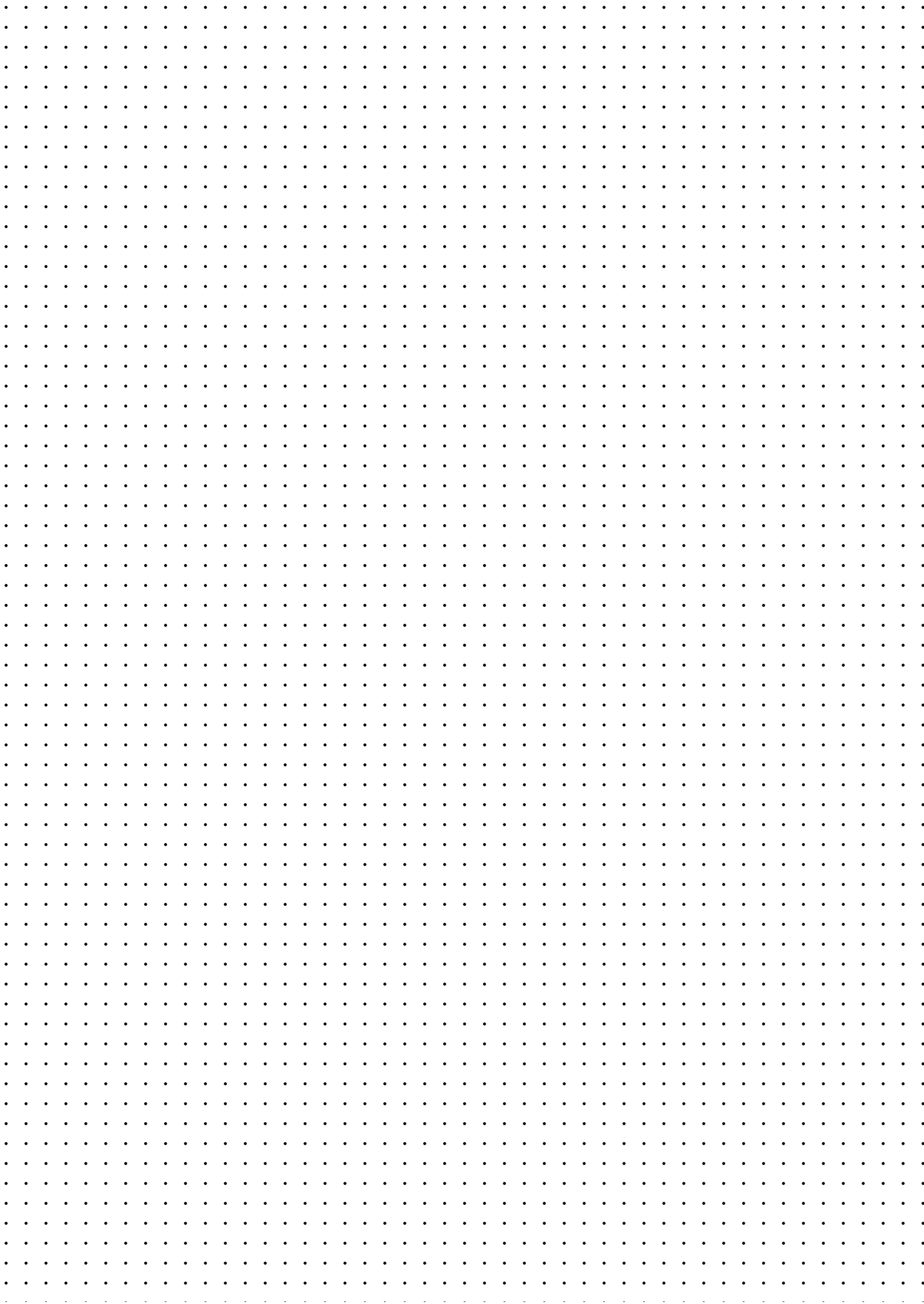
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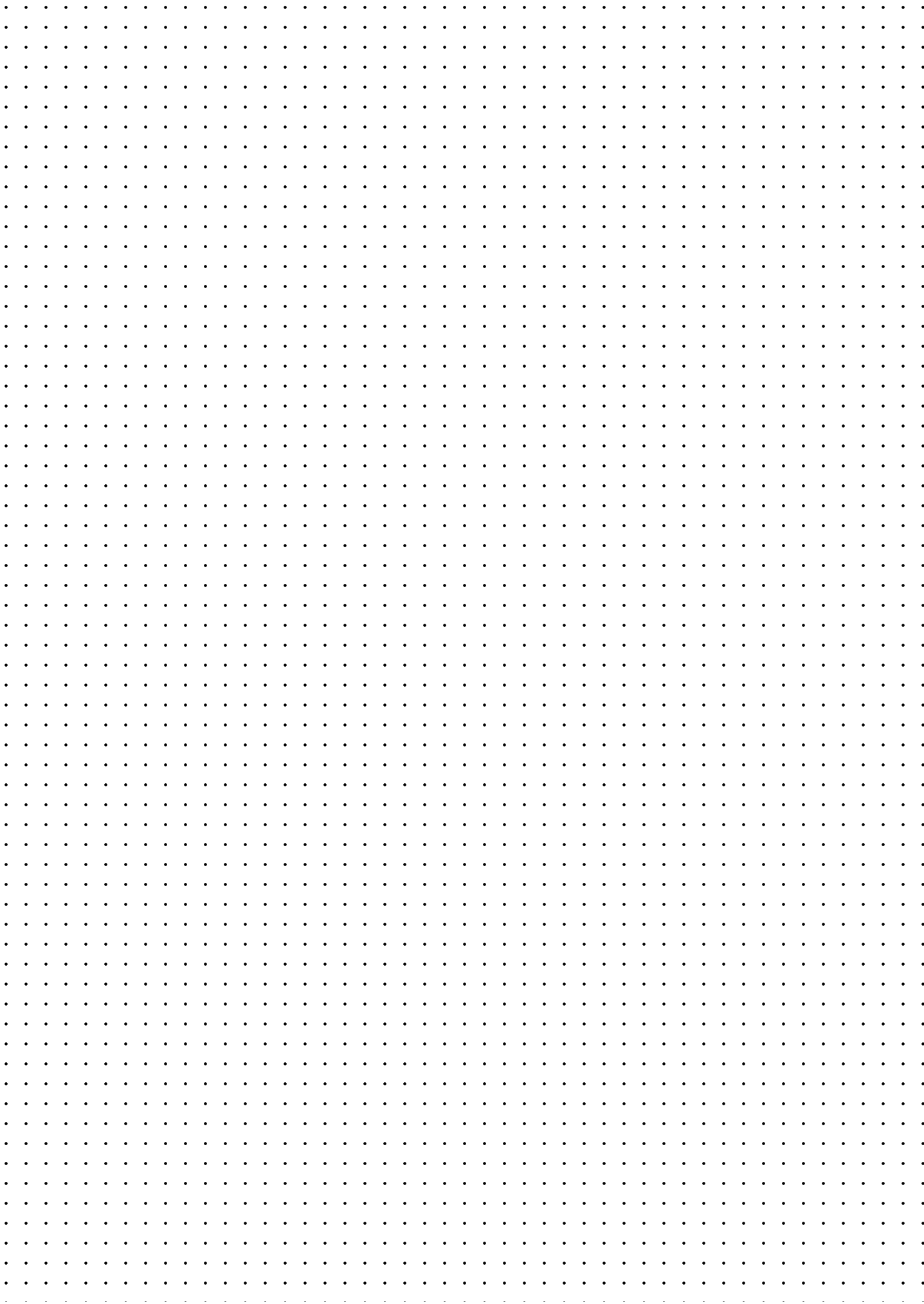
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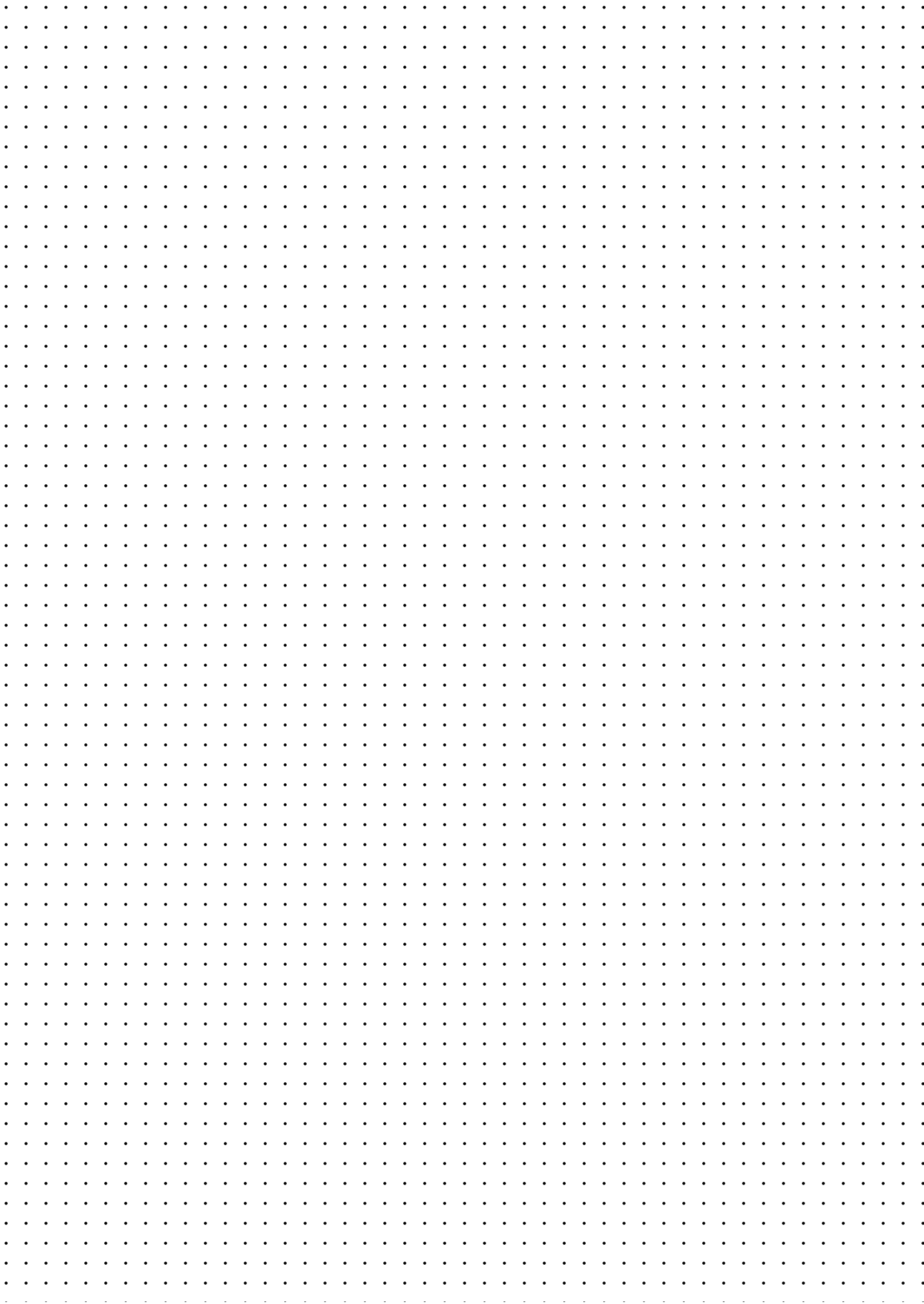
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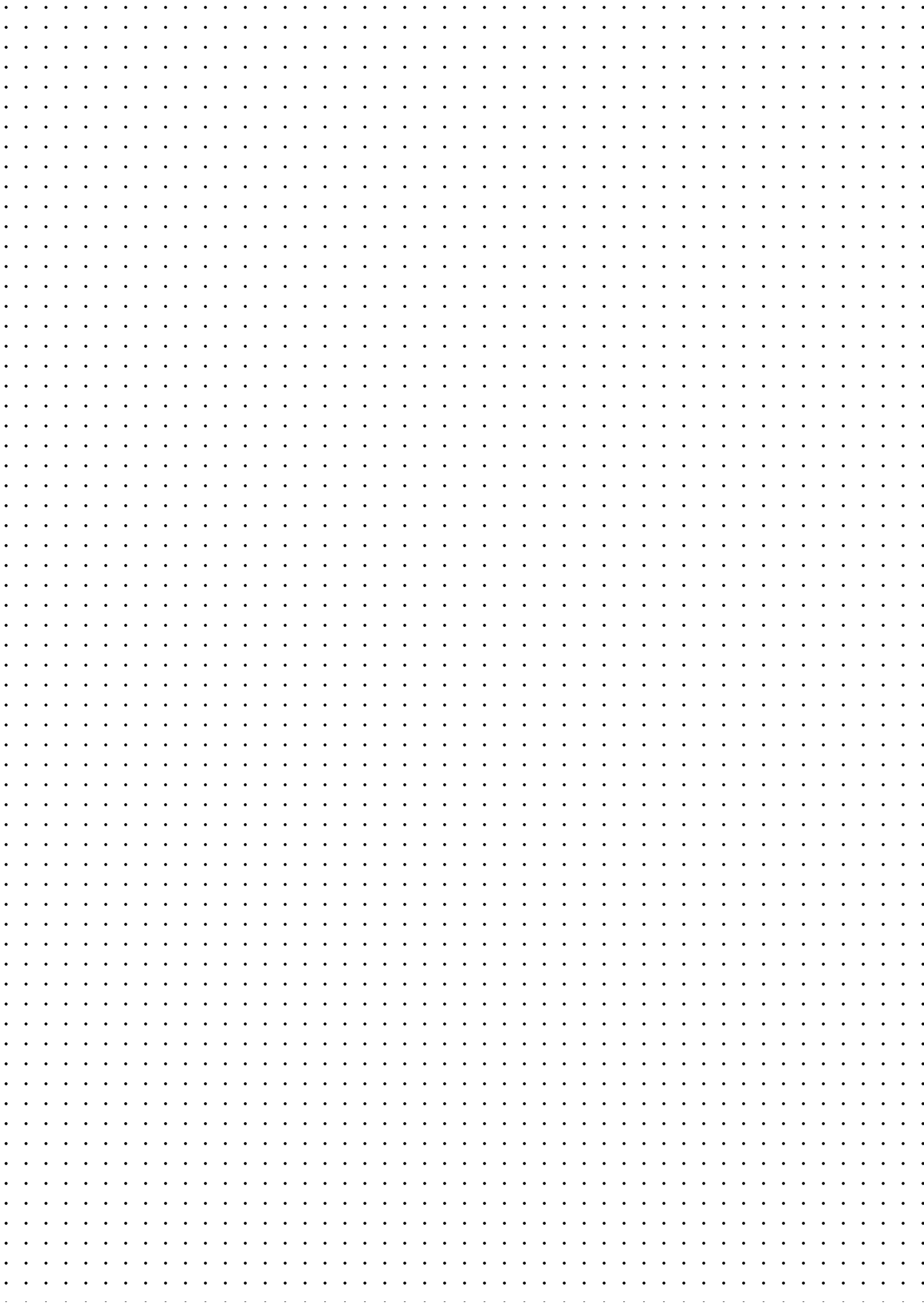
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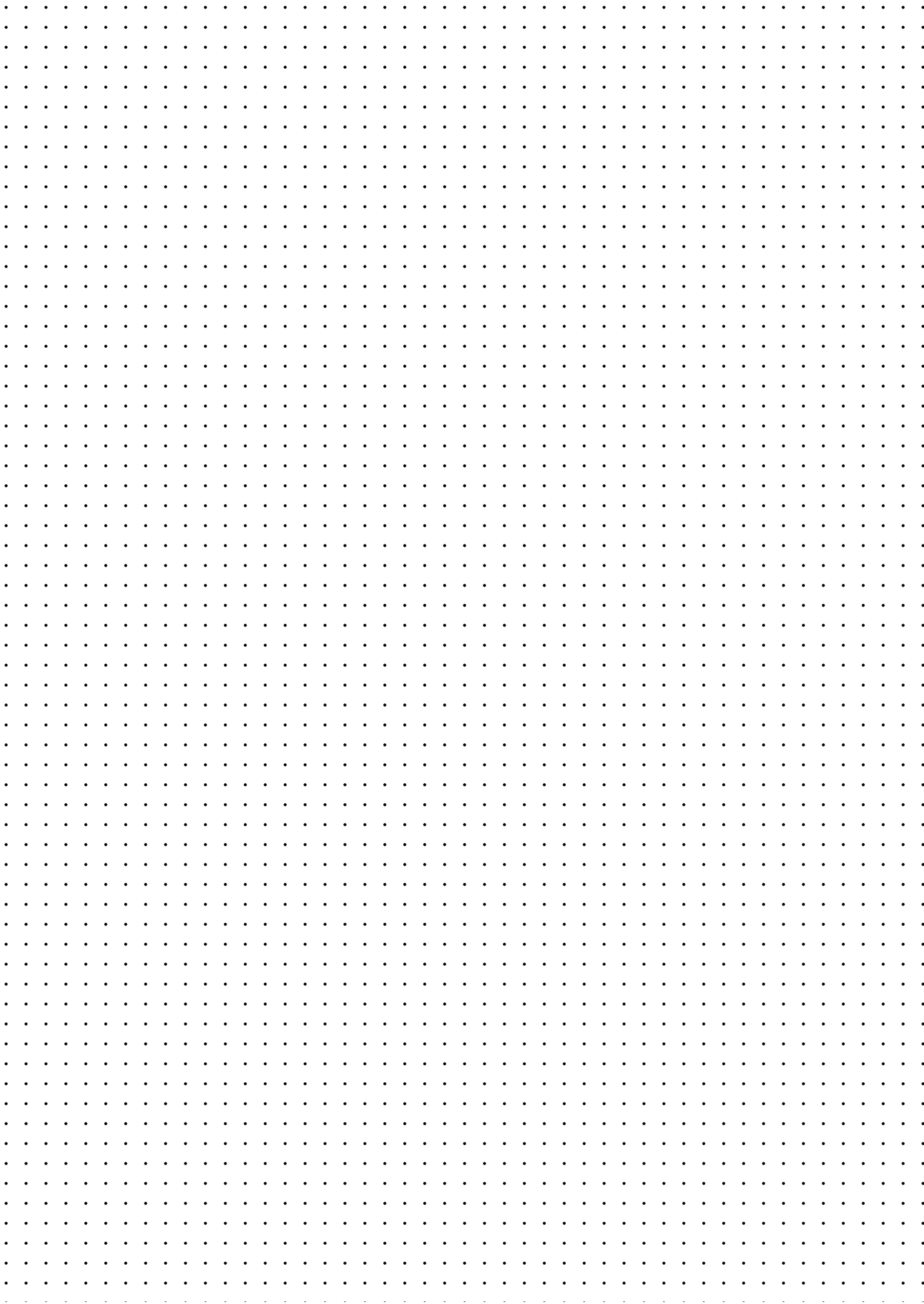
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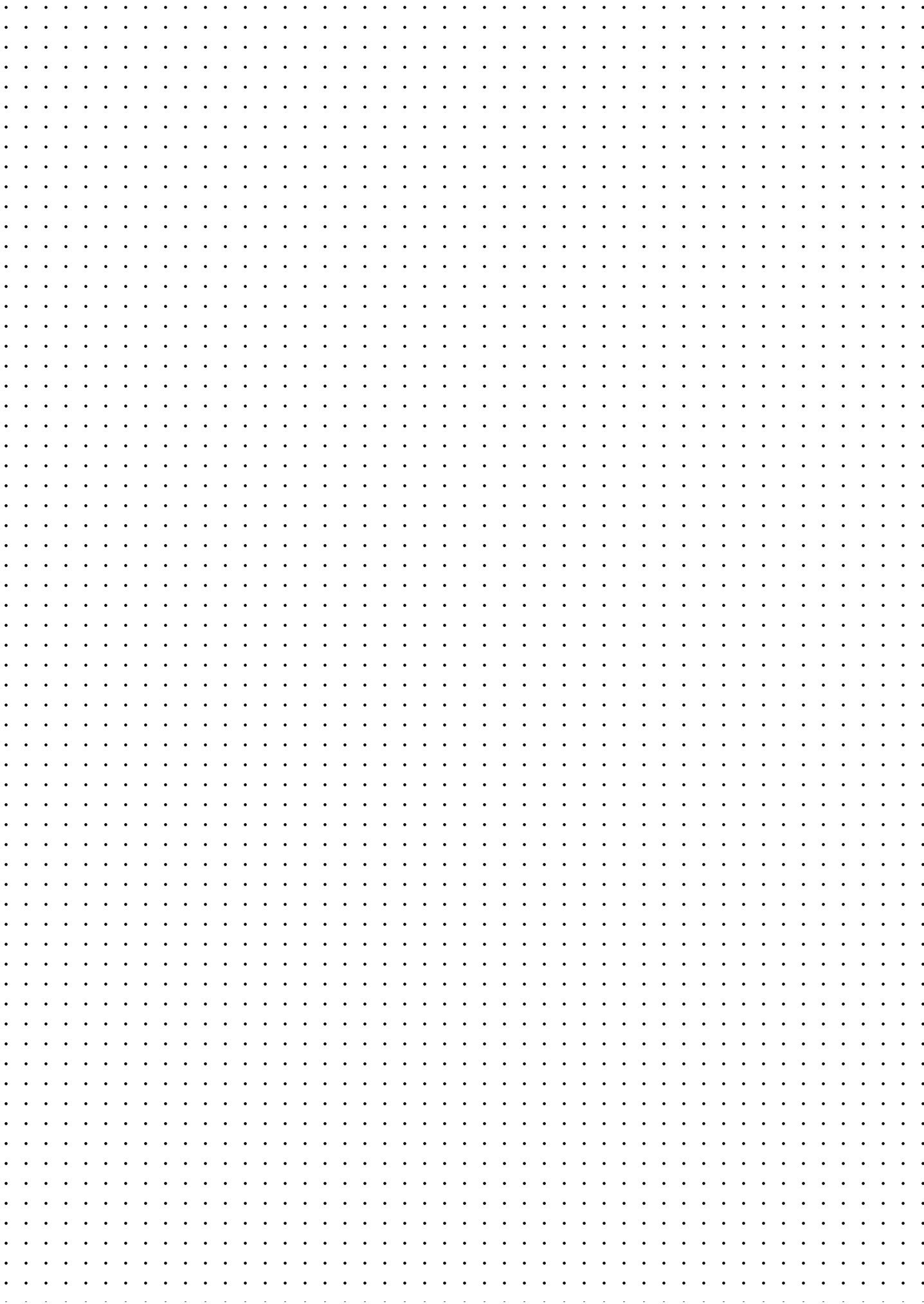
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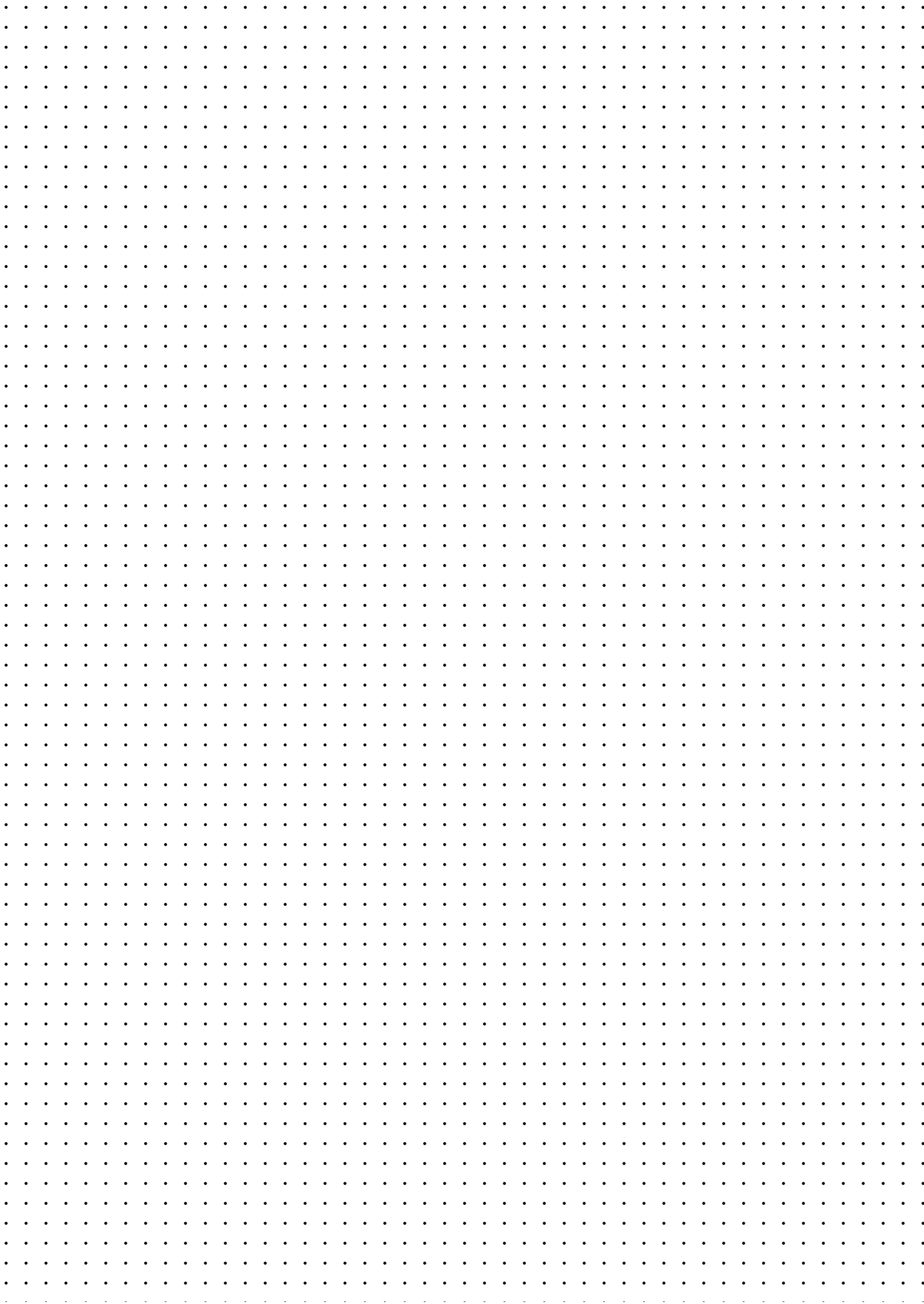
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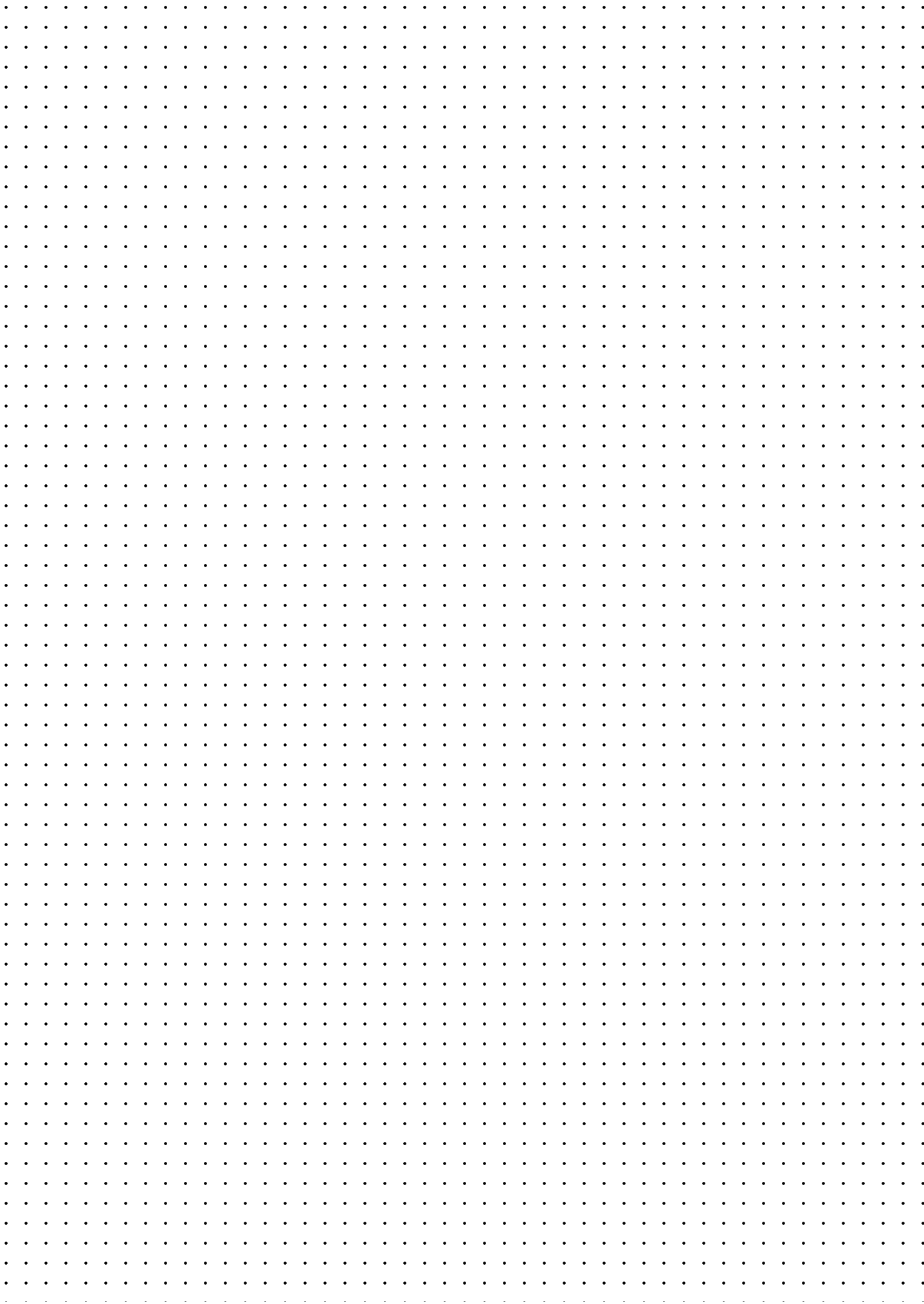
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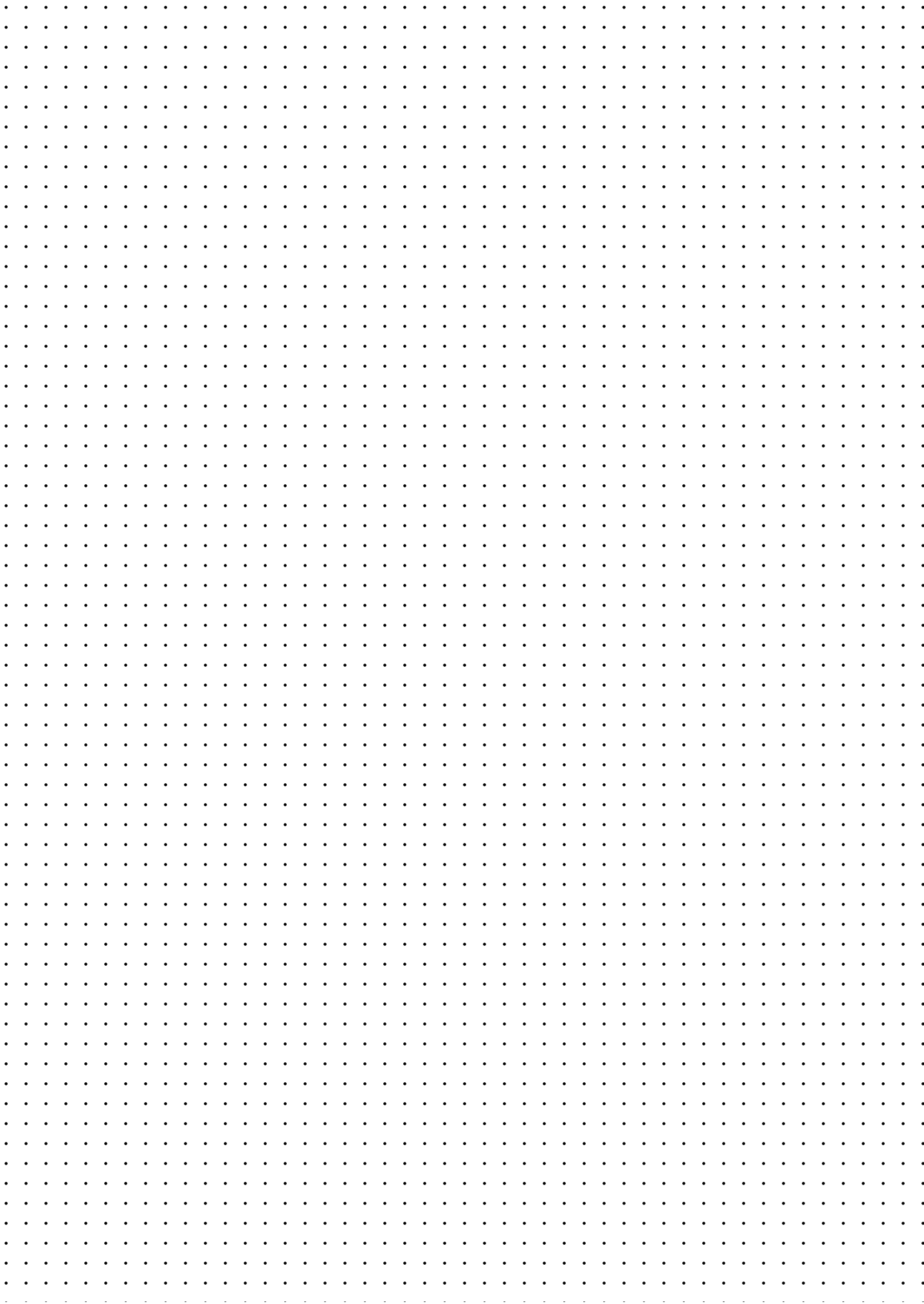
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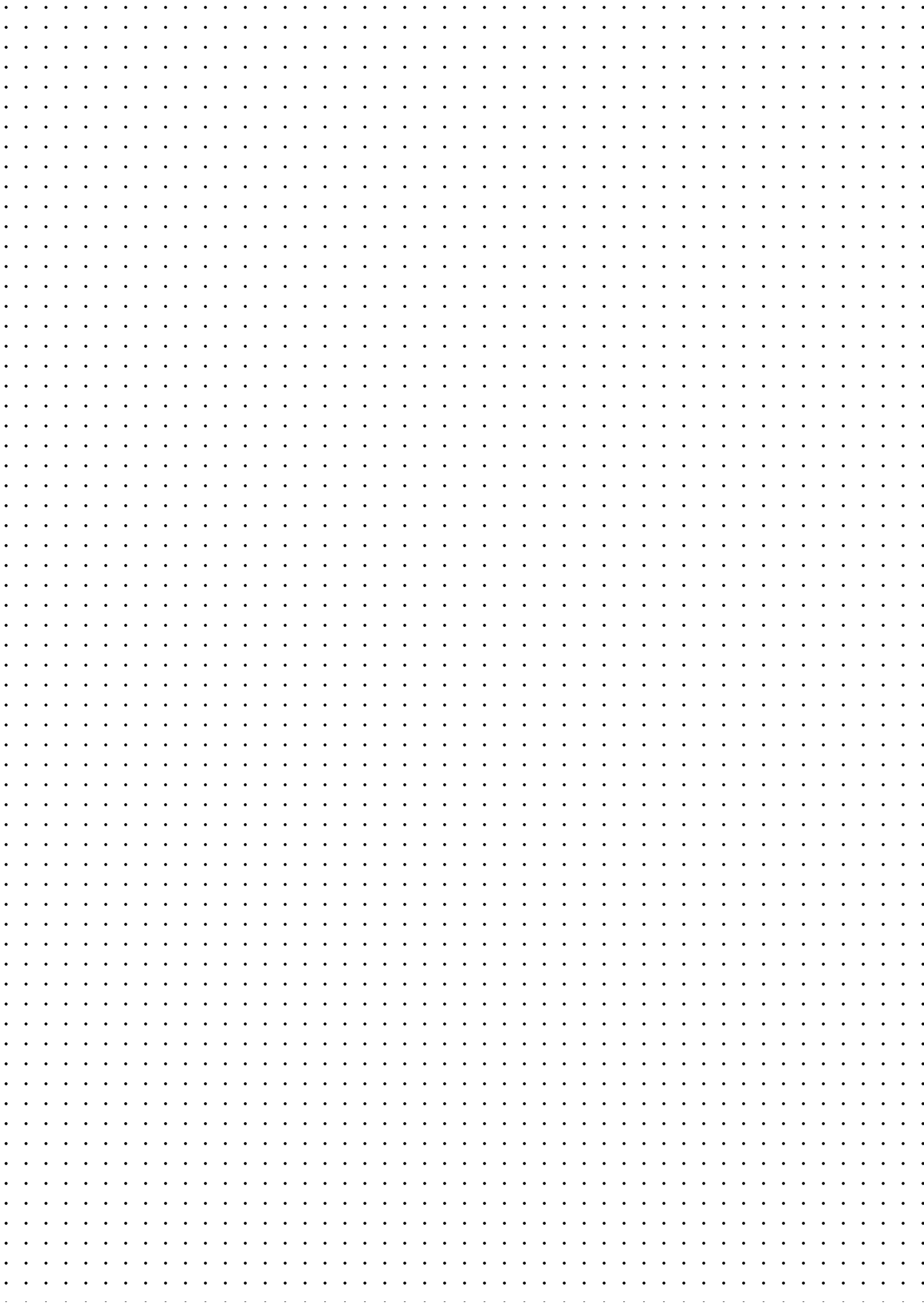
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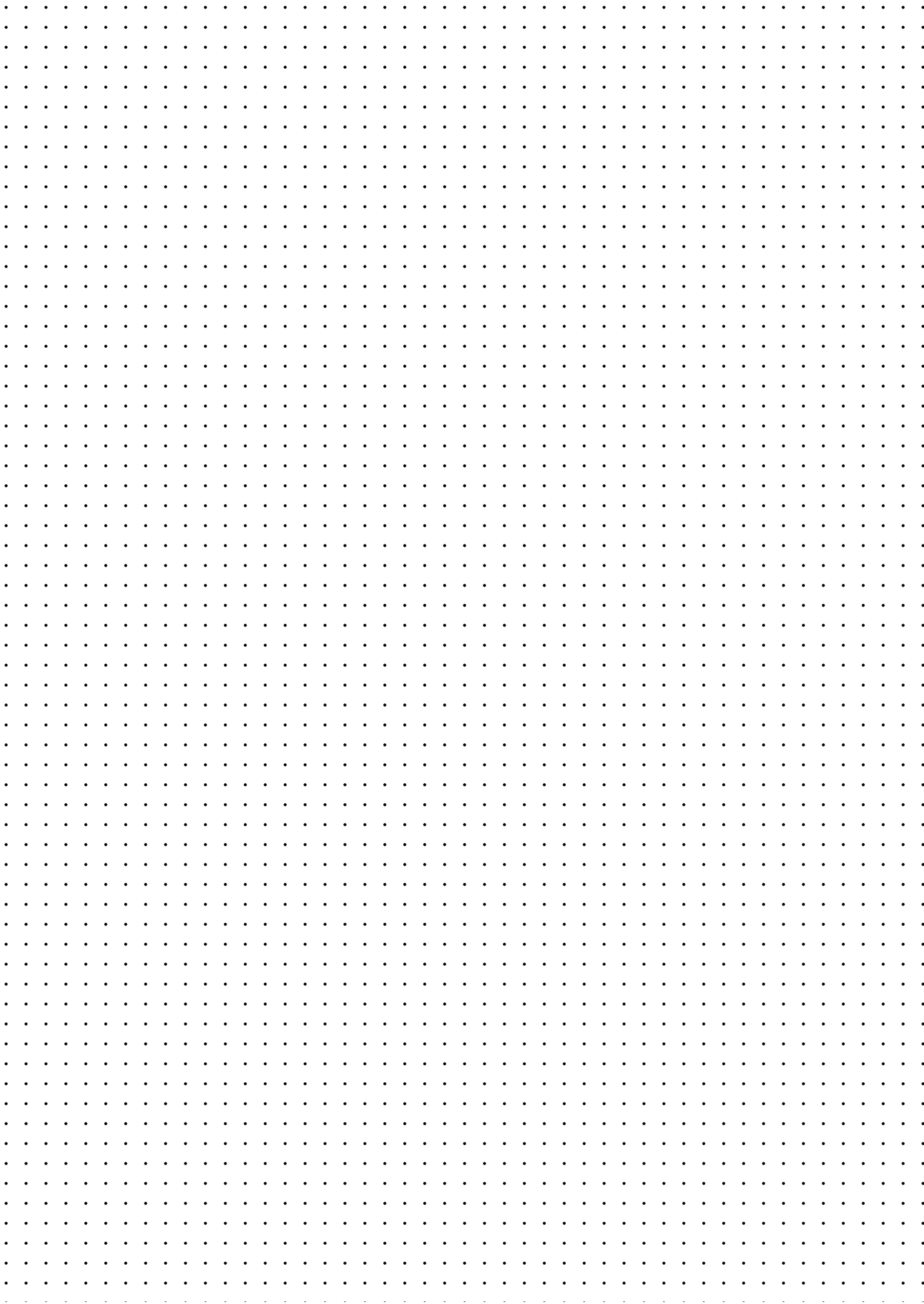


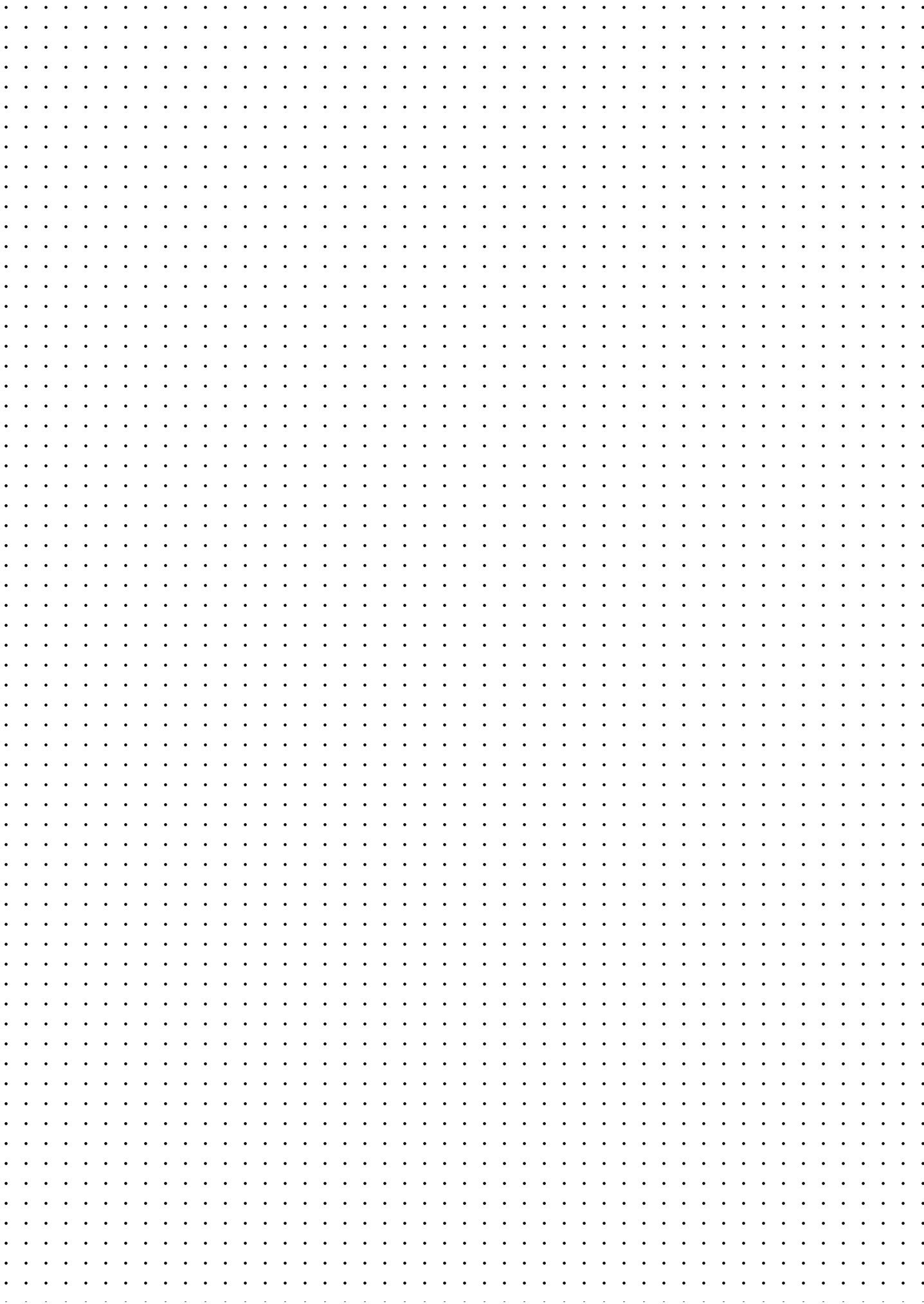


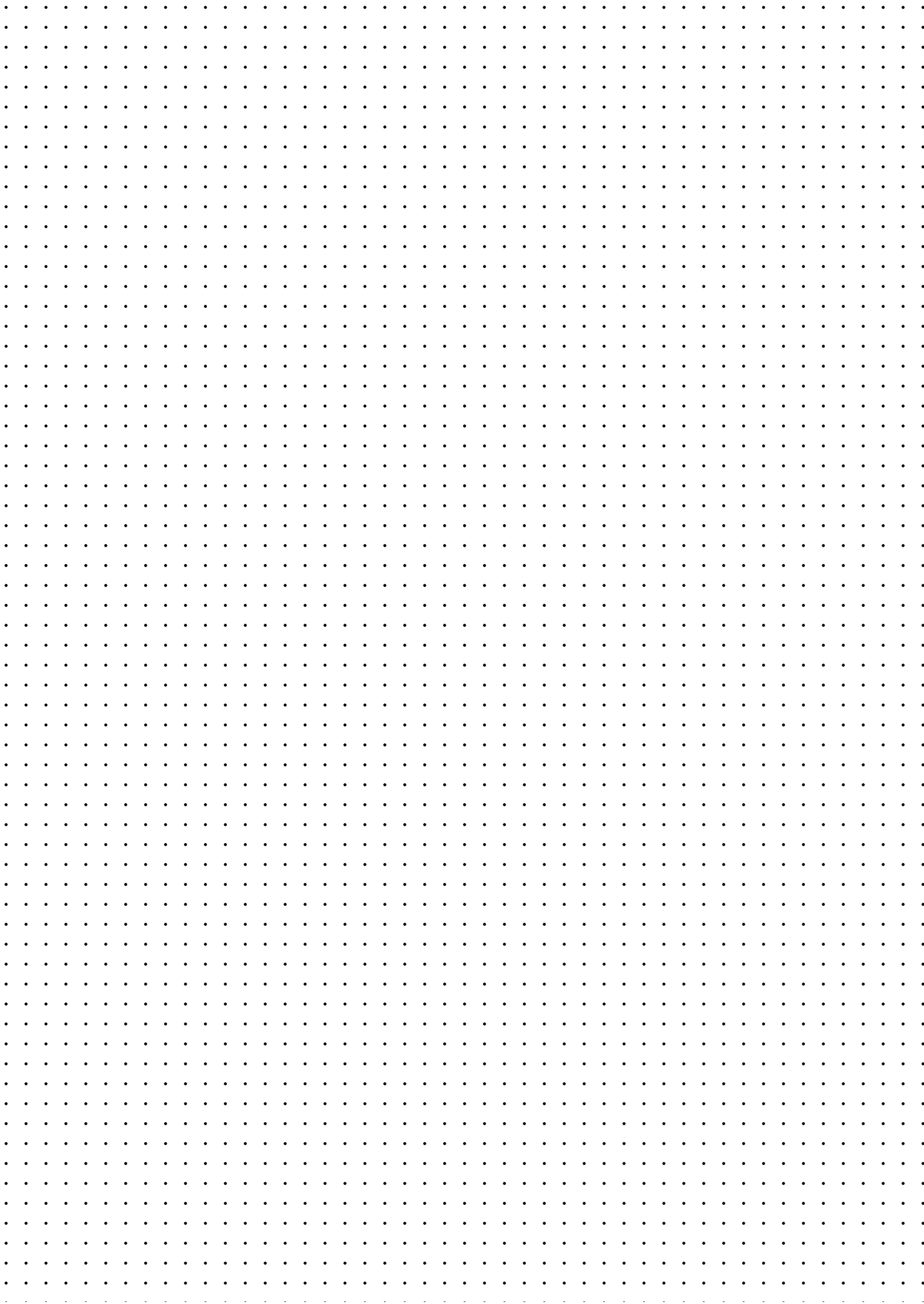
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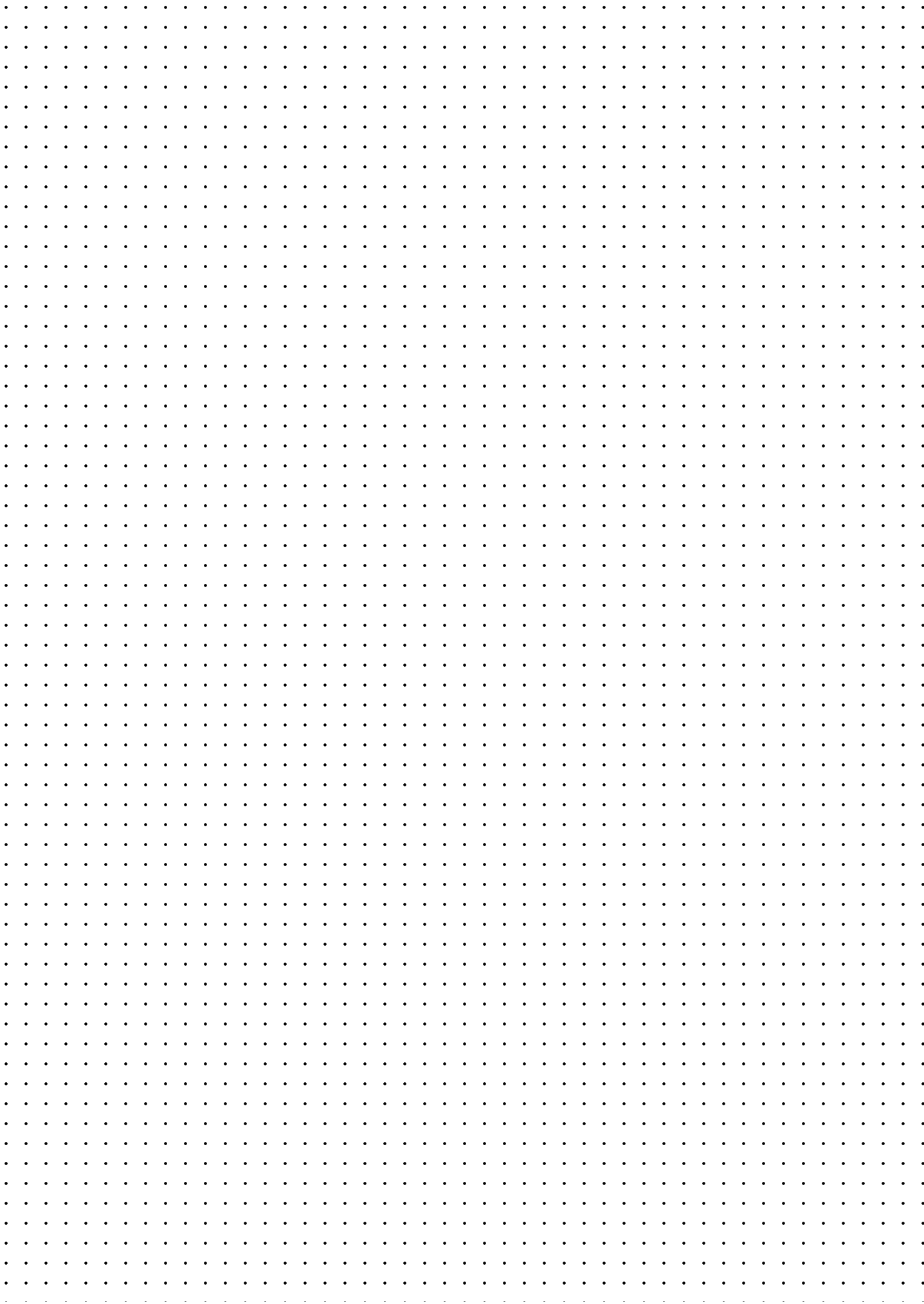


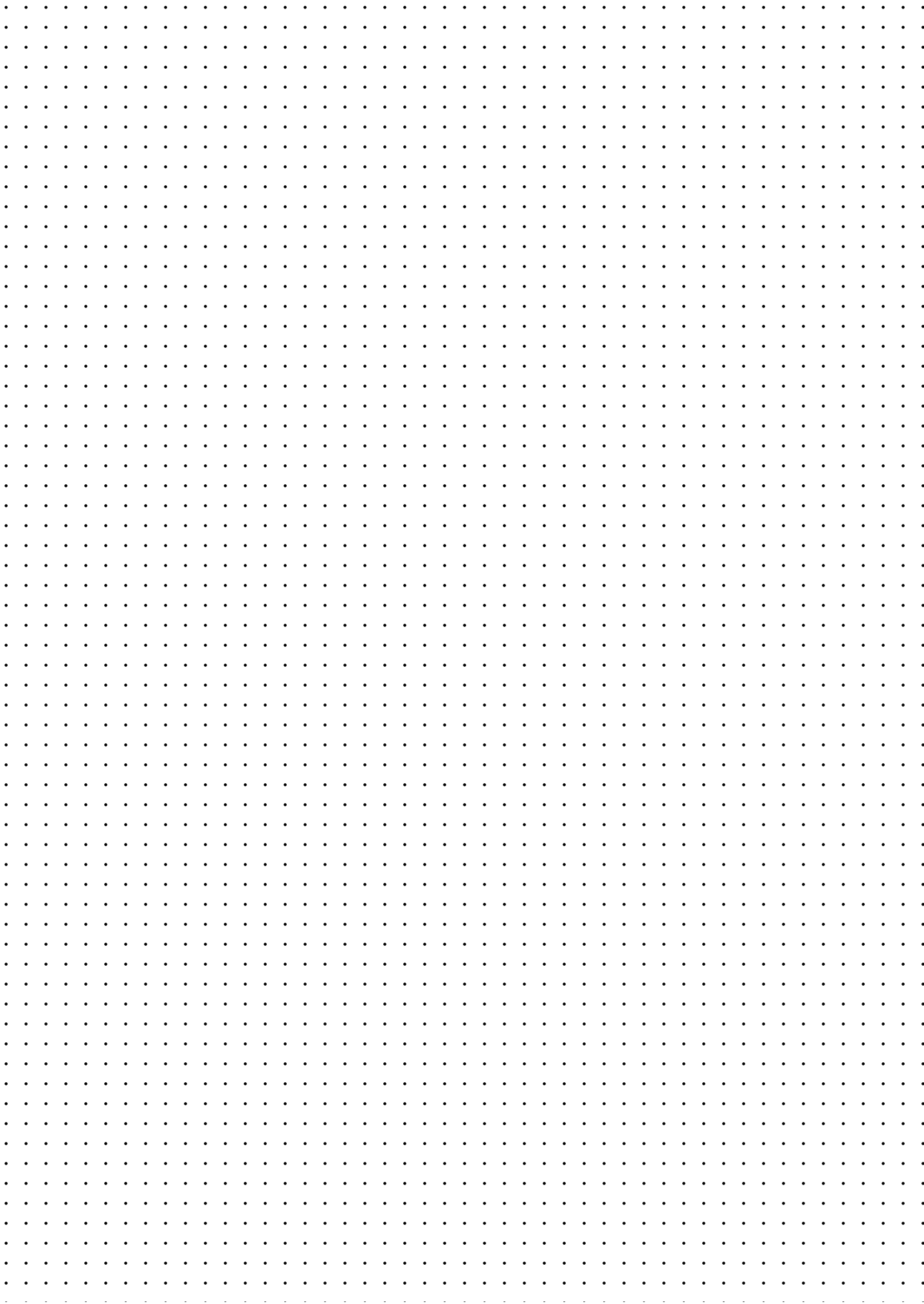


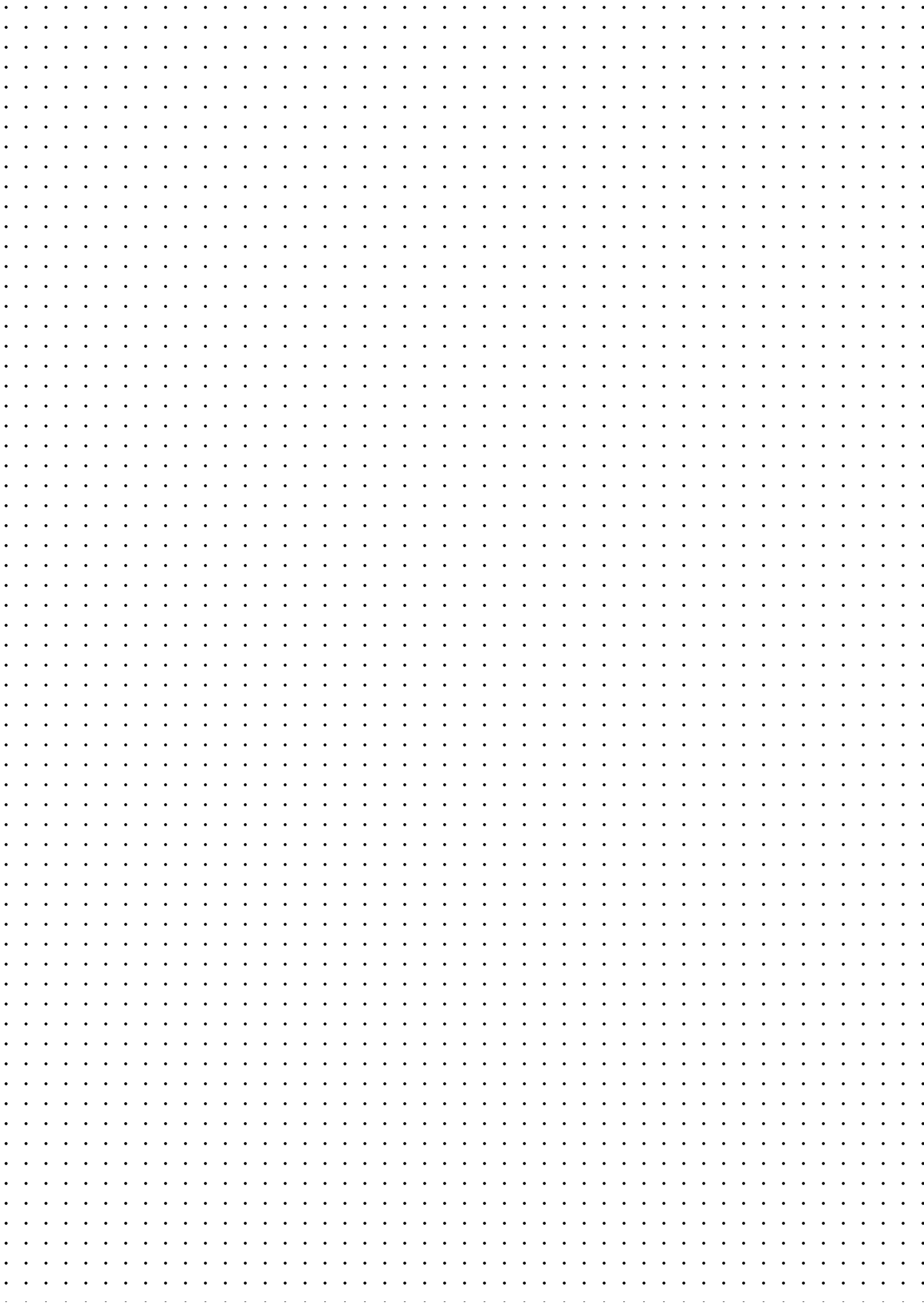
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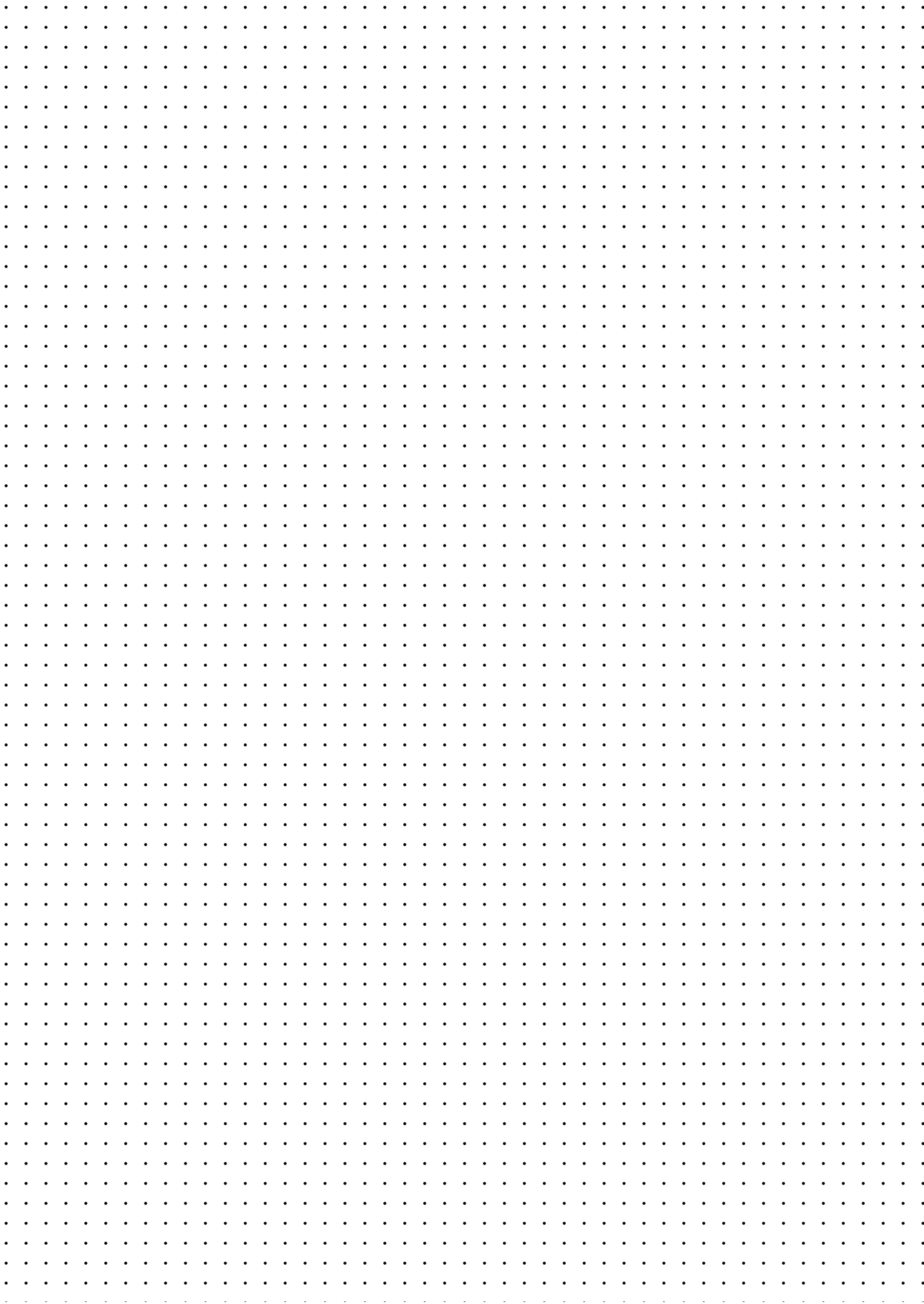
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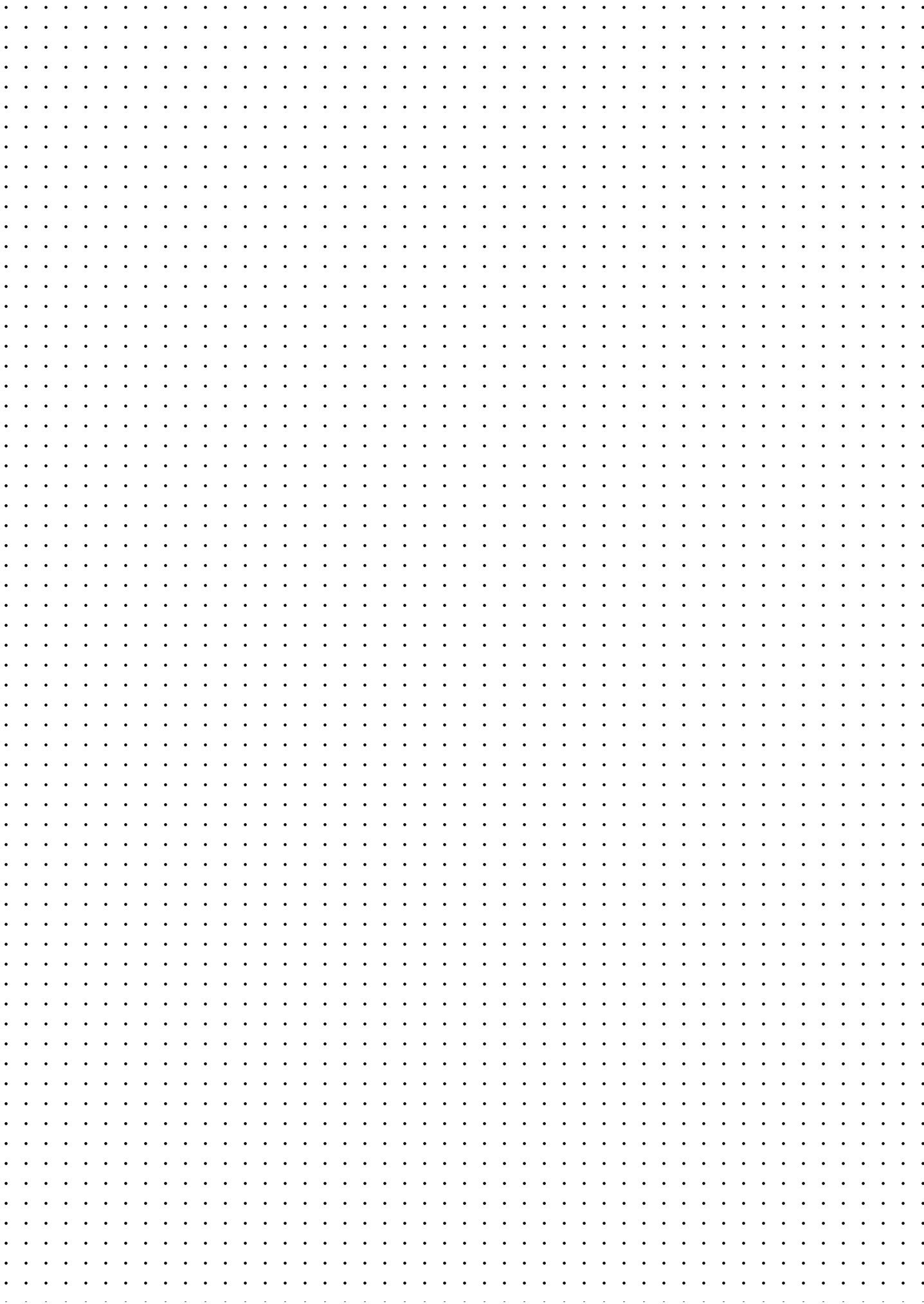
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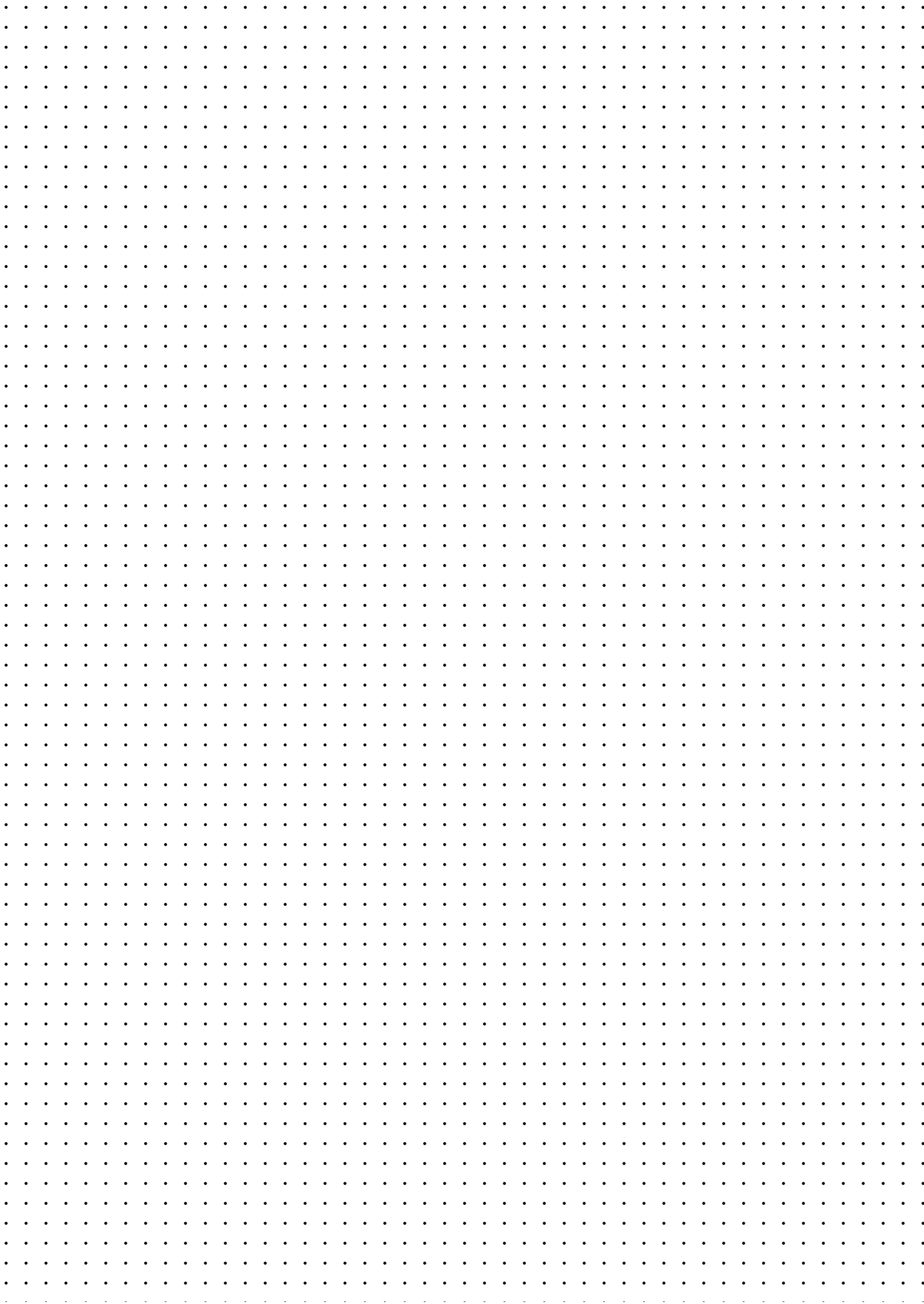
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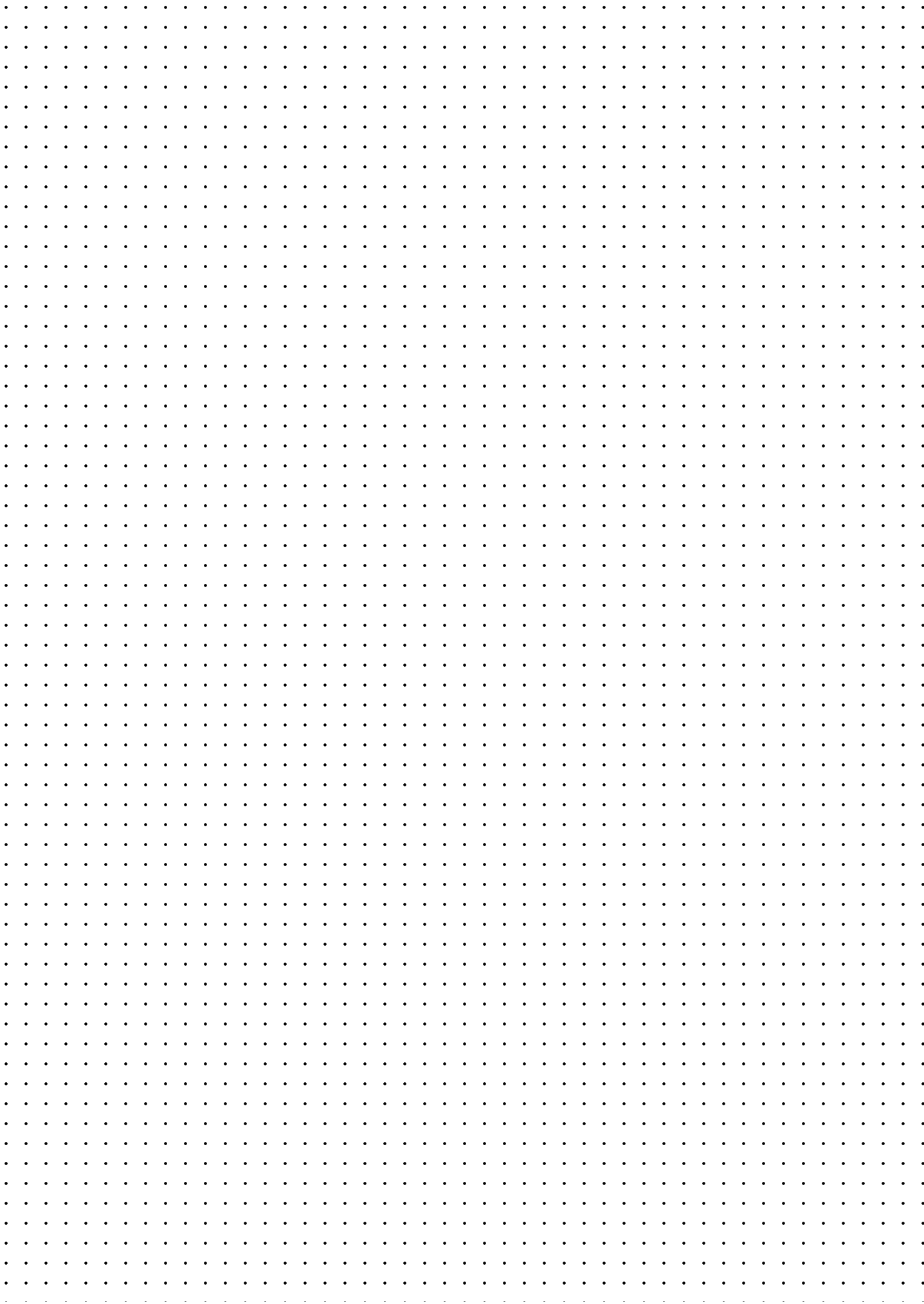
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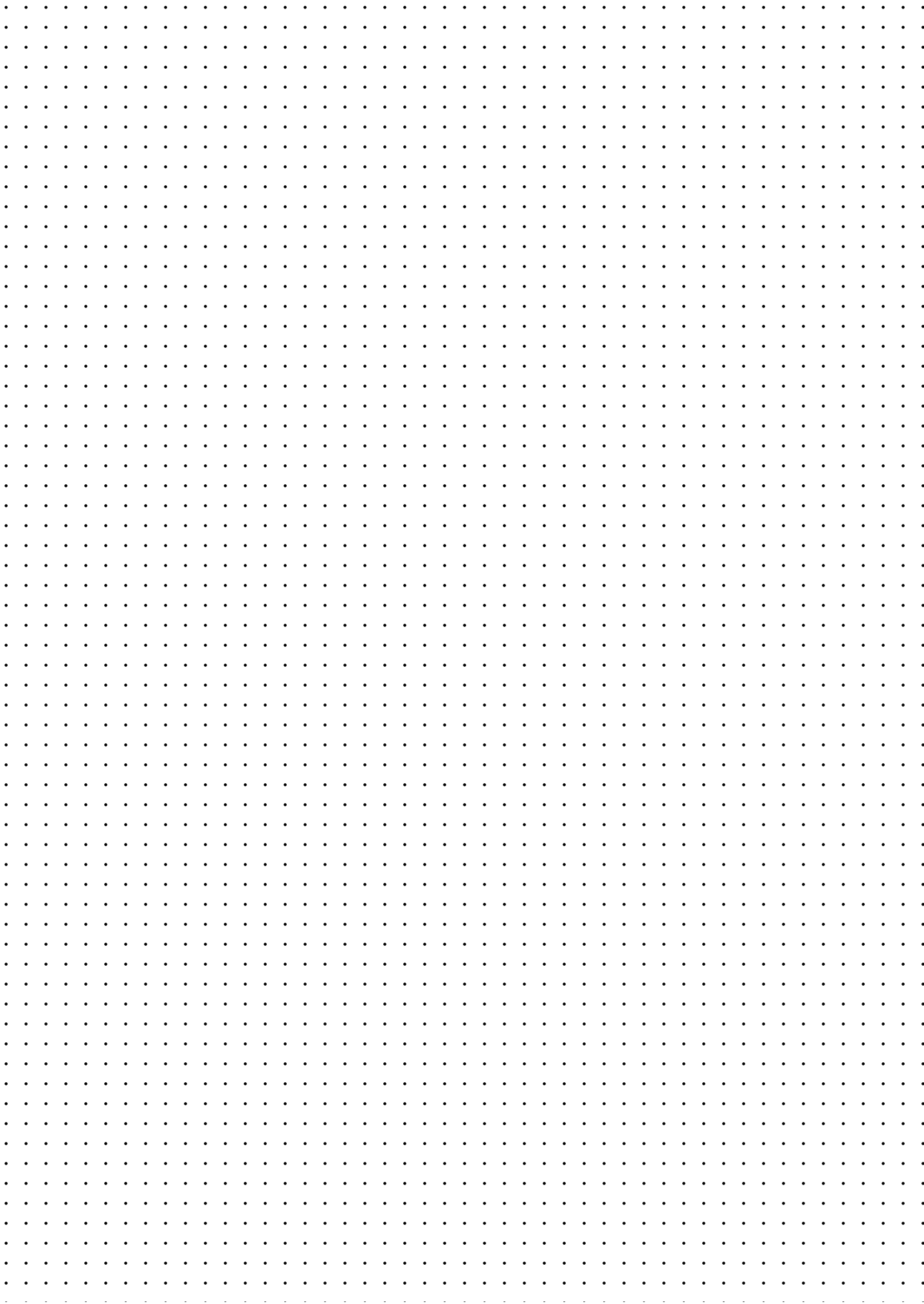
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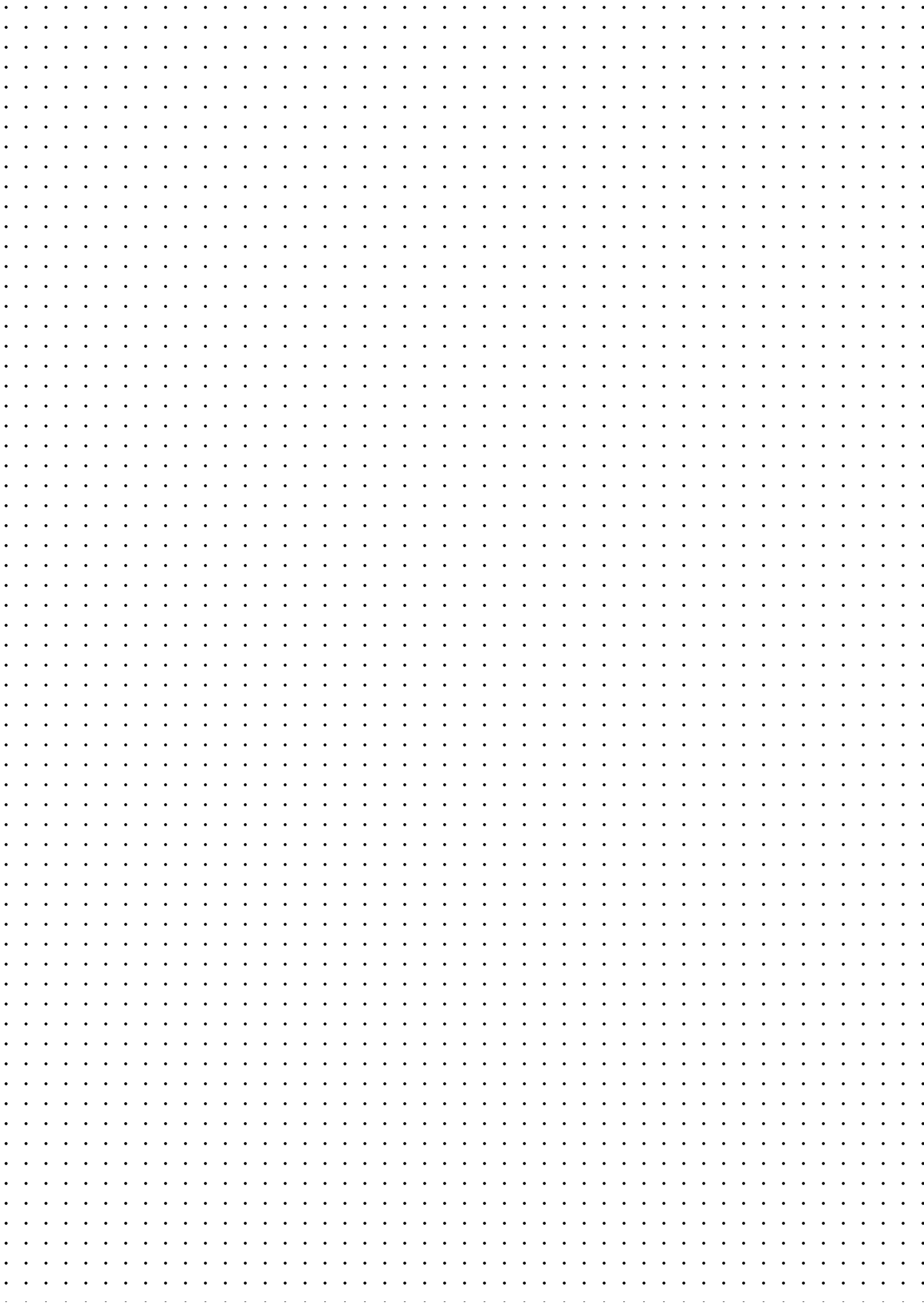
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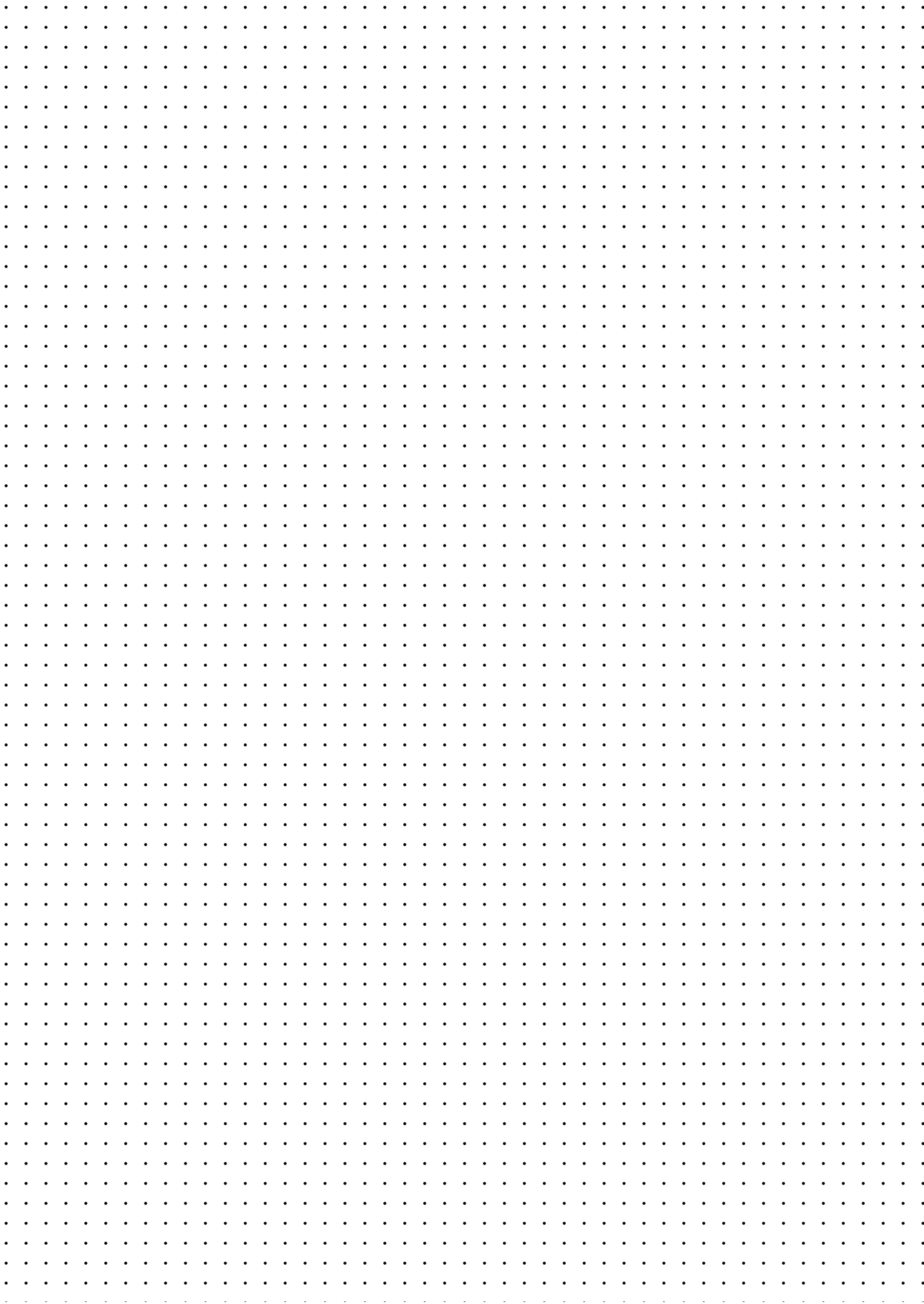
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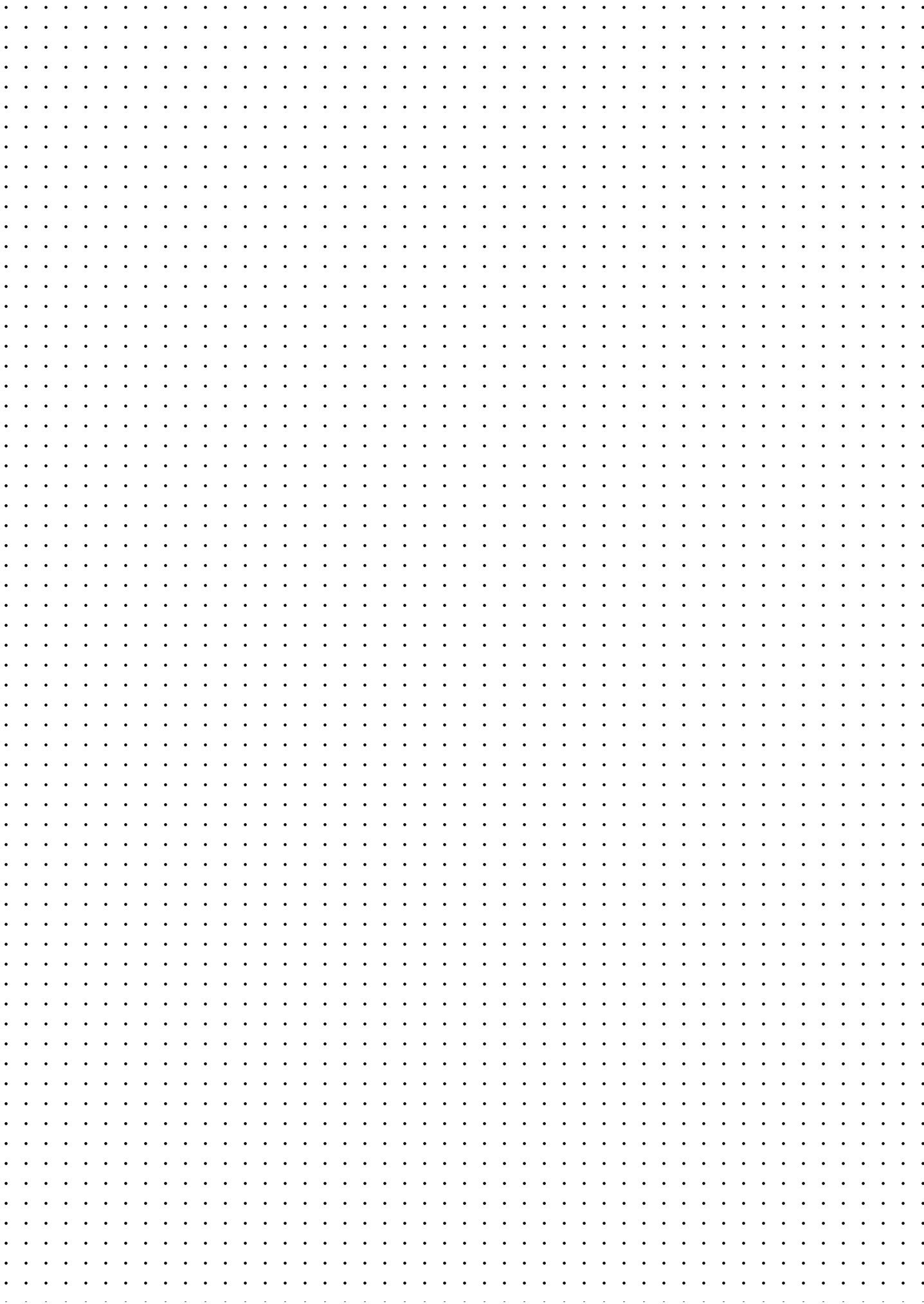
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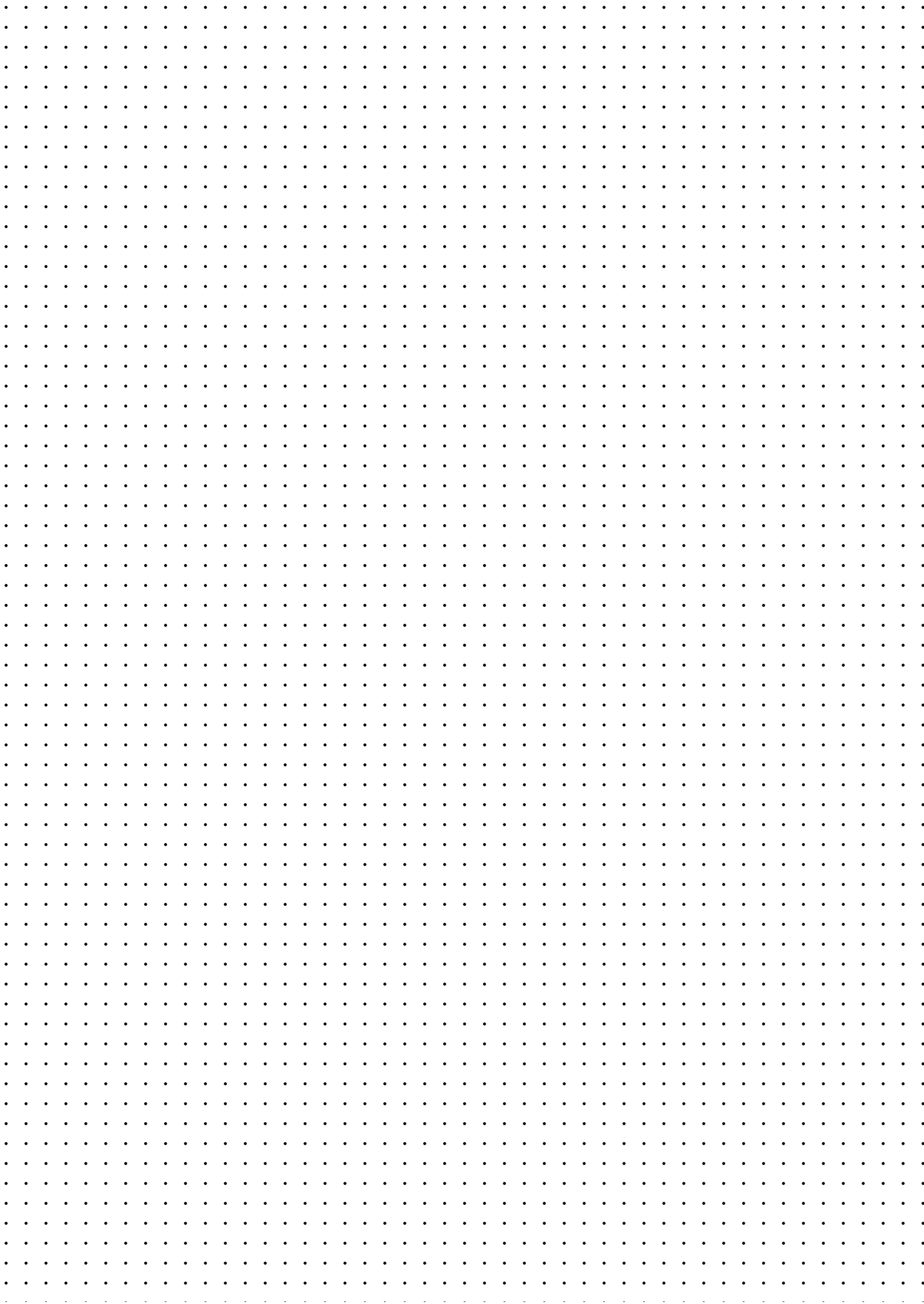
Metrology / Result

Deviations / Issues

Witness / Sign-off



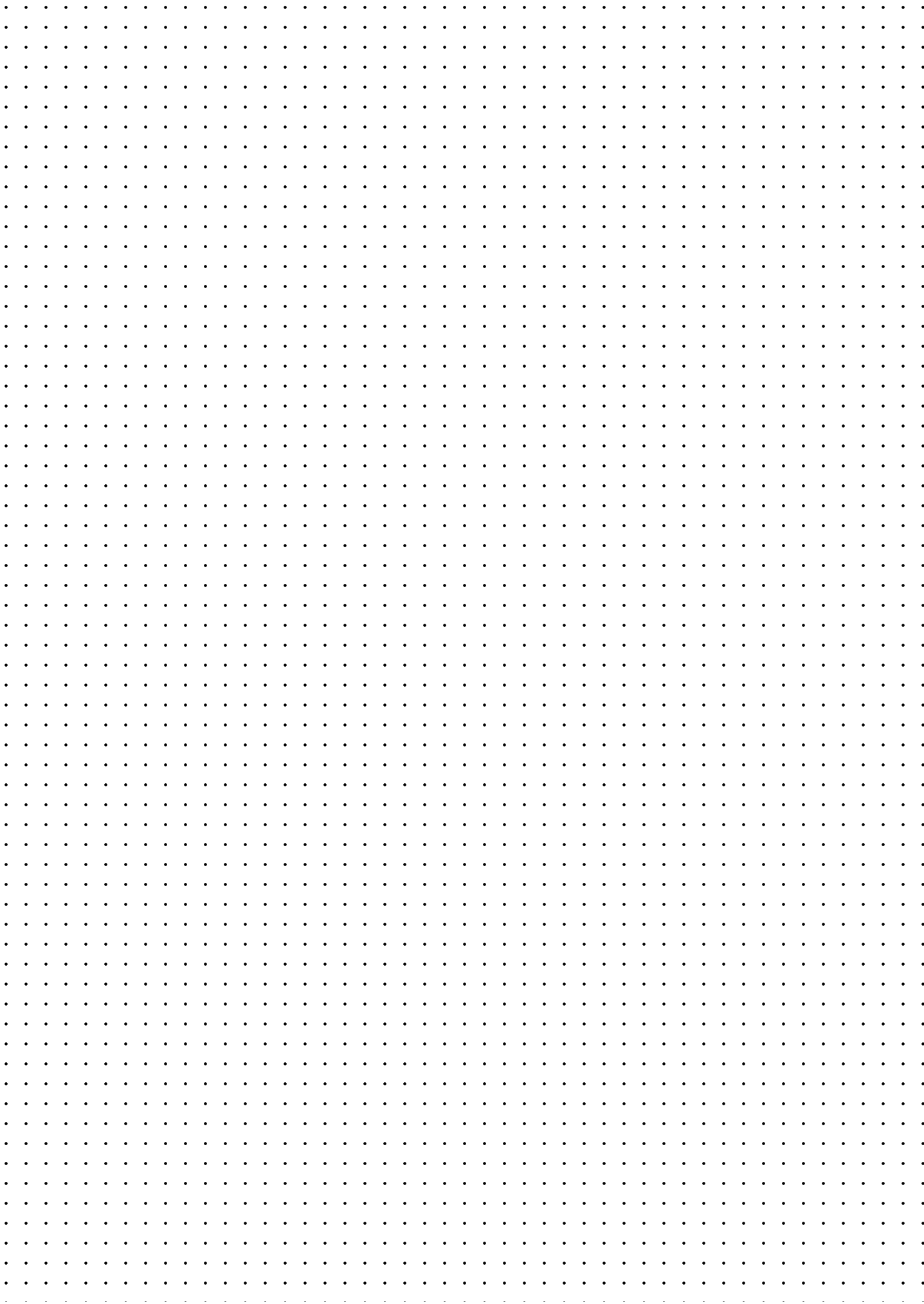


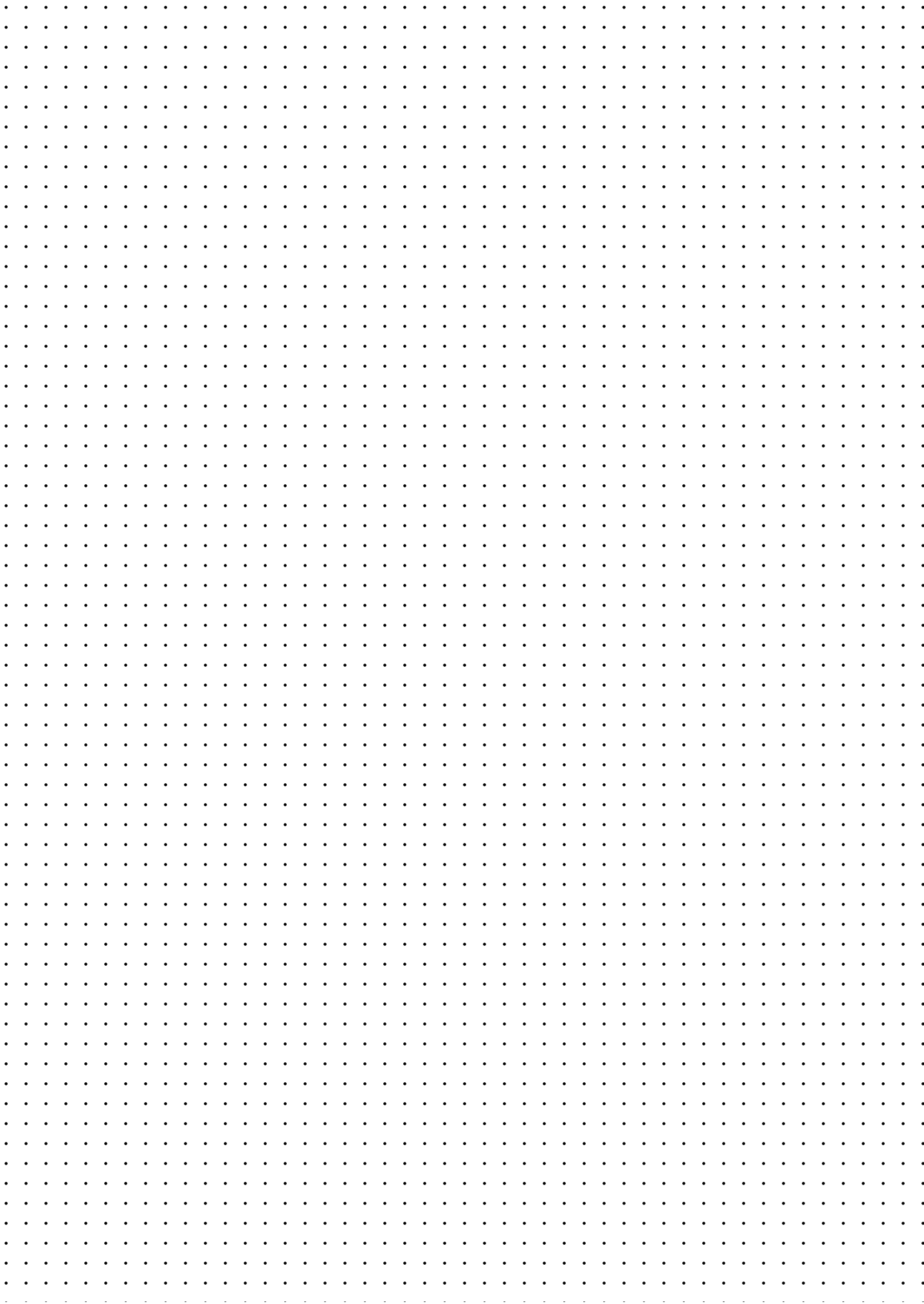


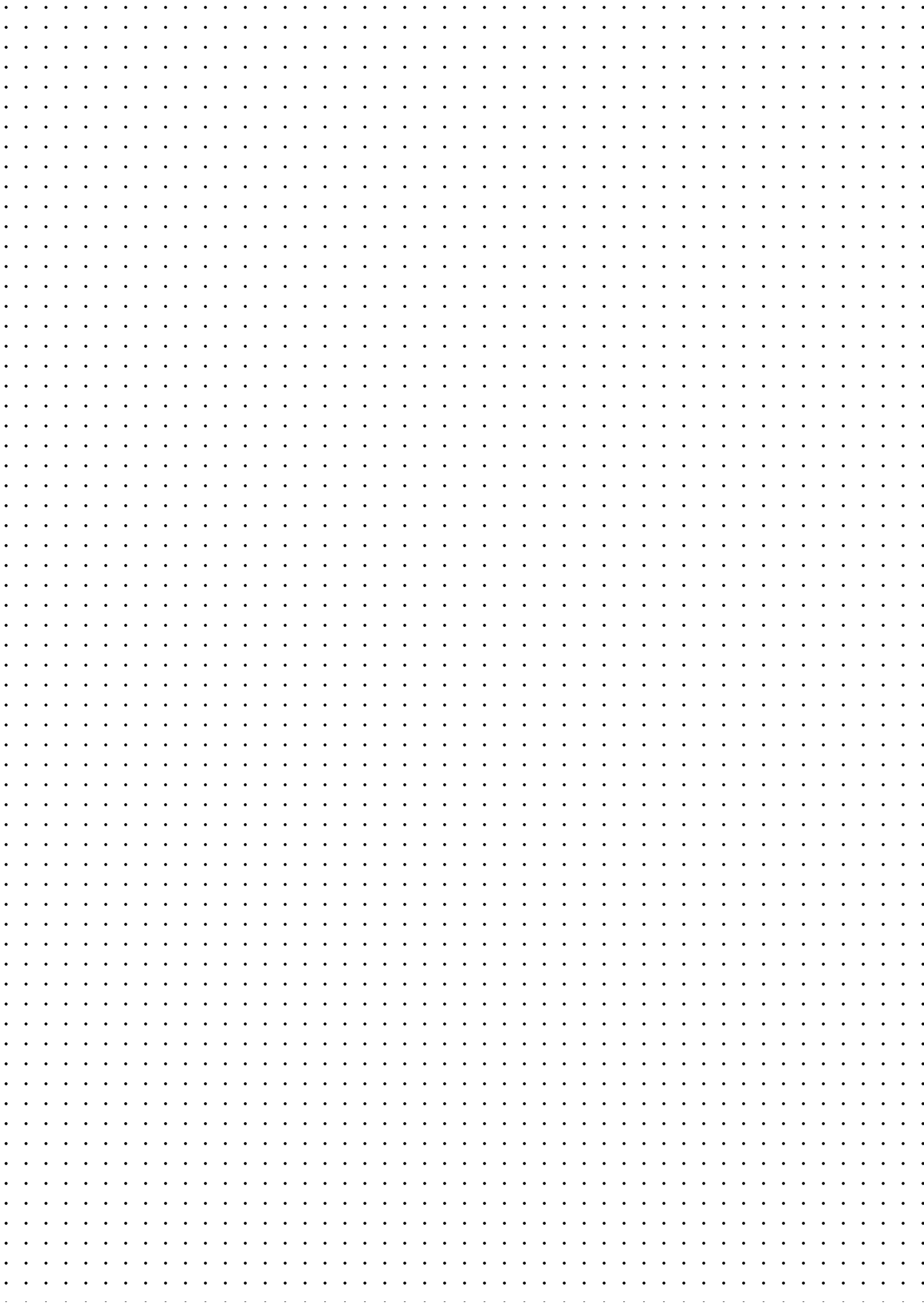
Semiconductor Process Record

No. _____

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Tool / Station			Process Step		
Objective					
Materials & Consumables					
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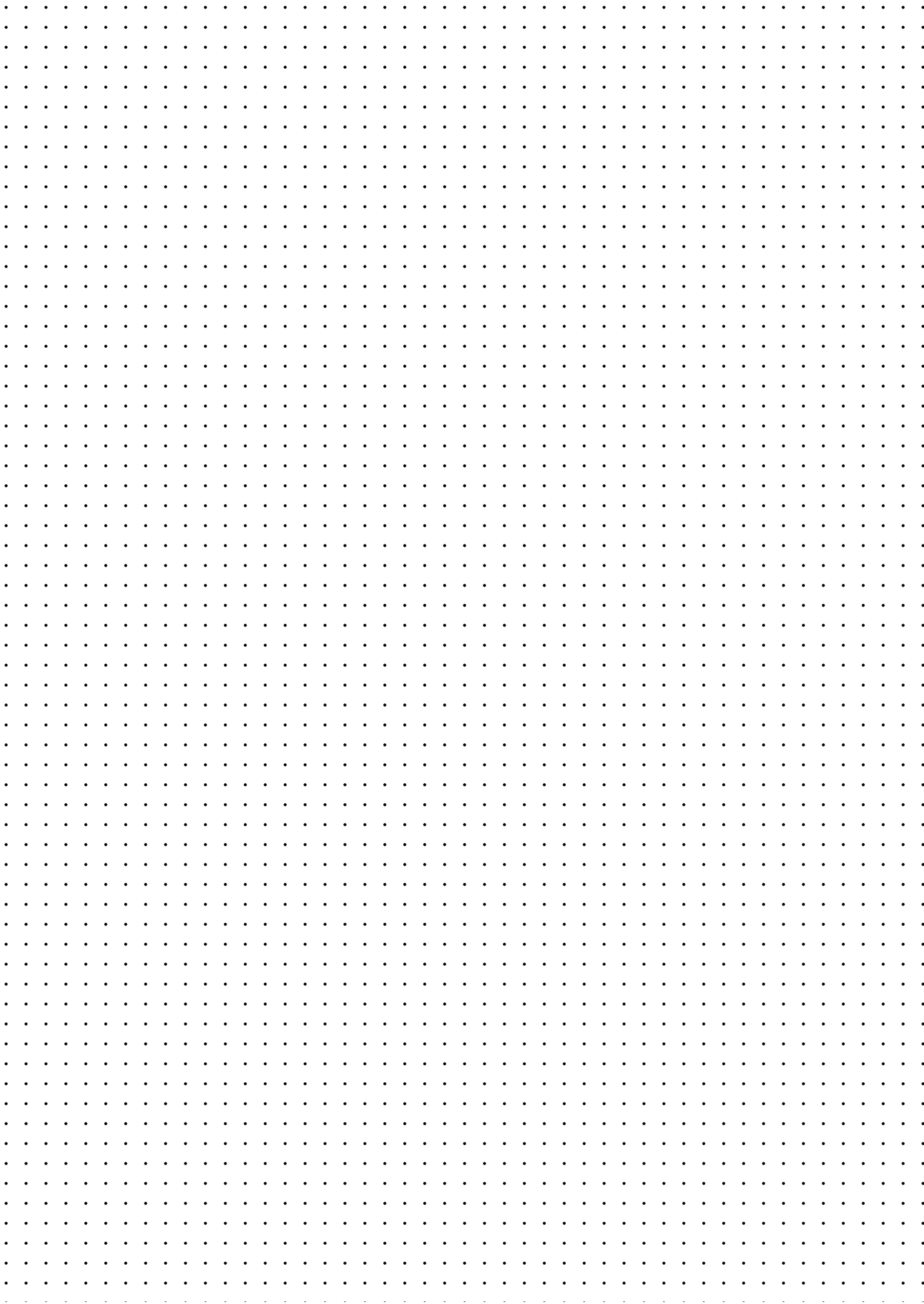


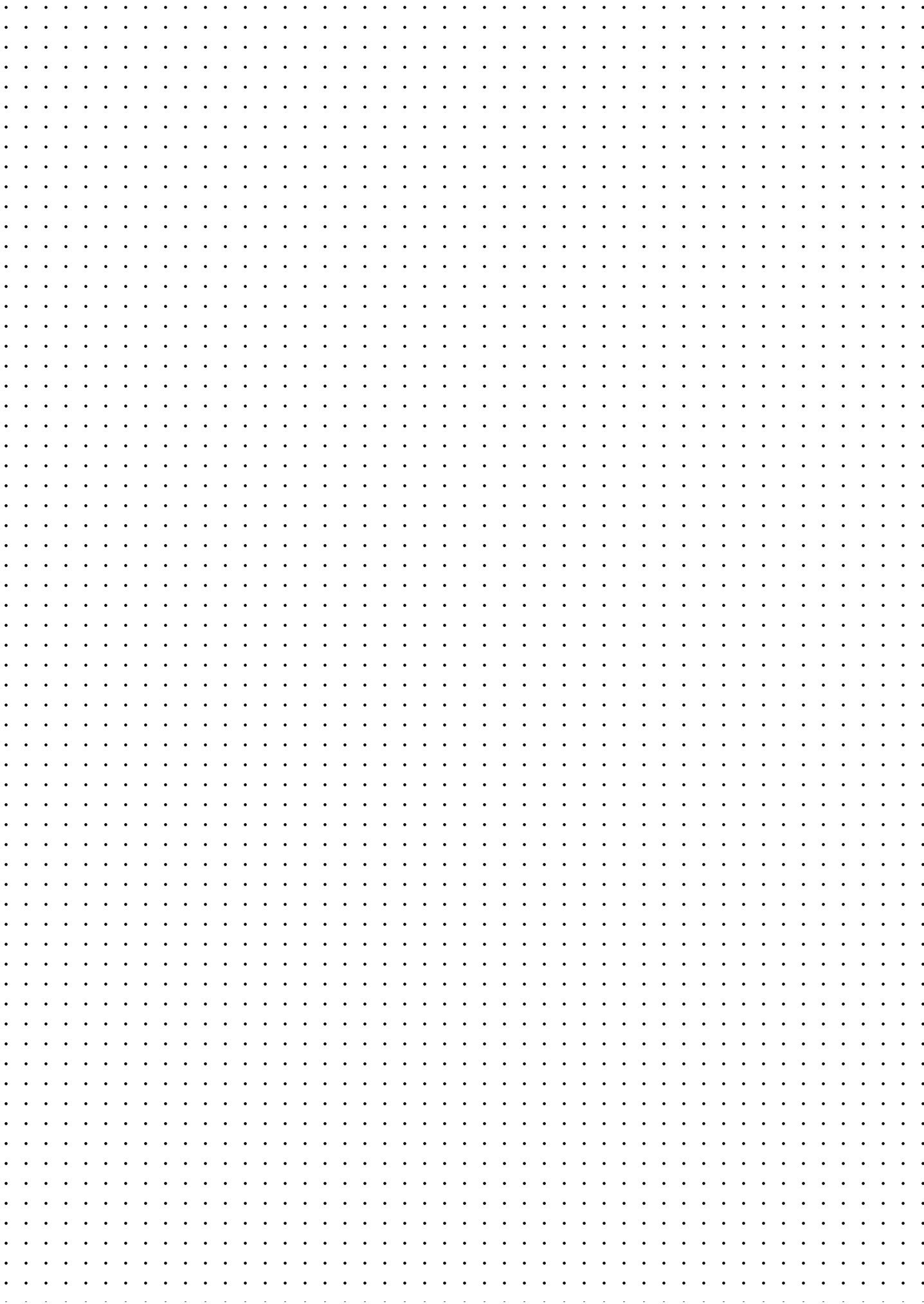


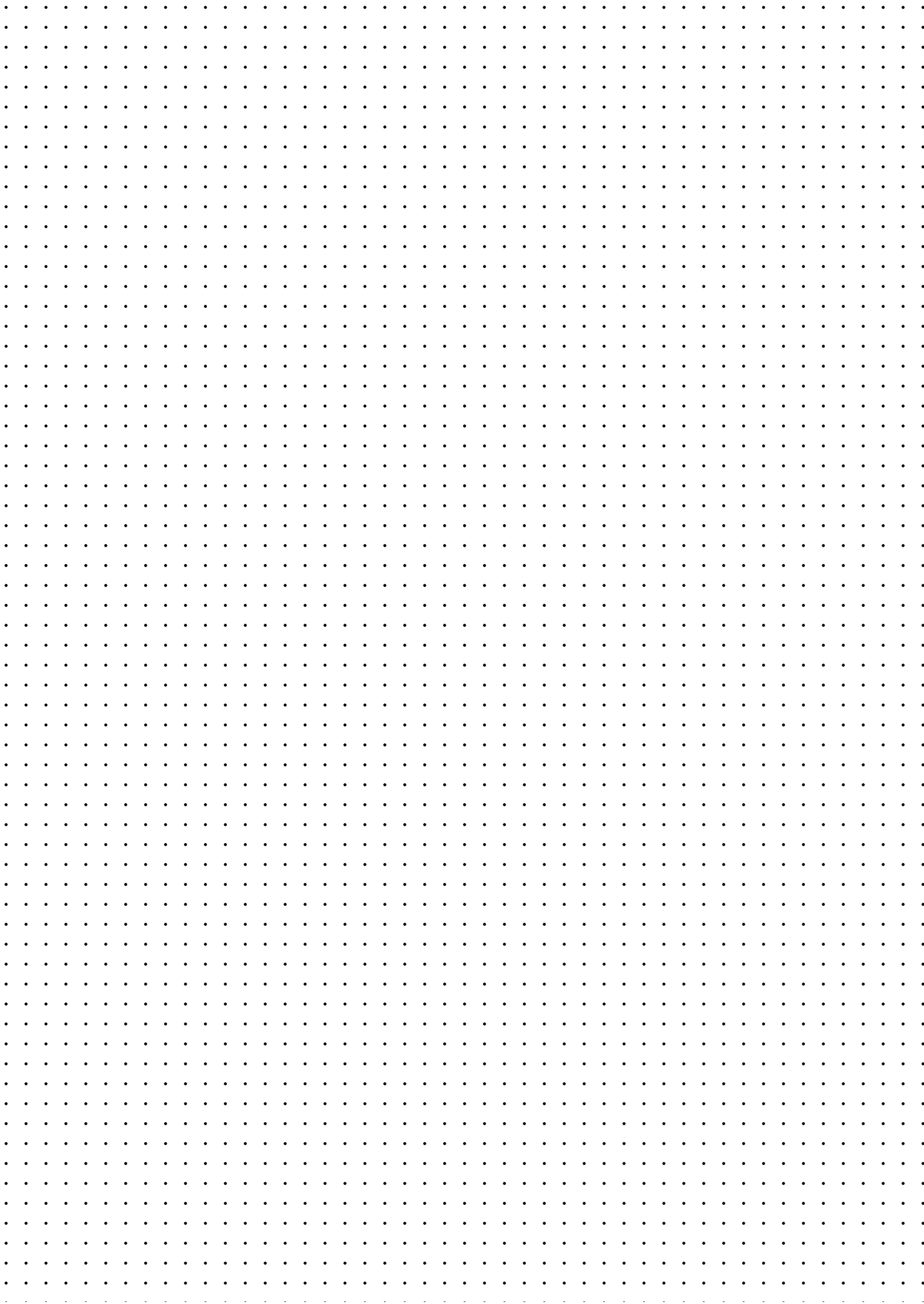
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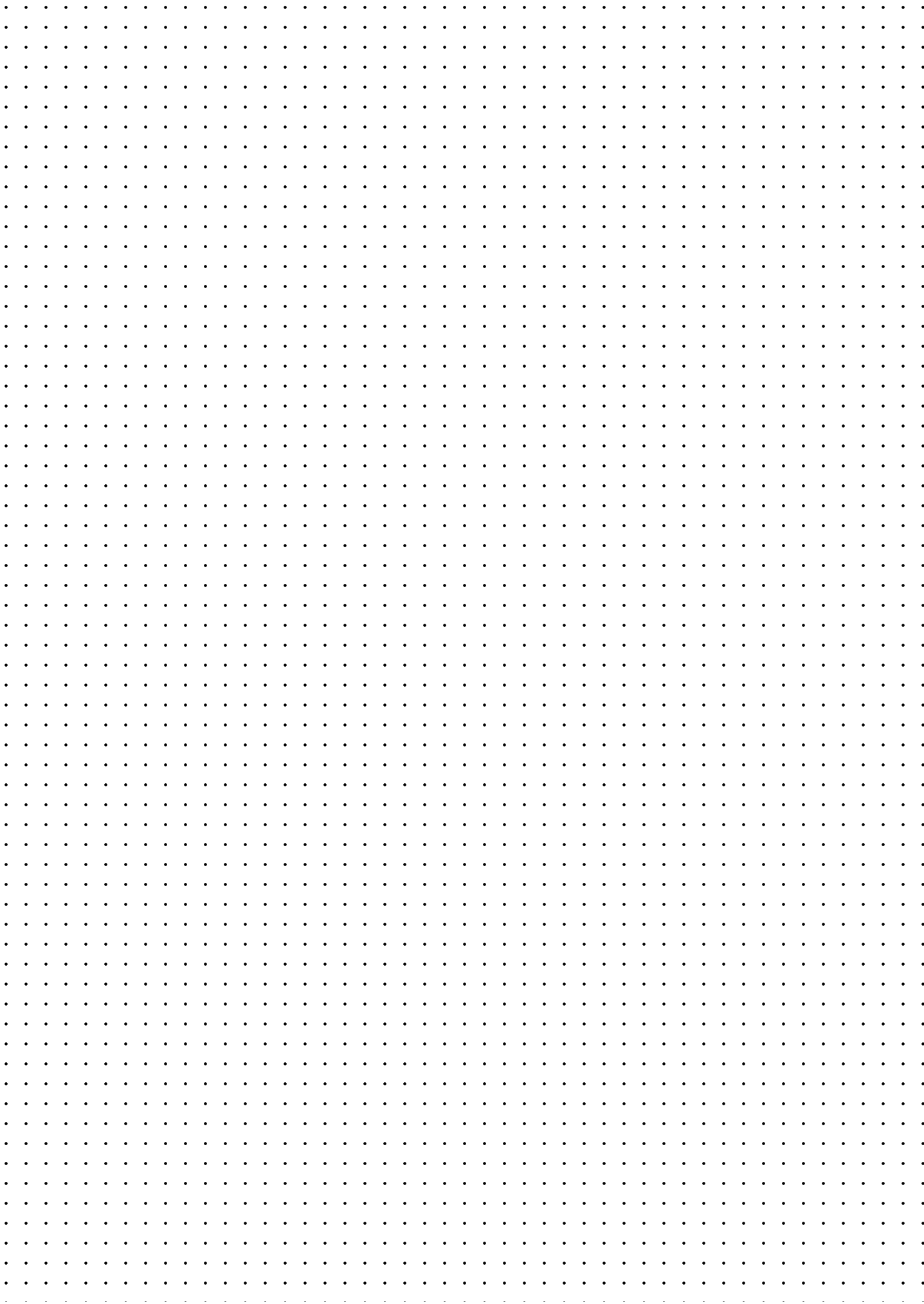
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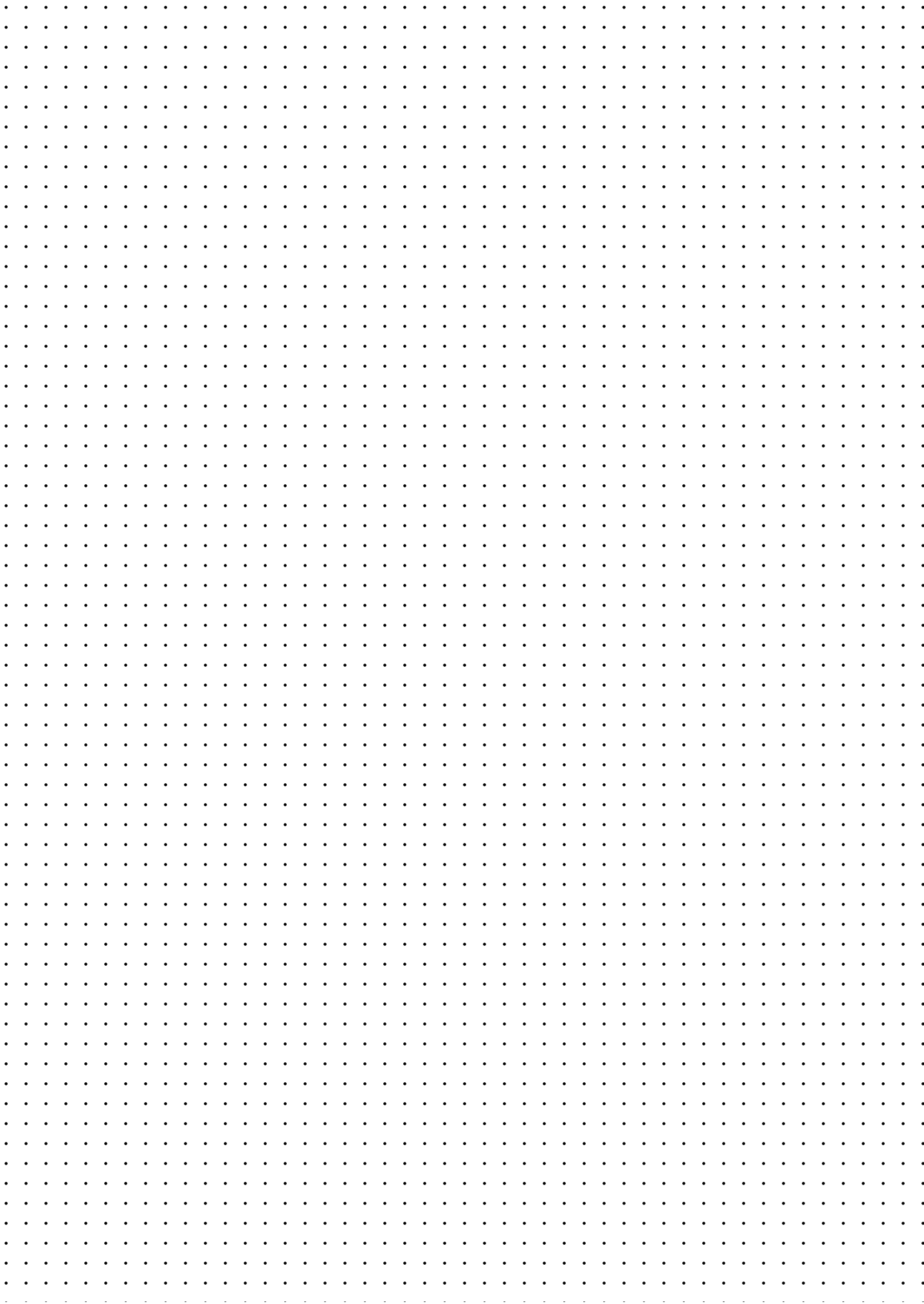
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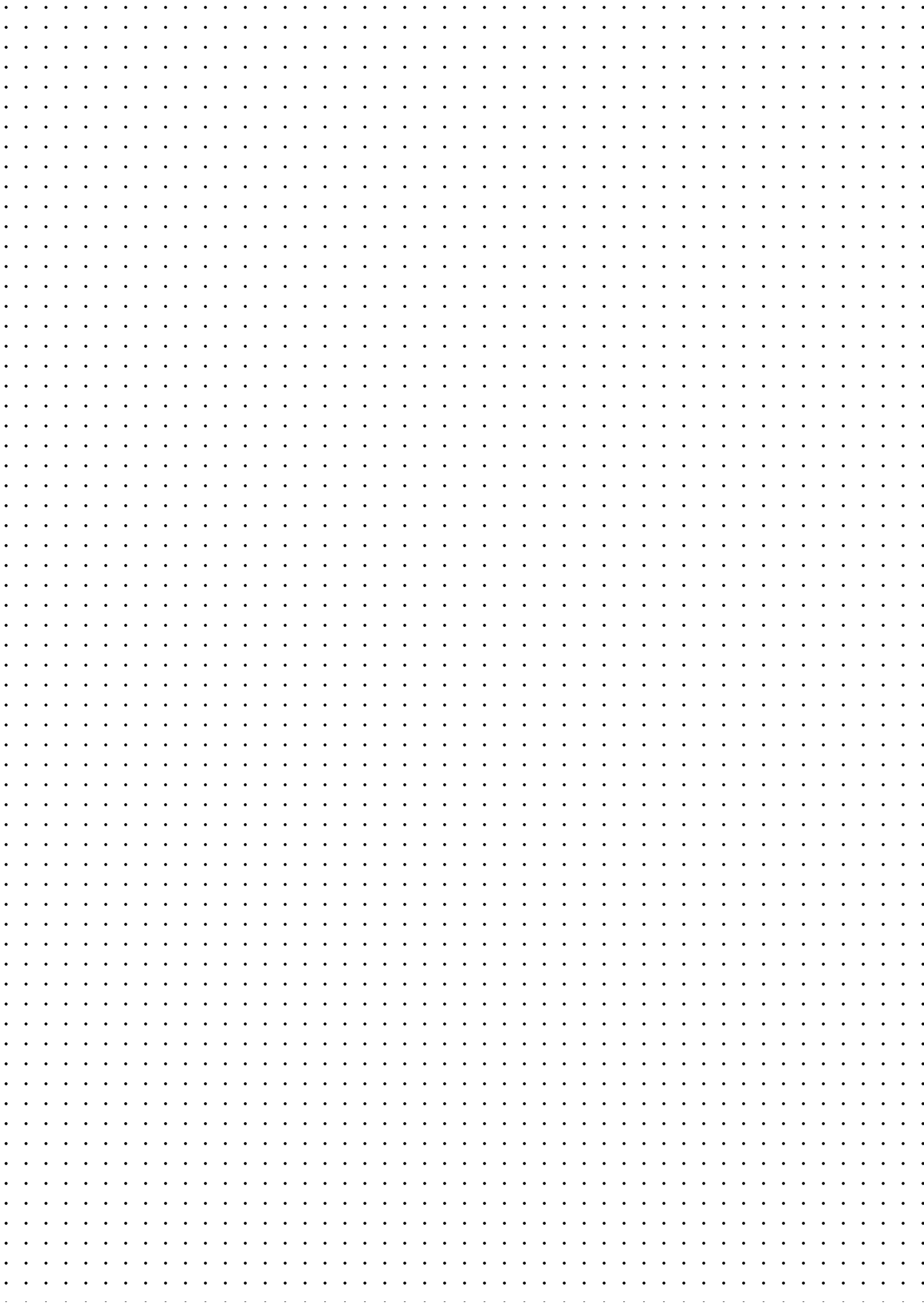
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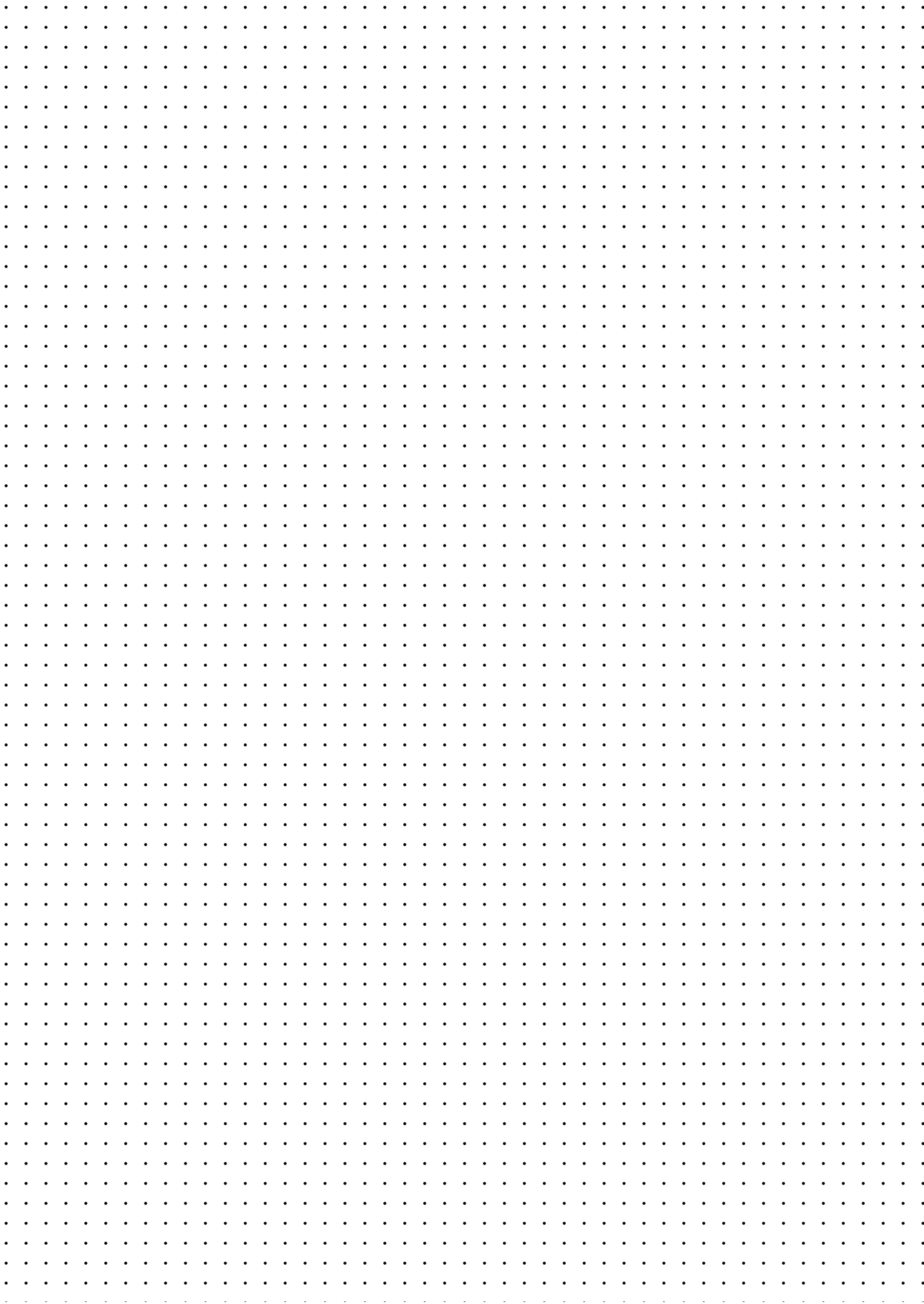
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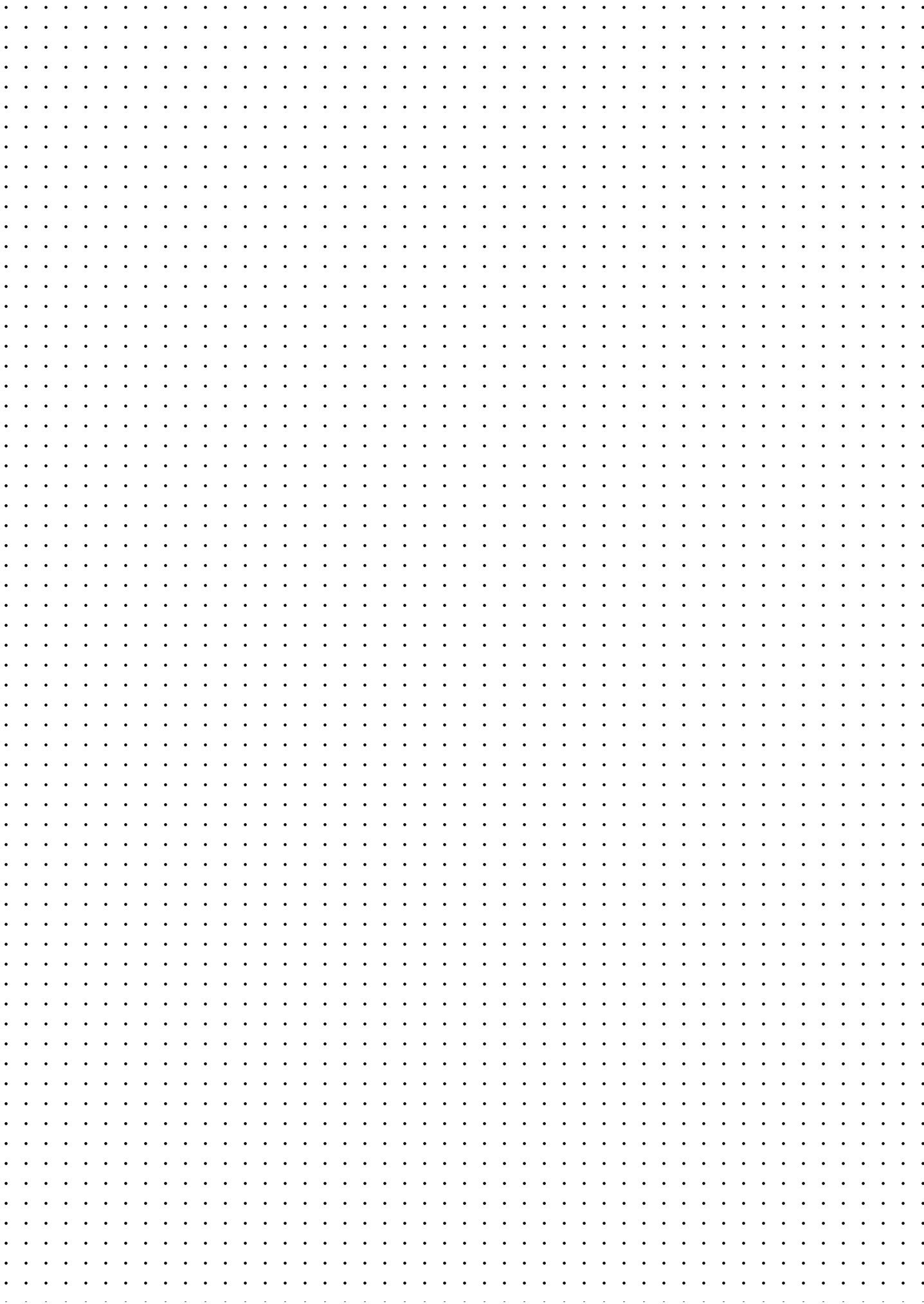
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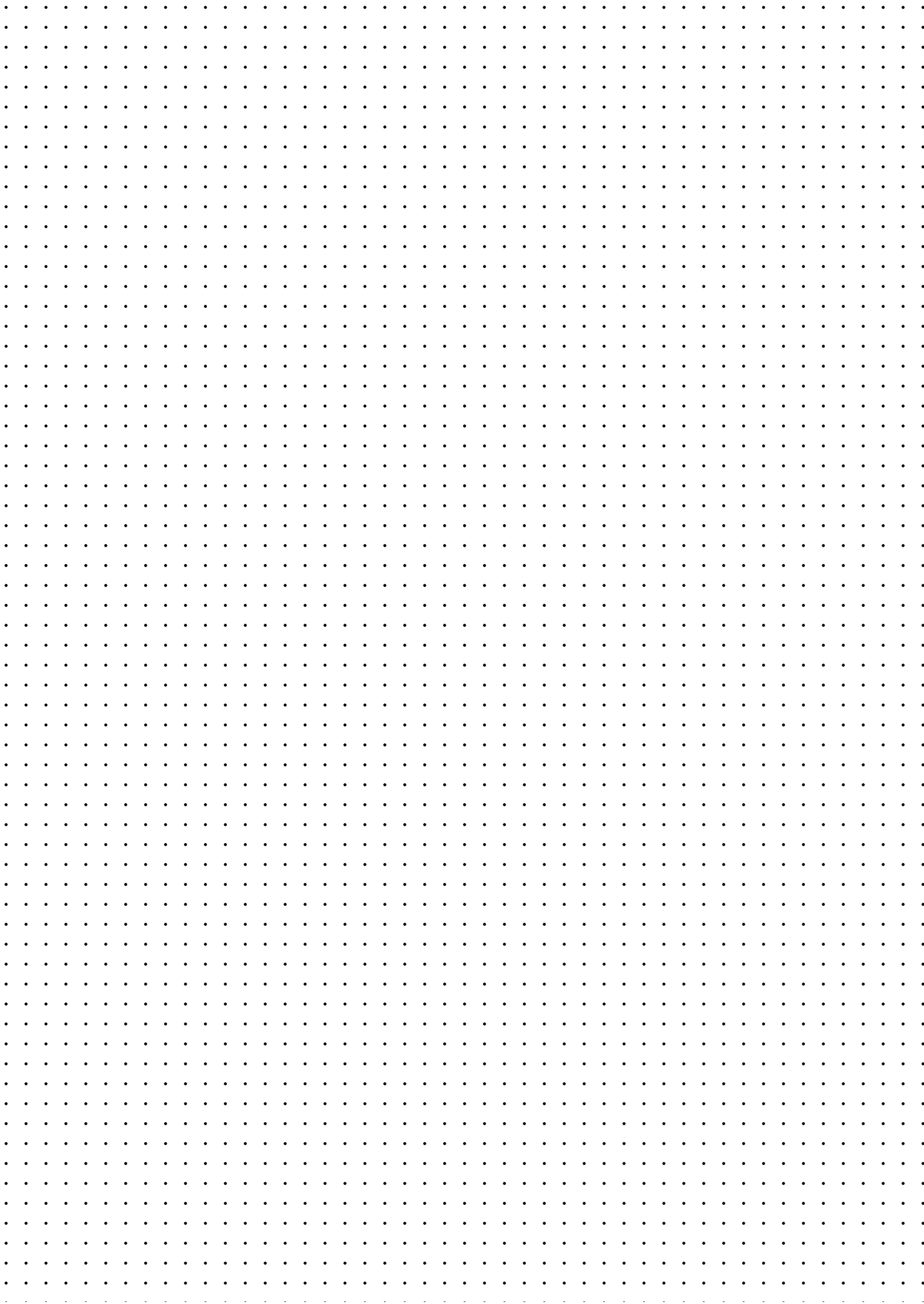
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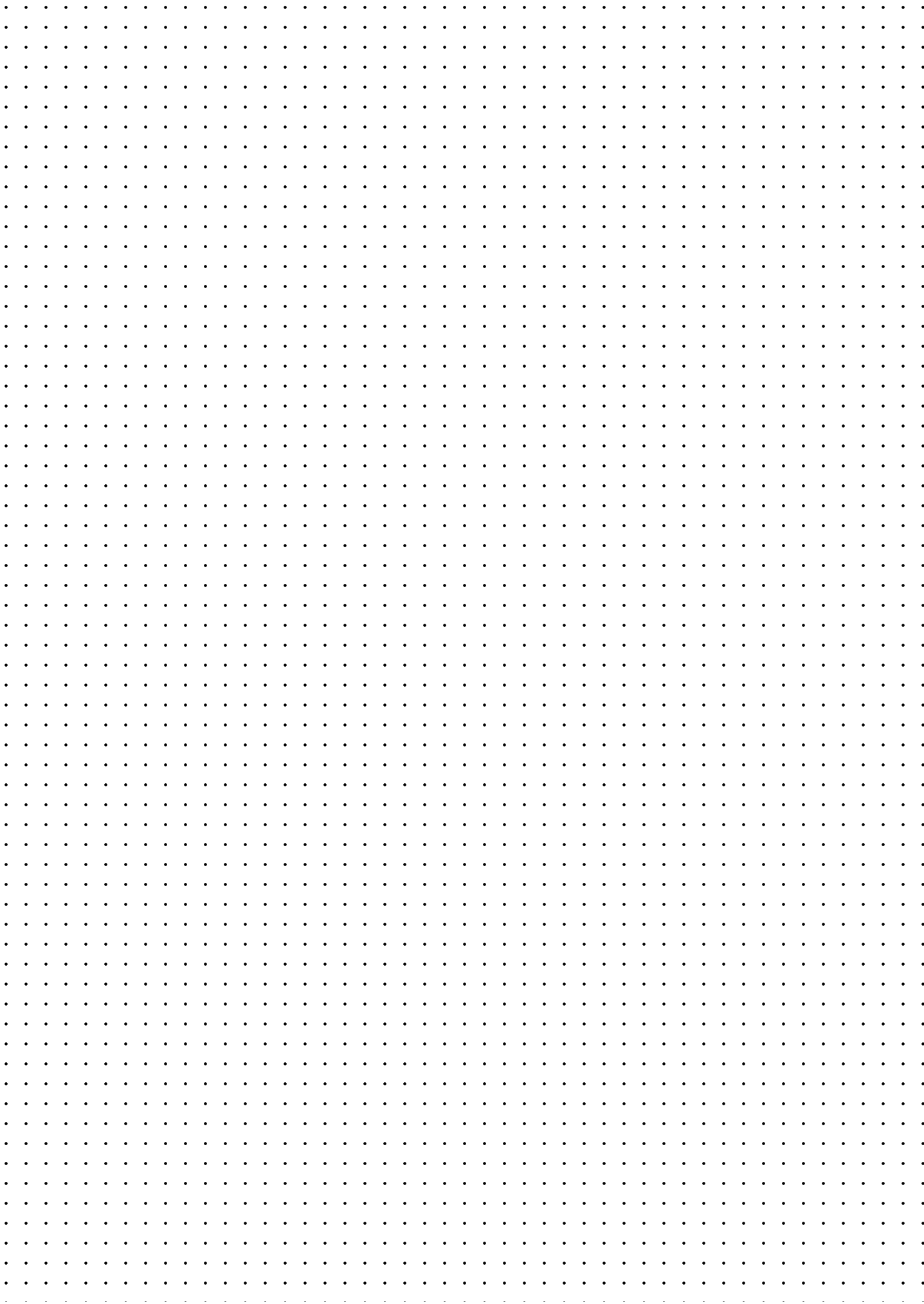
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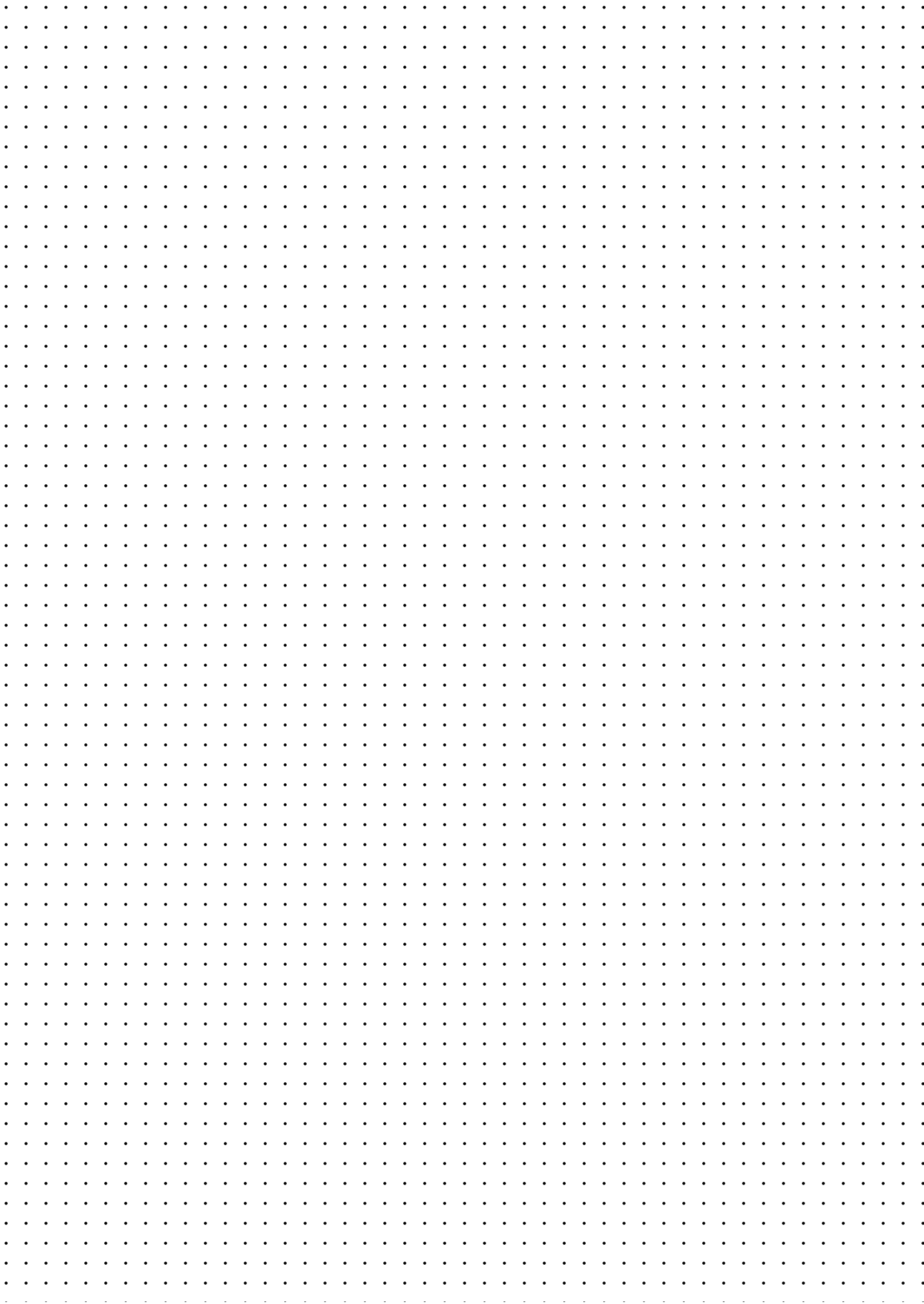
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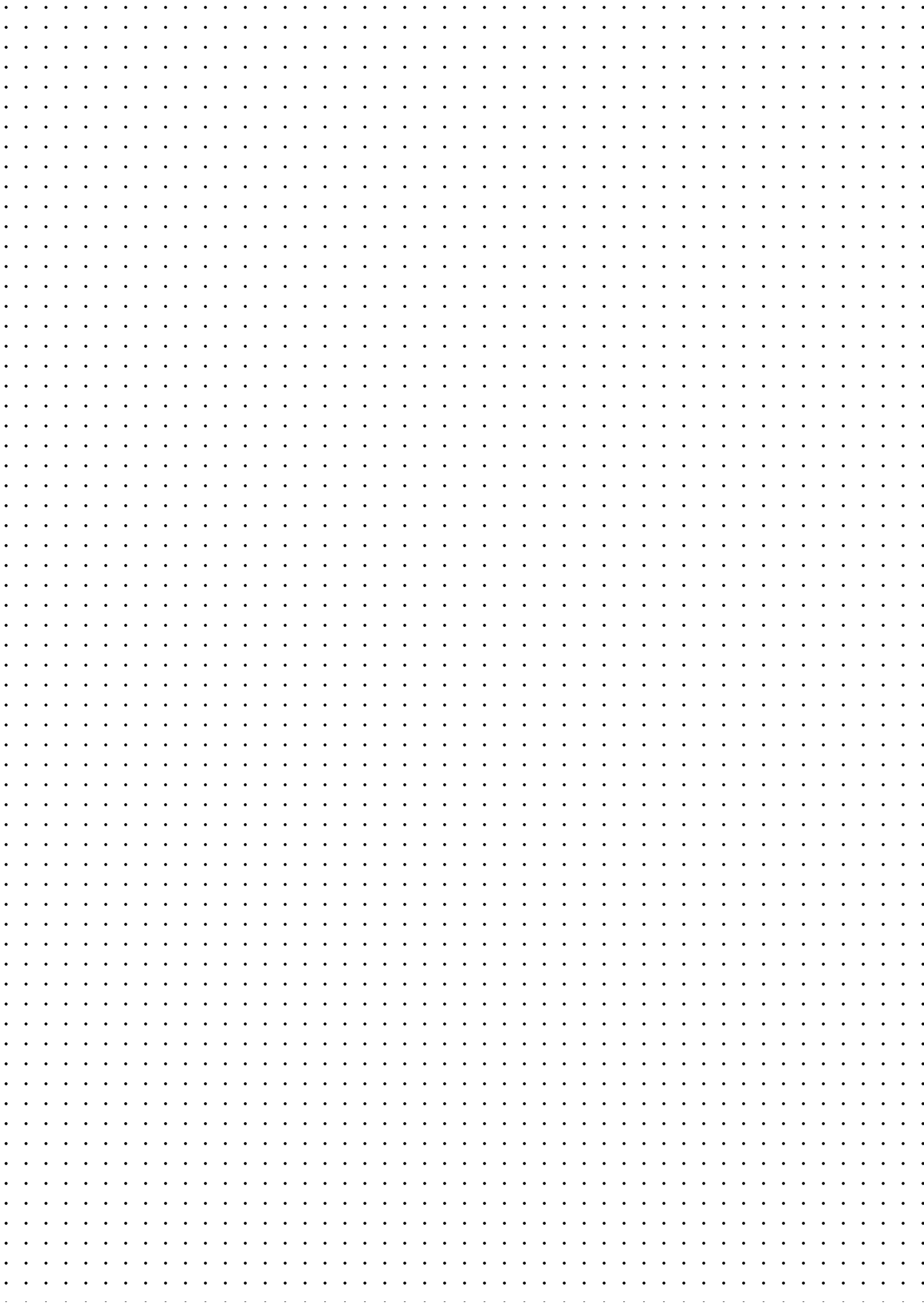
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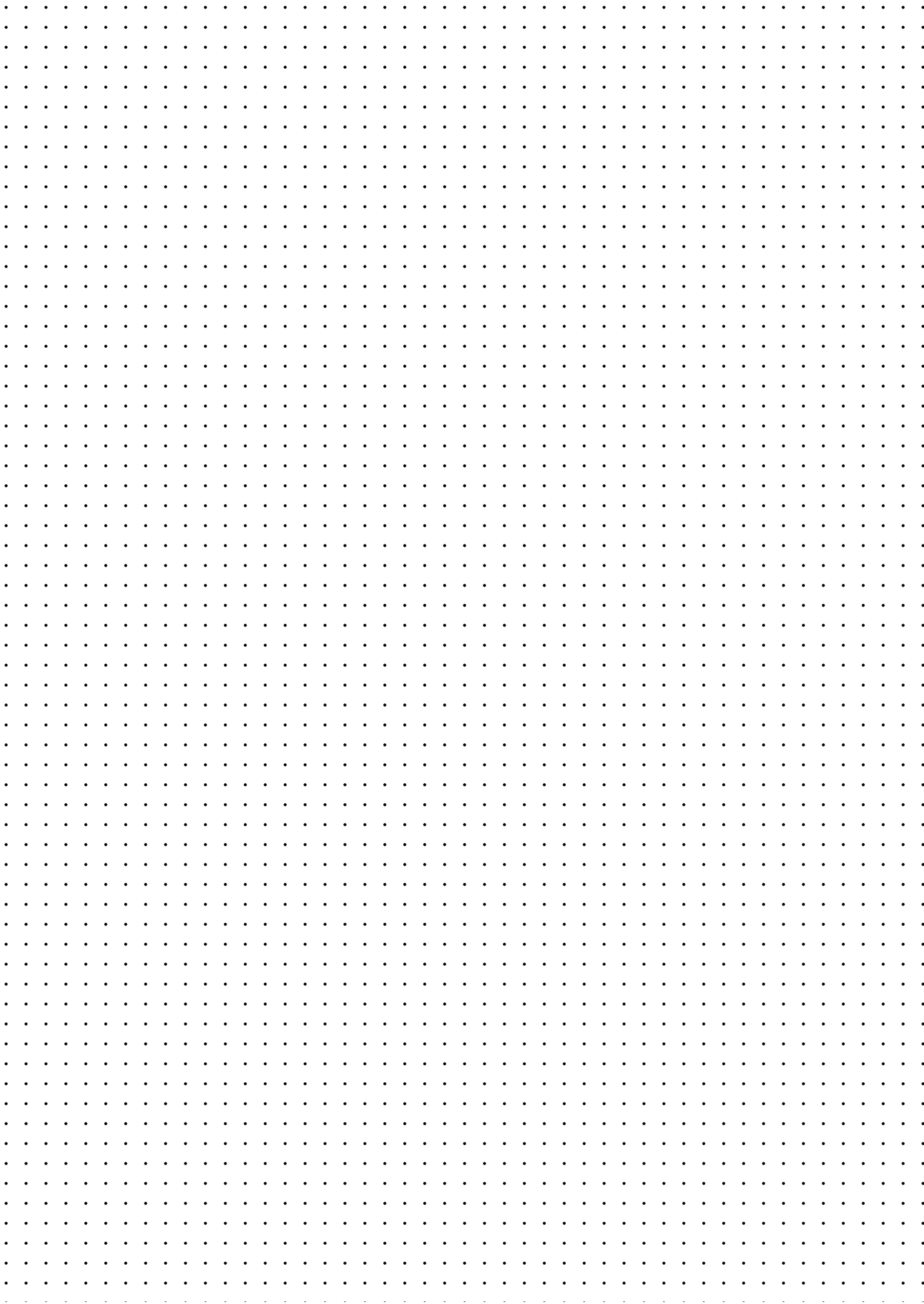
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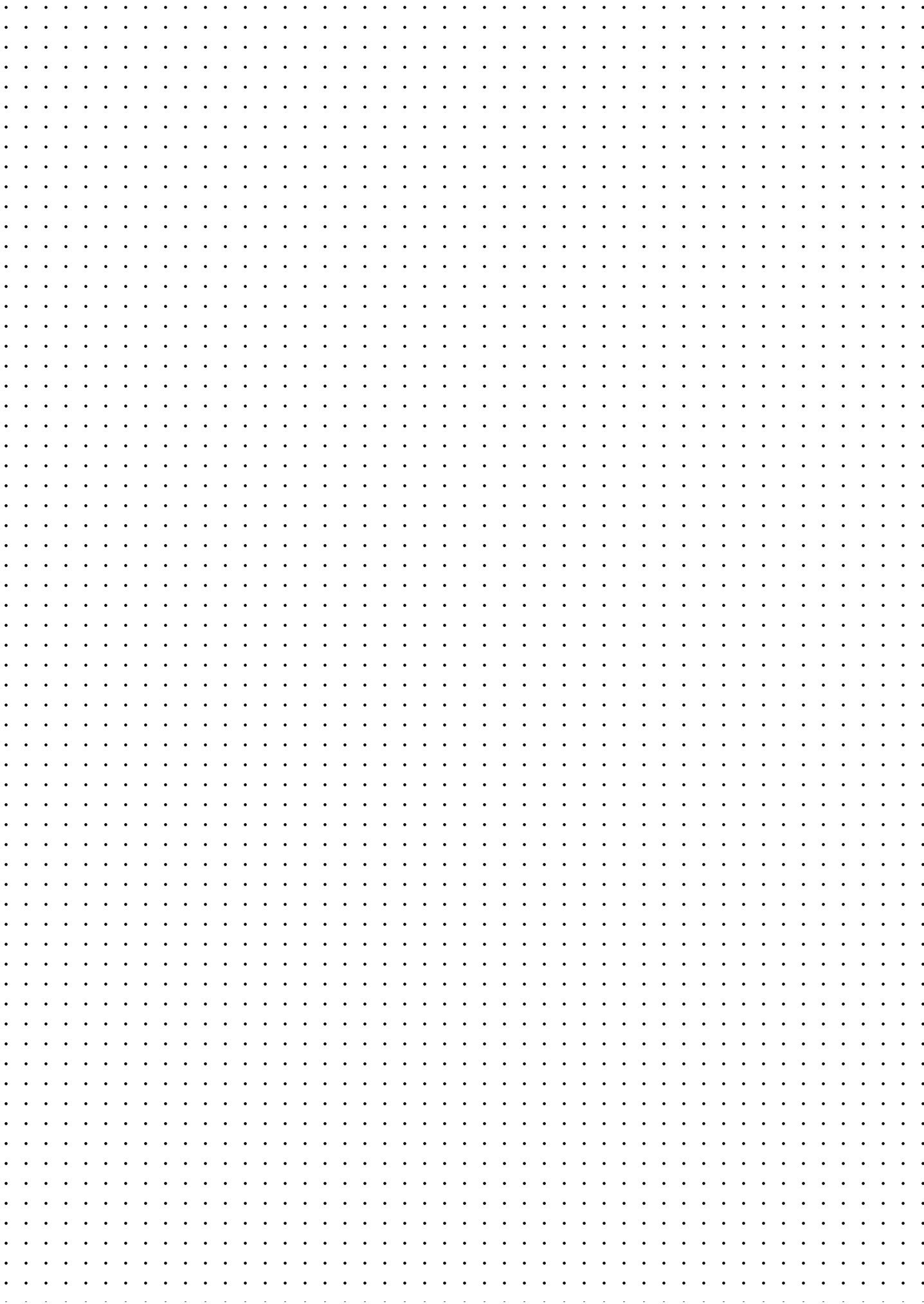
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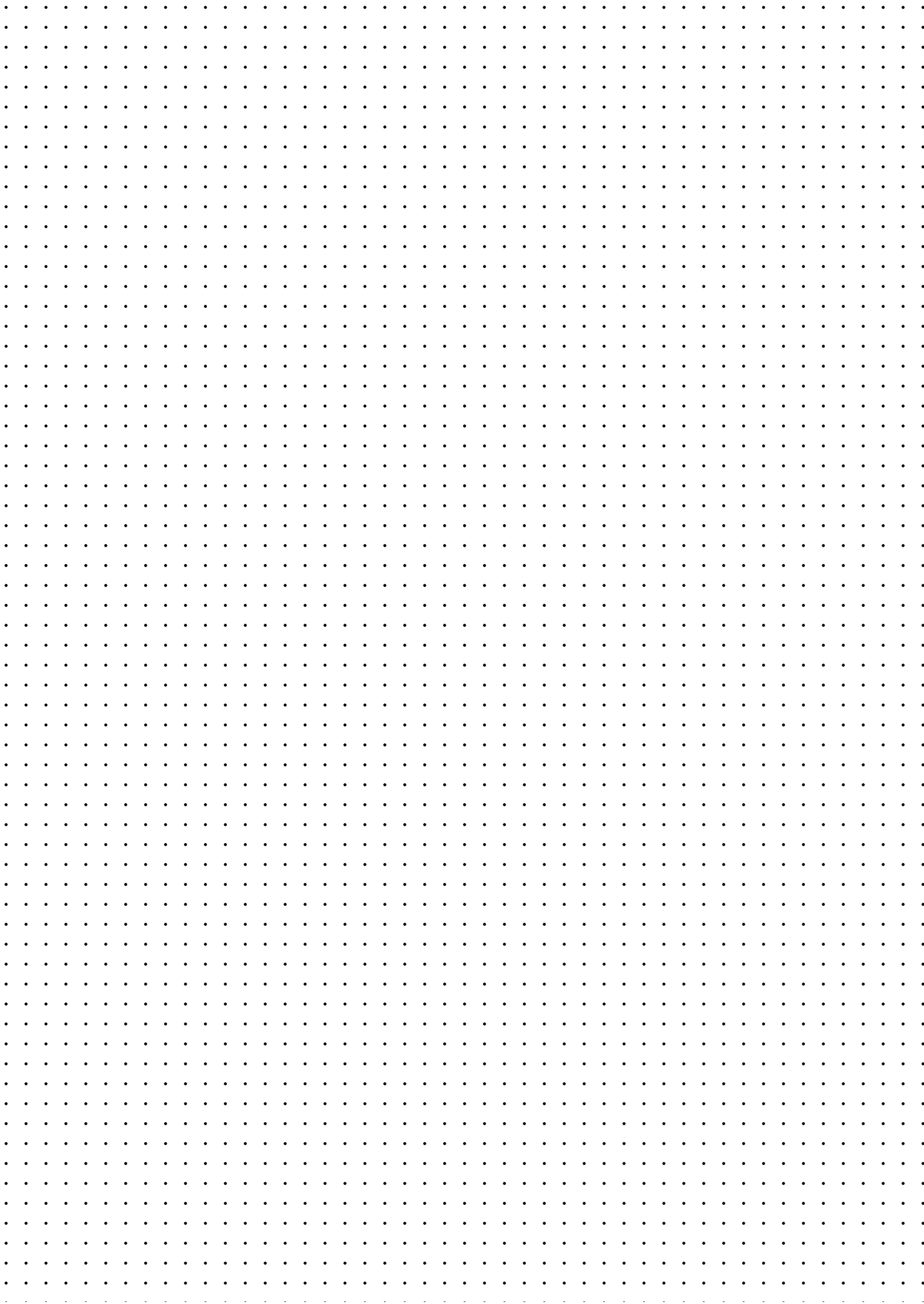
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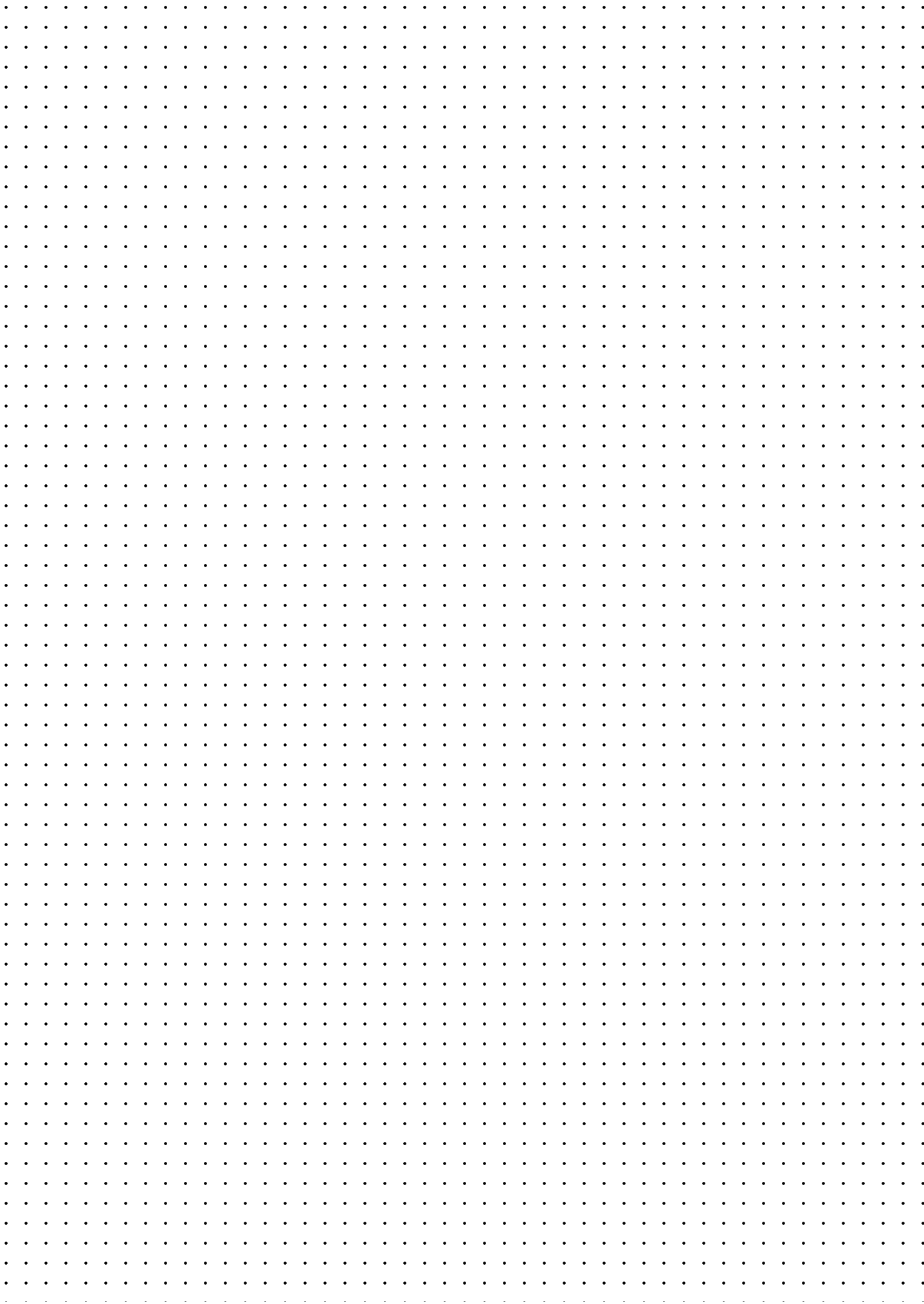


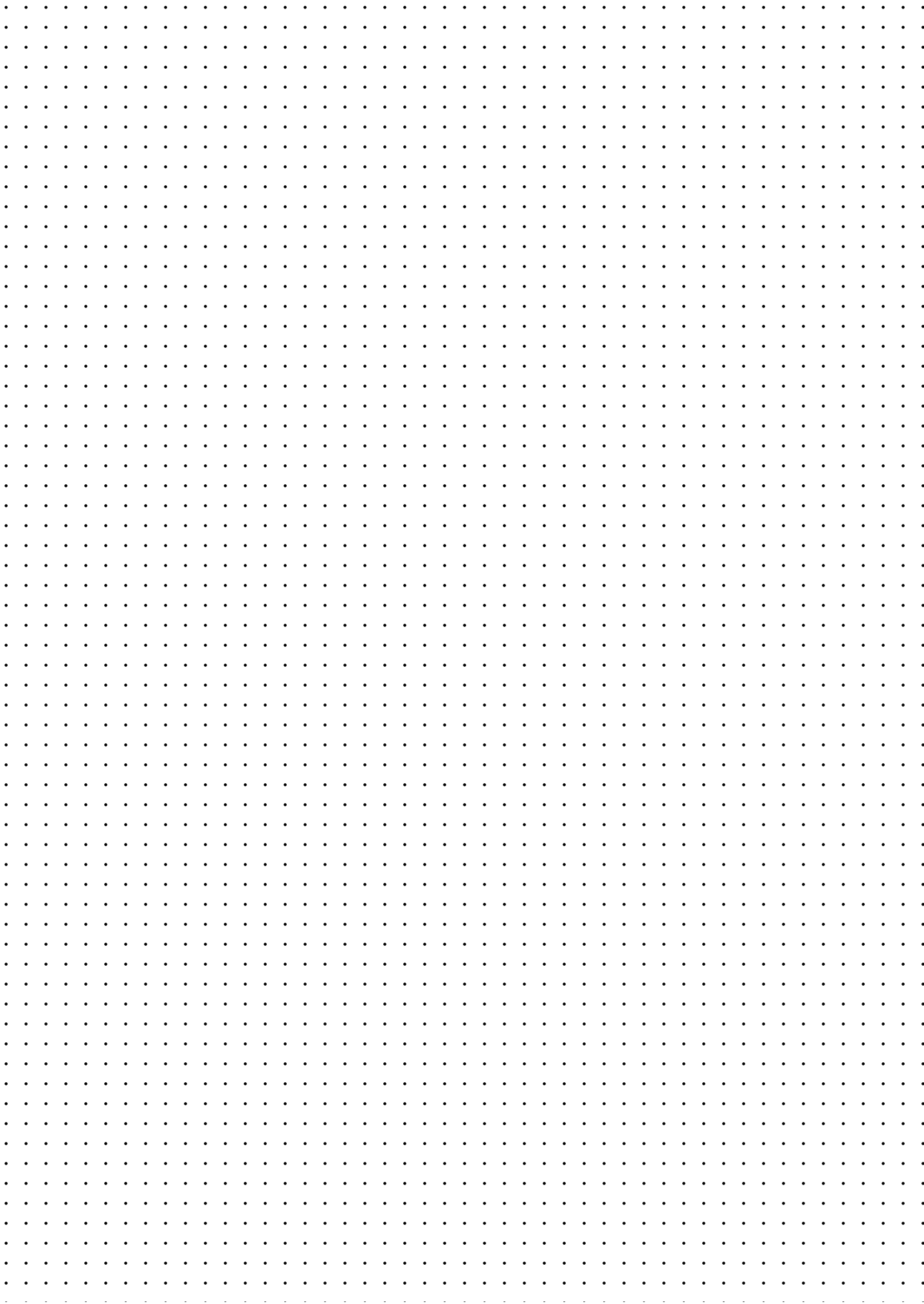


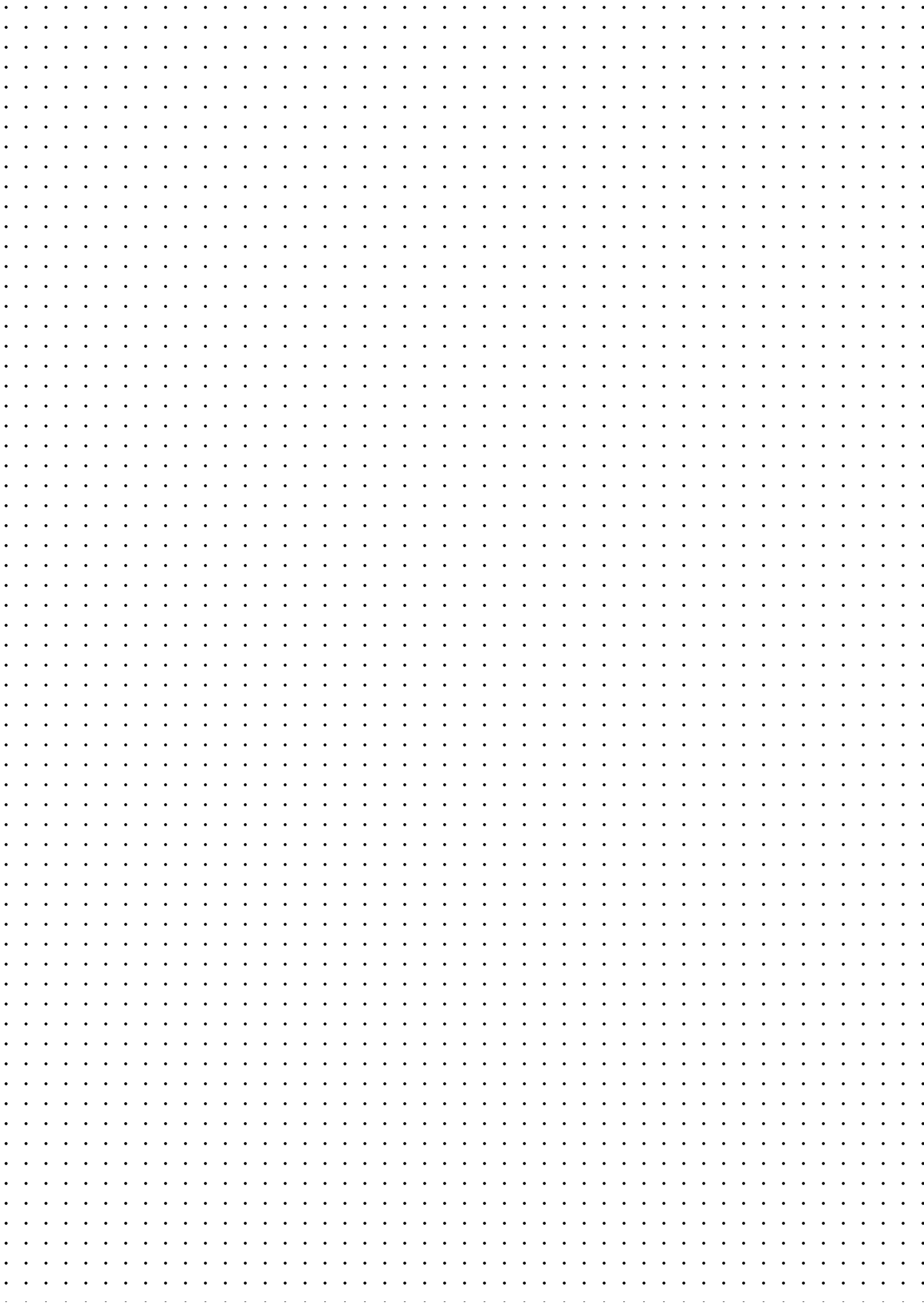
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